Take-home exam: December 19, 2012.

Course name: Network Optimization **Course no:** 42115

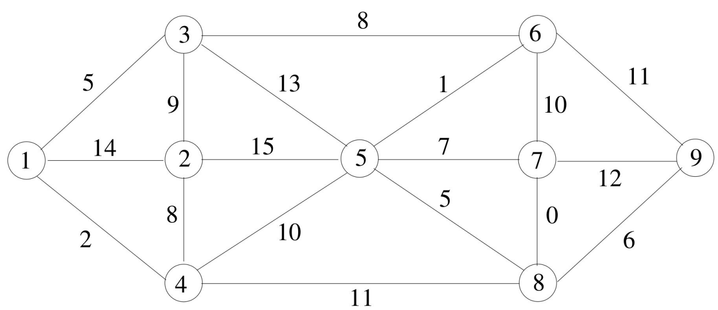
Rules: You are allowed to use your books, notes, slides, calculator, LP-solver etc. You are *not* allowed to get help from other people. You *must* fill in and sign the front page (last page in this document).

Assignment: The assignment consists of 15 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found at the last pages of this assignment. You can print the figures, and use them in your answer by drawing on the figure.

Points: Small problems give 4 points, medium problems give 6 points, large problems 10 points, and xlarge problems give 14 points. A total of 100 points can be obtained.

Minimum spanning tree

Given the following graph $G = (V, E)$

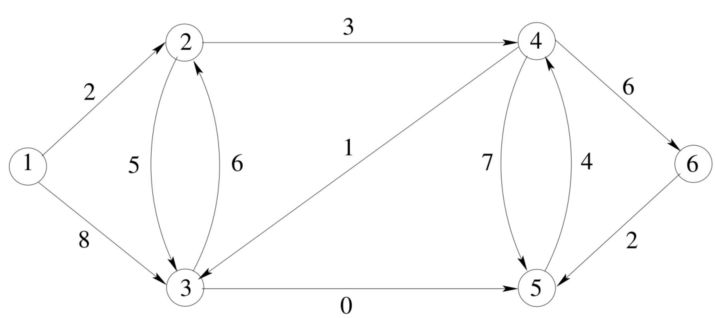


Problem 1: (small) Find the minimum spanning tree of G using Kruskal's algorithm ■

Problem 2: (small) Find the minimum spanning tree of G using Prim's algorithm ■

It is sufficient to indicate the chosen edges on the graph and to give each edge a sequence number (in which order they have been considered).

Shortest path



Problem 3: (small) Use Dijkstra's algorithm to solve the above shortest path problem starting from node 1. ■

It is sufficient to indicate the chosen edges in a shortest path tree on the graph and to give each edge a sequence number (in which order they have been considered). Moreover, the distance to each node should be indicated on the node.

Shortest path tree

Problem 4: (large) Are the following statements correct? (Prove or disprove each statement. To disprove a statement it is sufficient to show an example.)

a) The shortest path tree is unique when all edges have distinct weights.

b) In a directed graph with positive edge weights the shortest paths do not change when directed edges are replaced with undirected edges.

c) In a shortest path problem, if all edge weights are increased by adding a constant $k > 0$ then the cost of a shortest path is also increased by a multiple of k.

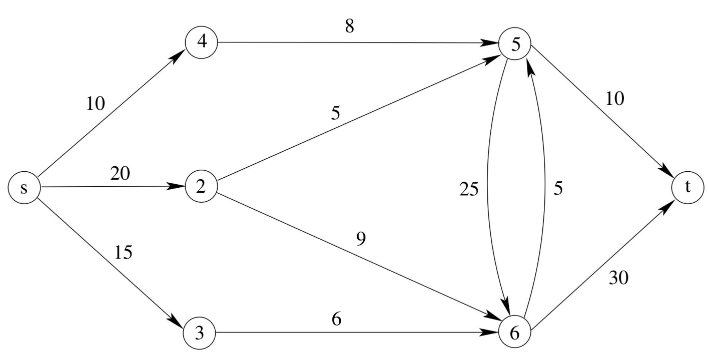
d) Among all shortest paths in $G = (V, E)$ the Dijkstra algorithm finds the shortest path with fewest edges.

e) Among all shortest paths in $G = (V, E)$ the Bellman-Ford algorithm finds the shortest path with fewest edges.

■

Maximum flow problem

Consider the maximum flow problem with source s and sink t

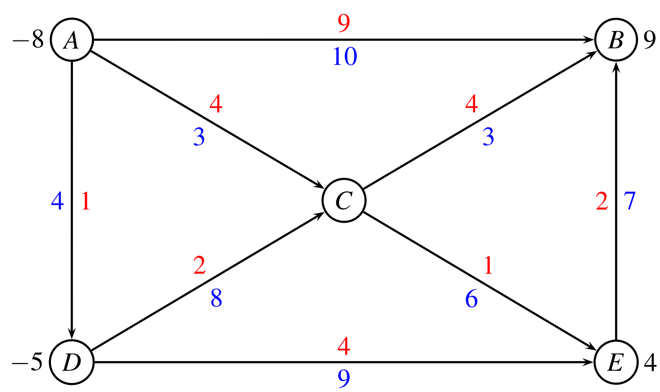


Problem 5: (medium) Solve the max flow problem using an augmenting path algorithm. In each iteration let the augmenting path be the shortest (measured in number of edges) path from s to t in the residual graph. In each iteration write down which augmenting path was used. ■

Problem 6: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Problem 7: (small) Find first a legal start solution by solving an appropriate max-flow problem. ■

It is sufficient to draw the max-flow problem and to show the solution. No details about how the max-flow problem is solved are needed.

Problem 8: (large) Use the augmenting cycle algorithm on the above network. In each iteration, write down which augmenting cycle was used. ■

No details about how you found an augmenting cycle are needed.

Text formatting

The well-known document processing program LATEX uses an optimization procedure to decompose a paragraph into several lines so that when lines are left- and right-adjusted, the appearance of the paragraph will be the most attractive. Suppose that a paragraph consists of n words and that each word is assigned a sequence number. Let $c_{ij} > 0$ denote the attractiveness of a line if it begins with the word i and ends with word $j-1$. Bigger values of c_{ij} are better. The program LATEX uses formulas to calculate the value of each c_{ij}, and we can assume they are given.

Problem 9: (large) Given the c_{ij} values, show how to formulate the problem of decomposing the paragraph into several lines of text in order to maximize the total attractiveness (of all lines) as a shortest path problem. ■

Problem 10: (small) Does the graph have any negative cycles? ■

Problem 11: (medium) Propose an algorithm for solving the above problem. Try to find an algorithm with the best possible running time (in big-Oh notation). ■

Hotel planning

A hotel has received a number of requests for the following week as shown in the following table

start\end	Wed	Thu	Fri
Mon	2	3	-
Tue	4	3	5
Wed	-	4	3

Each entry in the table gives the number of guests requesting a room from the ”start” day to the ”end” day. A dash indicates that no requests have been given. A stay may not be split.

The hotel gets 2$ for a 1-day stay, 5$ for a 2-day stay, and 8$ for a 3-day stay. The hotel wants to maximize the revenue. However, only the following number of rooms is available each of the nights:

Mon-Tue	Tue-Wed	Wed-Thu	Thu-Fri
5	6	8	7

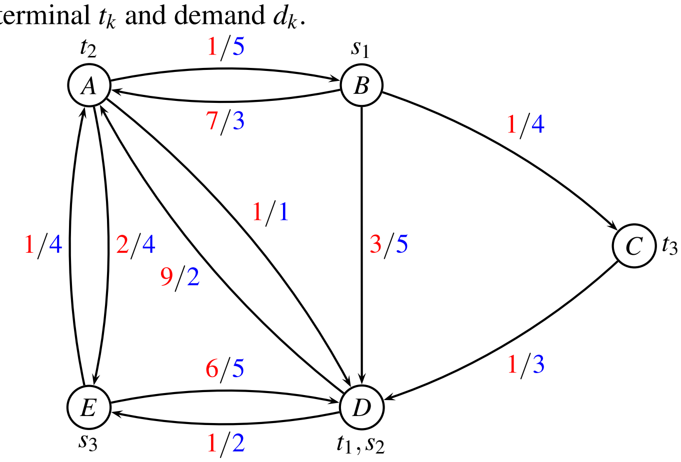
Problem 12: (xlarge) Formulate the problem as a minimum cost flow problem by drawing a graph $G = (V,E)$ where each edge (i, j) has a cost c_{ij}, and an upper bound on the flow u_{ij}. Moreover, every node i has a demand b_i which is negative for a source, and positive for a sink. ■

Hint: See application 9.4 in Ahuja, Magnanti, Orlin ”Network Flows”. (*Available on campusnet*).

Problem 13: (medium) Solve the problem with CPLEX or any other solver. Enclose a copy of the MIP model, and report the optimal solution. ■

Multicommodity flow

Consider the following unsplittable multicommodity flow problem where each edge has a cost and a capacity c_{ij}/u_{ij}. There are three commodities as indicated in the table, each commodity k having source s_k, terminal t_k and demand d_k.



k	s_k	t_k	d_k
1	B	D	1
2	D	A	2
3	E	C	2

A path formulation of the problem is given in the last page of the appendix.

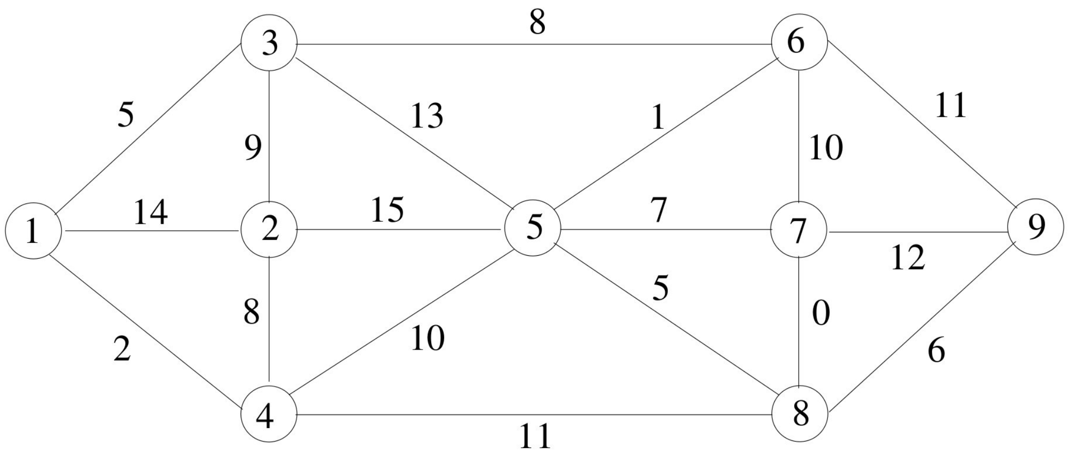
Problem 14: (small) Fill in the objective value in the path formulation, and solve the problem to LP-optimality and IP-optimality. Notice that the number of commodities flown is handled in the last constraints, as opposed to the project assignment. ■

Problem 15: (large) Run column generation starting from columns corresponding to the variables x_1, x_5, x_7. ■

Hint: In the last question you can add a constraint $x_i \leq 0$ to all columns not being used. It is sufficient to show the input (written as an LP-problem) and the primal/dual solution in each iteration. You may use CPLEX or any other LP-solve. When reporting the input it should be written as a normal LP-problem.

Appendix

The figures on the following pages can be used when solving the problem. It is allowed to write by hand on the figures in your answer.



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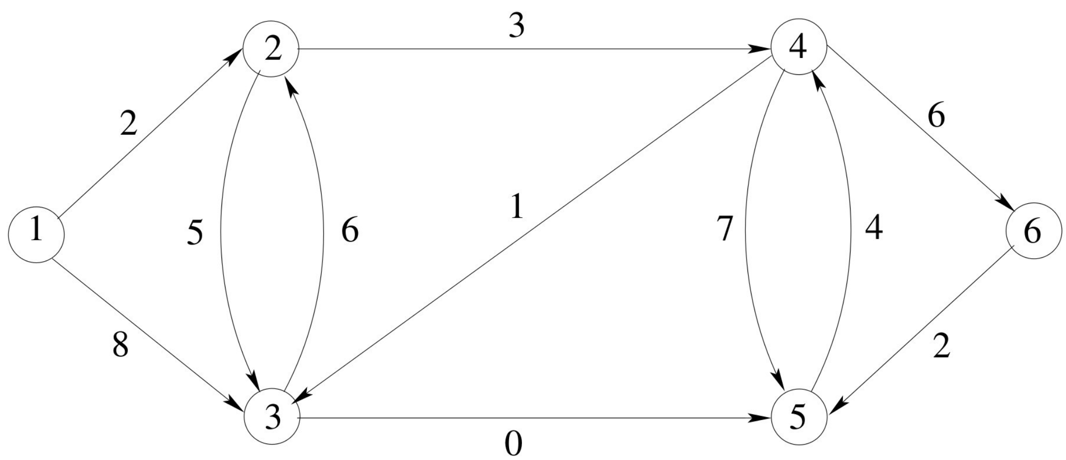
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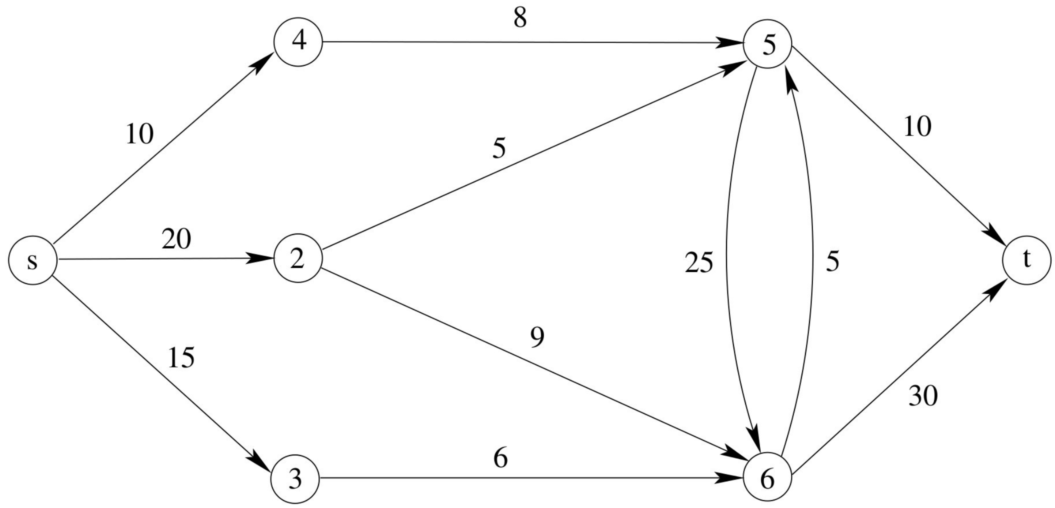

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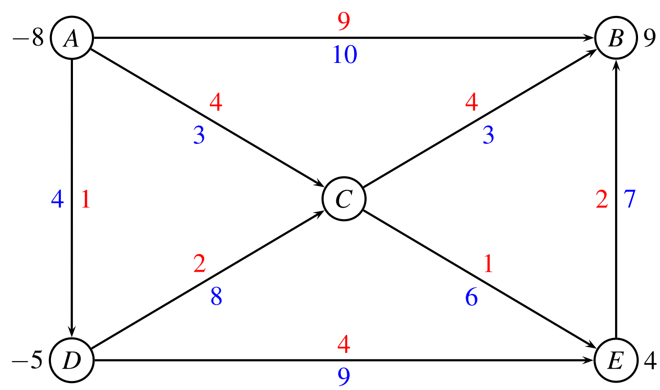

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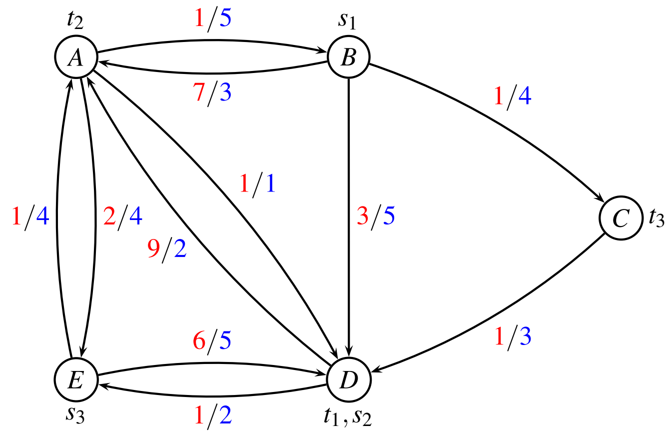

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1/4
2/4
3/5
C
t_3
9/2
6/5
1/3
E
D
s_3
1/2
t_1, s_2

```
minimize
 + x1+ x2+ x3+ x4+ x5+ x6+ x7+ x8
subject to
AB:                   +x7+x8 <= 5
AD: +x1                      <= 1
AE:    +x2                   <= 4
BA: +x1+x2                   <= 3
BC:       +x3         +x7+x8 <= 4
BD:          +x4             <= 5
CD:       +x3                <= 3
DA:             +x5      +x8 <= 2
DE:                +x6       <= 2
EA:                +x6+x7    <= 4
ED:    +x2               +x8 <= 5

K1: +x1+x2+x3+x4 >= 1
K2: +x5+x6 >= 2
K3: +x7+x8 >= 2

end
```

DTU Danmarks Tekniske Universitet **Front page**

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Course name: Network Optimization **Course no:** 42115

Name	study number

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Date Signature

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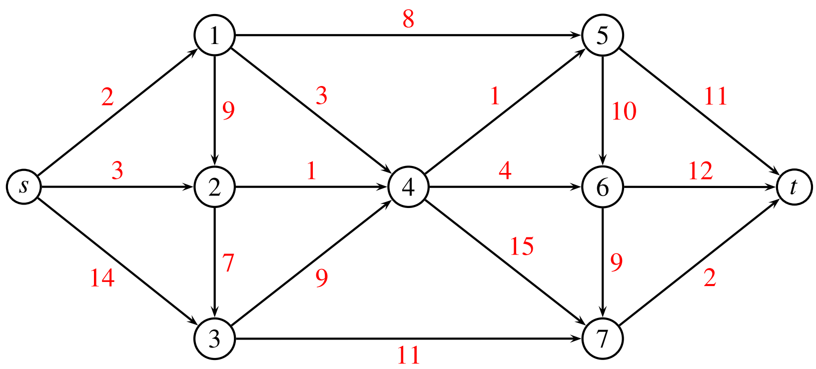
Maximum Flow

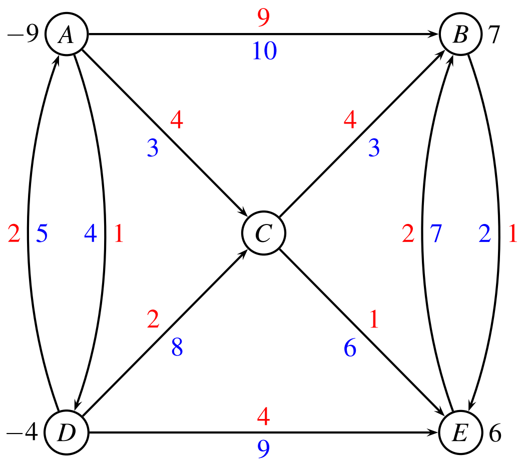


Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration write down which augmenting path was used, and how many units were sent. ■

Problem 2: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Problem 3: (small) Find first a legal start solution by solving an appropriate max-flow problem. ■

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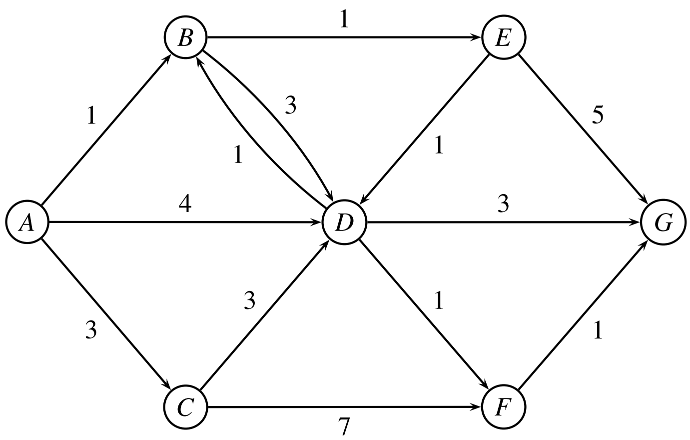
Problem 4: (large) Use the augmenting cycle algorithm on the above network. In each iteration, write down which augmenting cycle was used. ■

No details about how you found an augmenting cycle are needed.

Shortest path

Consider the following directed graph with distances w indicated on the edges:



Problem 5: (medium) Use Dijkstra's algorithm to solve the above shortest path problem starting at node A. Give sufficient details, so the algorithm can be followed. Draw the resulting shortest path tree. ■

We now consider a time constrained shortest path problem, where every edge counts as 1 time unit. So, for instance, the path $A \to D \to G$ has distance 2 since it passes two edges.

Problem 6: (small) Using the above shortest path tree, calculate the time consumption to every node. ■

We will now solve a time-constrained shortest path problem. For this purpose we use the Label Correcting algorithm from Slide 8 in the lecture "Resource Constrained Shortest Path Problem".

Problem 7: (medium) Initiate each node with a list of labels $L_i = \{(w,t)\}$ containing one label (w,t) which is the distance and time found in question 6.

Use the Label Correcting algorithm to find the shortest path to every node using at most 4 time units. Give sufficient details so that your solution approach can be followed. Indicate the found solution tree and the found distances. ■

Nurse scheduling

A hospital administrator needs to establish a staffing schedule for nurses that will meet the minimum daily requirements shown in the following table

shift	1	2	3	4	5	6
time	6:00-10:00	10:00-14:00	14:00-18:00	18:00-22:00	22:00-2:00	2:00-6:00
min nurses	70	80	50	60	40	30

A work-plan for a nurse is called a duty. A duty normally consists of two consecutive shifts, although a duty consisting of only shift 6 is also allowed. A duty consisting of shift 6 followed by shift 1 is not allowed. The cost of a shift 1 to 4 is set to 1 unit. The cost of night shifts 5 and 6 is set to 2 units. The cost of a duty is the sum of shift-costs.

The administration wants to schedule the employes so that a sufficient number of nurses is available for each period while minimizing the overall costs.

Problem 8: (medium) Formulate the problem as a minimum cost flow problem (i.e. a graph with indicated edge costs, capacities, sources, and sinks). Hint: see Application 9.6 in Ahuja, Magnanti, Orlin, "Network Flows". ■

Problem 9: (medium) Write up the network formulation from previous question as an LP-model ■

Problem 10: (small) Solve the model with CPLEX ■

Student assignment

In a academic program at DTU each master's degree student is required to undertake a 6-month internship project at a company site. Since the projects are such an important component of the student's educational program and vary considerably by company and by geography, each student would like to undertake a project of his or her liking. To assure that the project assignments are "fair", the students and program administrators have decided to use an optimiziation approach: Each student ranks the available projects in order of increasing preference (1 indicate most prefered project). If the student does not like a project, it is marked with a dash. The objective is to assign students to projects to achieve the best sum of total ranking of assigned projects. The project assignment has several constraints. Each student must work on exactly one project, and each project has an upper limit on the number of students it can accept. Each project must have a supervisor, drawn from a known pool of eligible faculty. Finally, each faculty member has an upper bounds on the number of projects that he or she can supervise. Also, each faculty member has preferences to which projects he or she wish to supervise (1 indicate most prefered project).

In a given semester we have 4 students, who have to be assigned to 3 projects. The preferences of the students are given in the following left table.

There are 4 faculty members for supervising the 3 projects. The preferences of projects are given in the right table. Notice, that all faculty members can supervise all projects.

student \ project	A	B	C
1	1	2	-
2	2	-	1
3	-	1	-
4	-	-	1
max students	2	1	2

project \ professor	U	V	W	X
A	1	1	3	2
B	2	2	2	3
C	3	3	1	1
max projects	2	1	2	1

Student preferences is the first priority, faculty member satisfaction second priority. Since all faculty members can supervise all projects, it is always possible to find a supervisor for a project. Every project needs to have a supervisor, but not all professors need to supervise a project. The supervisor uses the same resources for supervising a project with two students as a project with one student.

Problem 11: (large) Show how this problem can be solved using network flow techniques. (You should formulate it as two independant network problems. It is sufficient to draw the graphs and say which graph problem it is. You do not need to solve the graph problems). ■

Vehicle Routing Problem with time constraints

A delivery company has to visit 5 customers in some pre-described time-intervals $[a_i, b_i]$ as follows:

customer i	earliest arrival a_i	latest arrival b_i
A	1	1
B	2	4
C	1	2
D	3	4
E	2	3

Each customer should be visited at least once.

The delivery company has 5 vehicles available, but not all vehicles need to be used if this is cheaper. All vehicles start at the depot s and return back to the same depot t (we use different names for start and end depot to make the modeling easier).

A *location* is either a depot or a customer. The cost of traveling between two locations is given in the following table (left). The next table (right) shows the travel time between two locations.

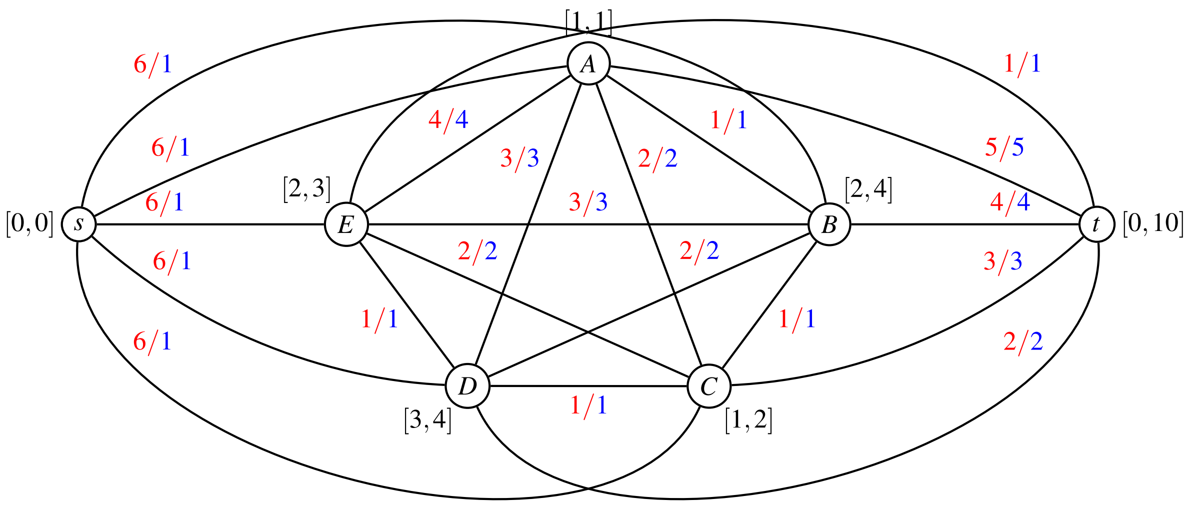
	cost						
	s	A	B	C	D	E	t
s	0	6	6	6	6	6	6
A	6	0	1	2	3	4	5
B	6	1	0	1	2	3	4
C	6	2	1	0	1	2	3
D	6	3	2	1	0	1	2
E	6	4	3	2	1	0	1
t	6	5	4	3	2	1	0

	time						
	s	A	B	C	D	E	t
s	0	1	1	1	1	1	1
A	1	0	1	2	3	4	5
B	1	1	0	1	2	3	4
C	1	2	1	0	1	2	3
D	1	3	2	1	0	1	2
E	1	4	3	2	1	0	1
t	1	5	4	3	2	1	0

It is assumed that the vehicle leaves depot s at time 0. Depot t has time window $[0,10]$ meaning that all routes must be finished within 10 time units. If a vehicle arrives to a location before the specified time window, the vehicle will wait. Waiting does not impose any cost. All loading/unloading times at the locations are included in the driving times.

As an example, a route for a vehicle can start at the depot s at time 0. The vehicle first drives to customer C, arriving at time 1. The vehicle next drives to customer D, arriving at time 2. However, the time window of cutsomer D is $[3,4]$ so the vehicle waits one time unit until time 3 where it can service customer D. The vehicle then returns back to depot t, arriving at time 5. The cost of the route is $6+1+2=9$.

The problem can be represented by the following graph. Every location is represented by a node, and possible transitions between locations are represented by edges.



The red values indicate the costs, while the blue values indicate the travel time. The numbers $[a_i, b_i]$ indicate the time windows of node i.

Problem 12: (small) Find a feasible solution to the problem, using at most 4 vehicles. The route of each vehicle should satisfy the time windows. Report the cost of the solution. ■

We will use a path formulation of the problem, hence we start by writing up all possible routes for the vehicles. We have the following possible routes

route	nodes	cost
1	s,A,B,D,t	?
2	s,B,D,t	?
3	s,C,B,D,t	?
4	s,C,D,t	?
5	s,C,E,t	?
6	s,C,E,D,t	?
7	s,D,t	?
8	s,E,D,t	?
9	s,E,t	?

Dominated routes have been deleted. A route is dominated, if it is more expensive than another route and visits a subset of the nodes.

Problem 13: (small) Calculate the costs of all routes. ■

Problem 14: (small) Calculate the waiting time in each route. The waiting time occurs if the vehicle arrives at a customer before the time window. For instance the route s, C, D, t has a waiting time of 1 unit. ■

Problem 15: (medium) Write up an IP model with all routes and solve it to optimality using CPLEX. It is sufficient to consider undominated routes. ■

Since the number of possible routes can be very big for real-life problems, we wish to solve the problem through column generation. To start the process, we choose the trivial routes where each vehicle only visits one customer on the route:

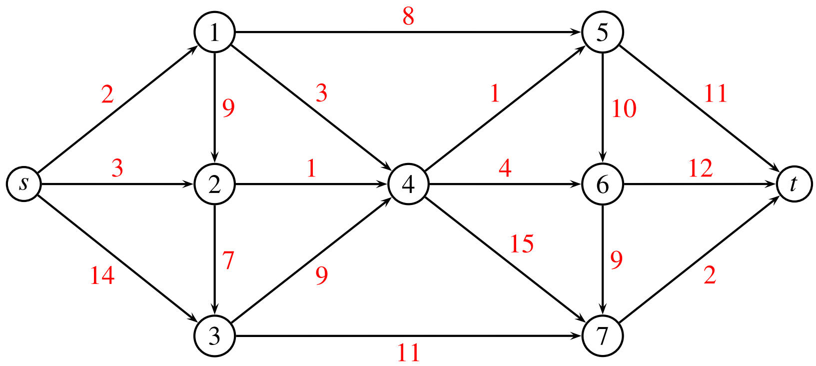
route	cost
s,A,t	11
s,B,t	10
s,C,t	9
s,D,t	8
s,E,t	7

which gives a solution of 45.

Problem 16: (xlarge) Use column generation to solve the problem to LP-optimality. In each iteration report dual values and show how the reduced costs are calculated. ■

Problem 17: (small) Compare the LP solution with the integer solution, and discuss the quality of the bound. ■

end of assignment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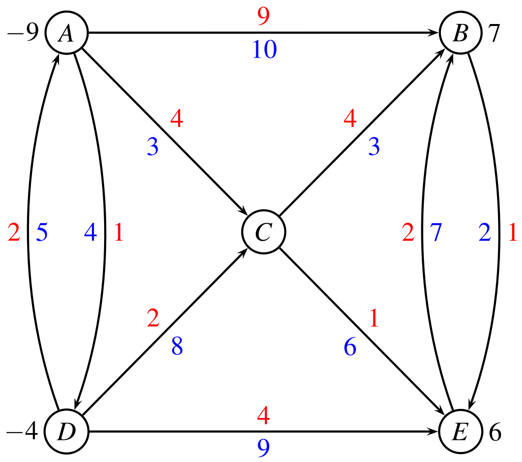

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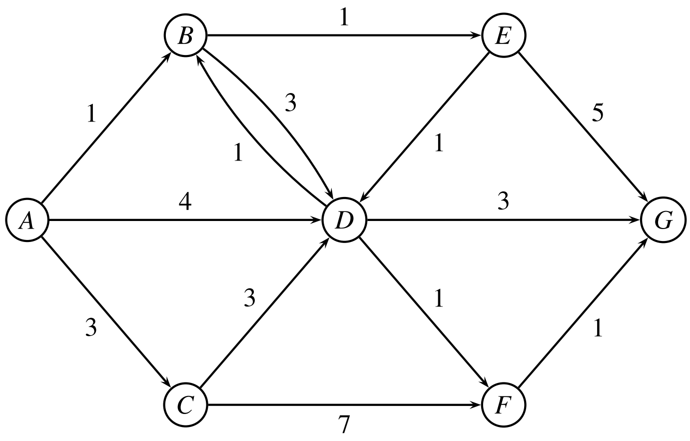

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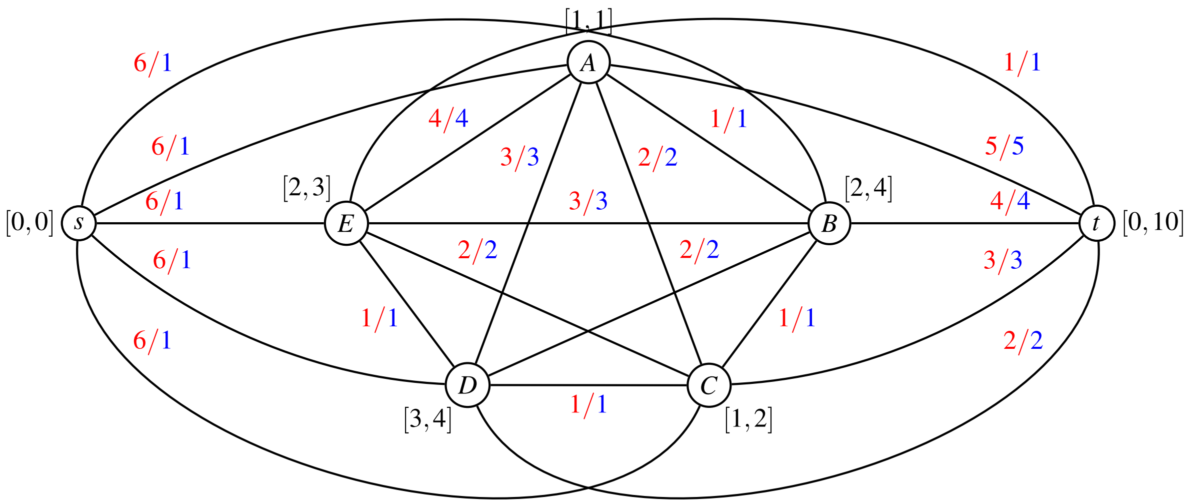

[1, 1]
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DTU Danmarks Tekniske Universitet **Front page**

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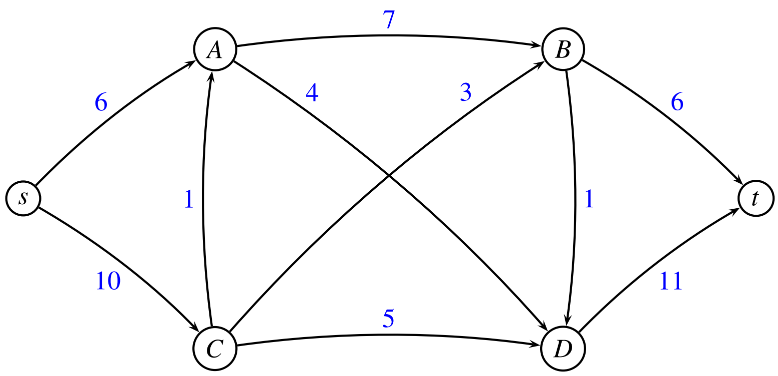
Maximum flow



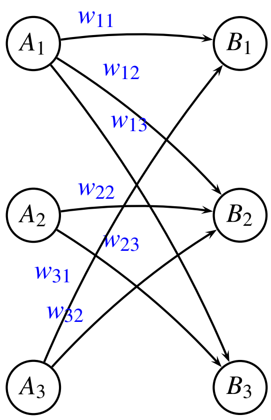
Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. ■

Problem 2: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Bipartite graph

A bipartite graph is a graph whose vertices can be divided into two disjoint sets A and B such that every edge connects a vertex in A to a vertex in B.

A perfect matching is a matching which matches all vertices of the graph, i.e. every vertex of the graph is incident to exactly one edge of the matching.



Problem 3: (medium) Suppose we have a bipartite graph $G = (V,E)$ in which each edge e has a weight w_e. Weights are positive integers. Explain how we can solve the problem to find a perfect matching of maximum total value, if it exists, with help of minimum cost flow algorithms. (Hint: draw a small example.) ■

Problem 4: (medium) Suppose that we want to find a matching of maximum total weight, i.e., we allow that some vertices are not matched. Is it possible to find such a matching with just one run of a minimum cost flow algorithm? Explain why (not). ■

Modeling a hub port

In the project assignment [1] five different ways of modelling a hub port are presented. The five models can also be seen in the appendix at the end of this assignment. It is assumed that all vessels arrive at different days, and that they stay in the port during day time. In model 5 we assume that the vessels are sorted according to their time of arrival.

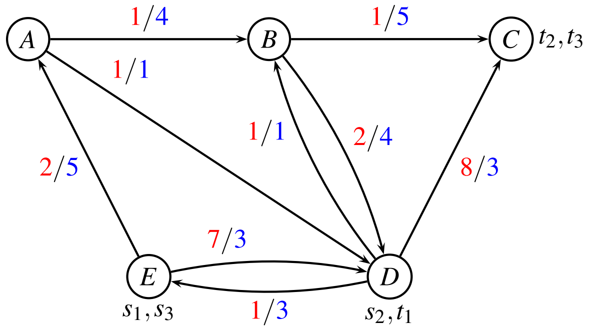
Problem 5: (small) Assume that at most C containers can stand on land from day to day during transshipments. The capacity C is shared among all commodities standing on land. Which of the five models can handle this constraint? Explain why (not). (It is not allowed to assign 0 capacity to some edges to "remove" them, since they may be carrying some other information). ■

Problem 6: (small) Assume that a tax has to be paid when containers are being transshipped from one vessel to another vessel. The tax depends on the nationality of the involved vessels. For instance a container being transshipped from a Danish vessel to a HongKong vessel need to pay a tax $T_{DK,HK}$. Which of the five models can handle such a tax? Explain why (not). ■

Unsplittable multicommodity flow problem

Consider the unsplittable multicommodity flow problem from the project assignment [1].

ASIANET **(Route net of a liner shipping company)**



k	s_k	t_k	d_k
1	E	D	2
2	D	C	2
3	E	C	1

For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. The right table describes the commodities k, with source s_k, terminal t_k and demand d_k.

We use *column generation* to solve the LP-relaxed path formulation of ASIANET, starting from the following routes:

commodity k	route	cost
1	E A B D	10
1	E D	14
2	D E A B C	10
3	E A B C	4

This leads to the following *restricted master problem*

$$
\begin{array}{llllllll}
\min & 10x_1^1 & + 14x_2^1 & + 10x_1^2 & + 4x_1^3 & & & \\
\text{s.t.} & 2x_1^1 & & + 2x_1^2 & + 1x_1^3 & \leq 4 & & (y_{AB}) \\
 & & & & & \leq 1 & & (y_{AD}) \\
 & & & 2x_1^2 & + 1x_1^3 & \leq 5 & & (y_{BC}) \\
 & 2x_1^1 & & & & \leq 4 & & (y_{BD}) \\
 & & & & & \leq 1 & & (y_{DB}) \\
 & & & & & \leq 3 & & (y_{DC}) \\
 & & & 2x_1^2 & & \leq 3 & & (y_{DE}) \\
 & 2x_1^1 & & + 2x_1^2 & & \leq 5 & & (y_{EA}) \\
 & & 2x_2^1 & & & \leq 3 & & (y_{ED}) \\
 & 1x_1^1 & + 1x_2^1 & & & \geq 1 & & (\overline{y}_1) \\
 & & & x_1^2 & & \geq 1 & & (\overline{y}_2) \\
 & & & & x_1^3 & \geq 1 & & (\overline{y}_3) \\
\end{array}
\qquad (1)
$$

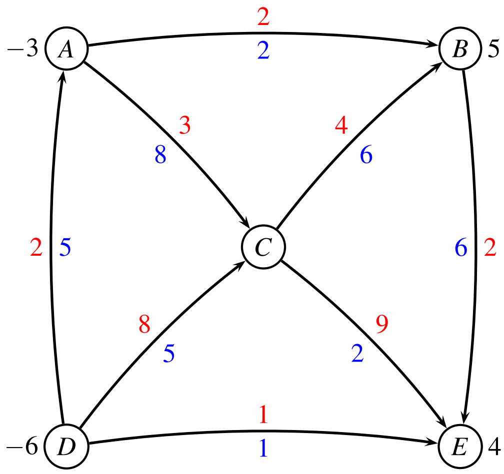
$$0 \leq x_1^1, x_2^1, x_1^2, x_1^3 \leq 1$$

Solving the above problem we get the primal solution $x_1^1 = \frac{1}{2}, x_2^1 = \frac{1}{2}, x_1^2 = 1, x_1^3 = 1$. The dual variables are $y_{AB} = -2$, and $\overline{y}_1 = 14$, $\overline{y}_2 = 14$, $\overline{y}_3 = 6$. All other dual variables are 0.

Problem 7: (medium) What is the next column that should be added to the model? (give sufficient details to justify your choice) ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i in every node is positive for a sink, and negative for a source.



Having solved a maximum flow problem to find a feasible solution we are told that $L = \{(D,C)\}$ and $U = \{(D,E),(A,B),(C,E)\}$ and $T = \{(D,A),(A,C),(C,B),(B,E)\}$. The flow on edges in U is at the upper bound, the flow in edges L is at the lower bound, while the edges in T form a tree.

Problem 8: (small) Calculate the flow corresponding to the above feasible solution defined by T,U,L. ■

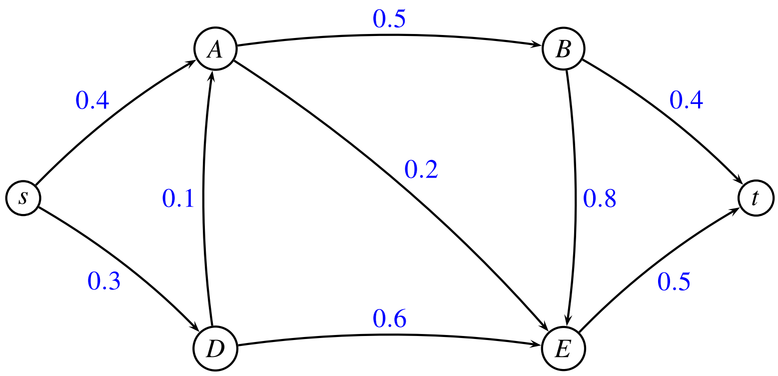
Problem 9: (large) Use the network simplex algorithm to solve the min cost flow problem to optimality. ■

Shortest path

In a directed communication network, a message is sent from a node s to another node t. At each arc (i,j) of the network there is a probability p_{ij} that the message will get lost on this arc. The probabilities p_{ij} are fixed and independent. An internet provider wants to choose a path from s to t, such that the probability that the message gets lost on this path is minimal. The probability that a message is *not* lost along a path using two edges (i,j) and (j,k) is $(1-p_{ij})\cdot(1-p_{jk})$.

Problem 10: (medium) Argue that a path from s to t minimizing the probability of being lost can be found by solving a shortest path problem from s to t where the edge weights w_{ij} are chosen as $w_{ij} = -\log(1-p_{ij})$ for edge (i,j). Hint: $\log(x\cdot y) = \log x + \log y$. ■

Problem 11: (small) Use the transformation for the following example where probabilities p_{ij} are marked with blue on the edges. It is sufficient to use two decimal digits precision in the transformation.



■

Problem 12: (medium) Choose an appropriate shortest path algorithm (argue why you choose it), and find the shortest path from s to t in the resulting graph. ■

end of assignment

References

[1] David Pisinger, "Flowing containers in liner shipping", *42115 Network Optimization, Project Assignment, DTU 2014*. can be downloaded from DTU CampusNet.

DTU Danmarks Tekniske Universitet **Front page**

Take-home exam: December 18, 2014.

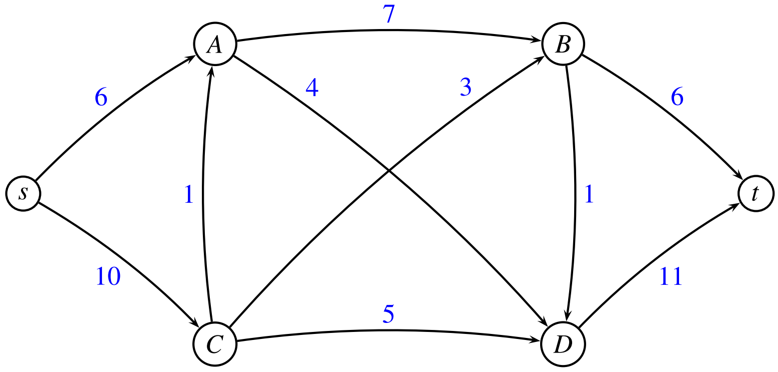
Course name: Network Optimization **Course no:** 42115

Name (printed letters)	study number

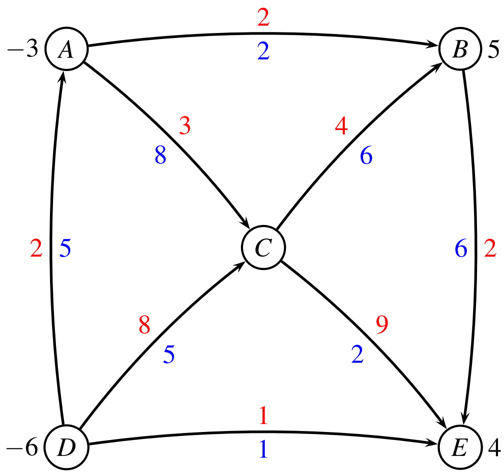
I hereby declare that I have worked alone on the exam and not discussed it with anybody else.

Date Signature Number of pages

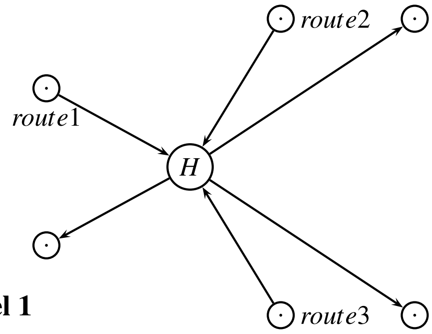
Maximum flow



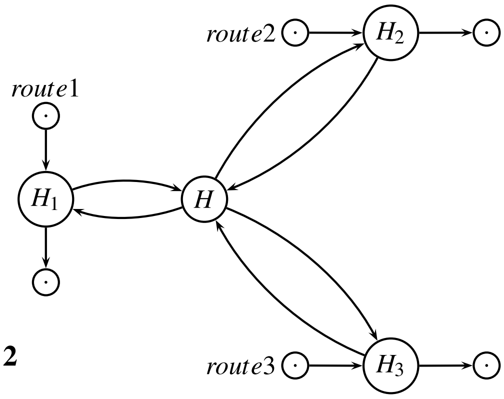
Min cost flow



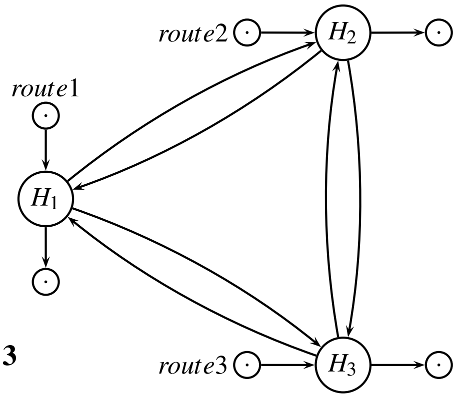
Modeling a hub port



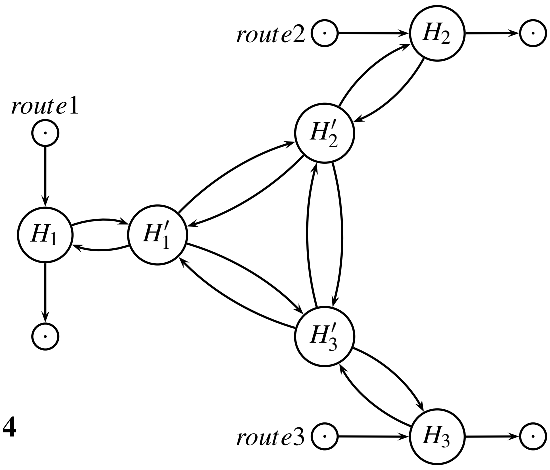
Model 1



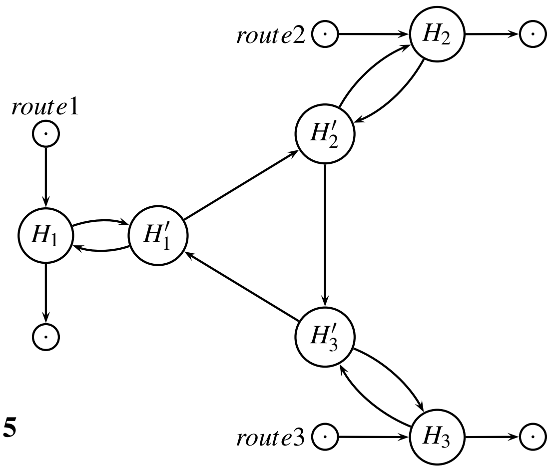
Model 2



Model 3

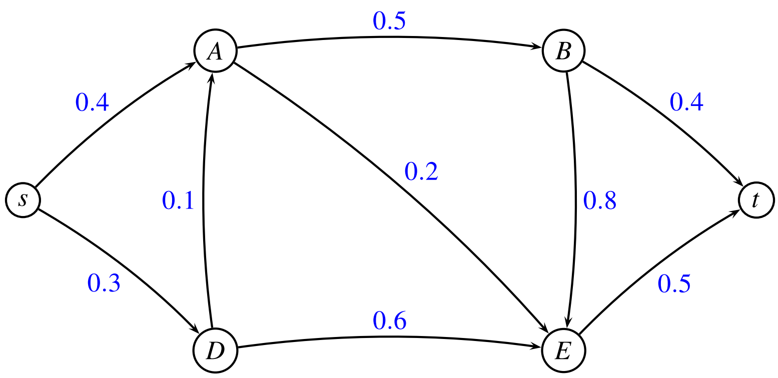


Model 4



Model 5

Shortest path



x	$\log x$
0.1	−1.00
0.2	−0.70
0.3	−0.52
0.4	−0.40
0.5	−0.30
0.6	−0.22
0.7	−0.15
0.8	−0.10
0.9	−0.05
1.0	0.00

DTU
Danmarks Tekniske Universitet

Written exam December 17, 2015.

Course name: Network Optimization **Course no:** 42115

Rules: You are allowed to use your books, notes, slides, computer (but no internet access). You can write in Danish or English.

Assignment: The assignment consists of 12 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Points: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 32 points can be obtained.

Critical path problem

The below table shows the precedence relations and durations for a nine activity project. The task is to schedule all activities so as to minimize the overall completion time.

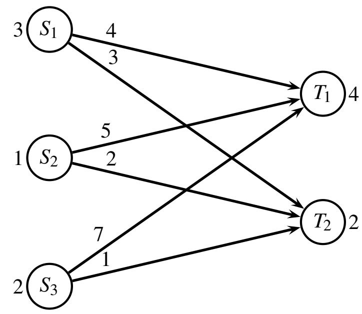
Activity	Description	Predecessors	Duration
A	Site clearing		4
B	Removal of trees		3
C	General excavation	A	8
D	Grading general area	A	7
E	Excavation for trenches	B,C	9
F	Placing formwork and reinforcement for concrete	B,C	12
G	Installing sewer lines	D,E	2
H	Installing other utilities	D,E	5
I	Pouring concrete	F,G	6

Problem 1: (medium) For each activity, find the earliest starting time and the latest starting time. ■

Problem 2: (small) Mark the critical path. ■

Fixed charge transportation problem

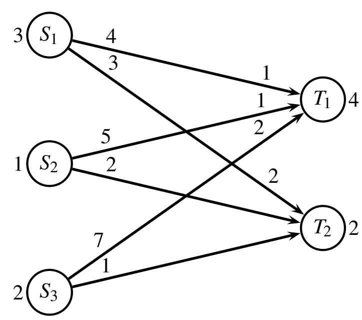
Three factories S_1, S_2, S_3 are transporting items to two customers T_1, T_2. The *transportation costs* of sending one item is indicated on the graph below. The production at factories are respectively $p_1 = 3, p_2 = 1, p_3 = 2$, while the demands at the customers are $d_1 = 4, d_2 = 2$. There is a *fixed charge* of cost 5 associated with each transportation. The fixed charge is to be paid if one or more items are sent along an edge, however, the fixed charge is independant of the number of items sent.



Since fixed charge transportation problems are known to be difficult to solve, we use a Dantzig-Wolfe decomposed formulation, where we write up all patterns of sending the p_i items from node S_i. The cost of a pattern consists of the transportation costs, and the fixed charges for using edges.

Problem 3: (medium) Write up the Dantzig-Wolfe decomposed model. The first constraint should make sure that the d_1 items arrive at customer T_1, and the second constraint should make sure that the d_2 items arrive at customer T_2. ■

Since the Dantzig-Wolfe decomposed model may be quite large, it is solved through column generation. The initial three columns are chosen by solving the transportation problem by a greedy method, getting the following solution. Values at the start of an edge indicate the edge cost, while values at the end of an edge indicate the flow.



Problem 4: (small) Which three columns in the Dantzig-Wolfe formulation correspond to the greedy solution? ■

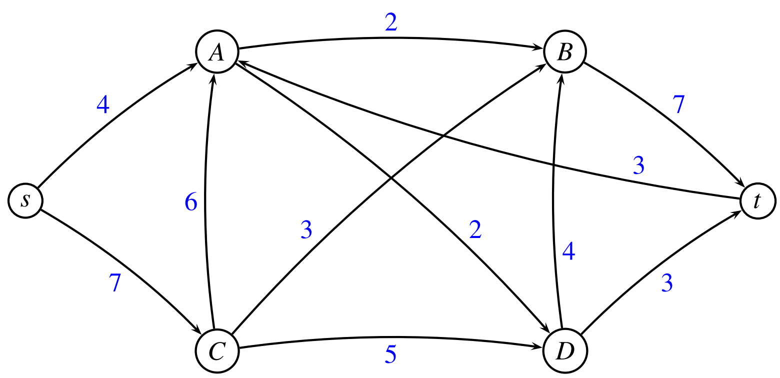
The restricted master problem problem is solved with only these three columns. Let u_i be the dual variables corresponding to the constraints saying that d_i items arrive at customer T_i. Let v_j be the dual variables corresponding to the constraints saying that only one pattern may be used for factory S_j.

The dual variables for the restricted master problem are $u_1 = 0$, $u_2 = 10$, $v_1 = 0$, $v_2 = 10$, $v_3 = 19$.

Problem 5: (small) Calculate the reduced costs of all patterns (which are not currently used in the restricted master problem). Which pattern should be added to the restricted master problem (and why?). ■

Maximum flow

Consider the following maximum flow problem where edge weights denote the capacity of an edge.



Problem 6: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 7: (small) Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Minimum cost flow problem

A caterer provides napkins on a 5-day schedule. The demand for each of the five days is:

Day	1	2	3	4	5
No. napkins	300	200	400	100	500

The caterer begins with no napkins, and to supply napkins on any particular day she has three options:

- Purchase new napkins at 8 kr per napkin
- Quick-Launder dirty napkins for the following day at 3 kr per napkin
- Slow-Launder dirty napkins for 2 days following at 1 kr per napkin

After the fifth day all napkins are discarded. How should the caterer go about providing napkins for the 5-day period?

Problem 8: (large) Formulate the problem as a minimum cost flow problem. It is sufficient to draw the graph, indicating demands on nodes, and costs/capacities on all edges. You do not need to solve the problem. ■

Arbitrage opportunity

A given day the exchange rates are as follows. Each row says how much you get for 1 unit of the given currency.

	USD	EUR	DKK	CHF	GBP
USD	1.00	.88	6.54	.96	.65
EUR	1.14	1.00	7.46	1.09	.74
DKK	.15	.13	1.00	.15	.10
CHF	1.05	.92	6.84	1.00	.68
GBP	1.54	1.35	10.05	1.47	1.00

Assume that you can convert 1 USD to 0.88 EUR, and then 0.88 EUR to some other currencty, for finally converting it back to USD such that you end up having more than 1 USD. This is call an *arbitrage opportunity*.

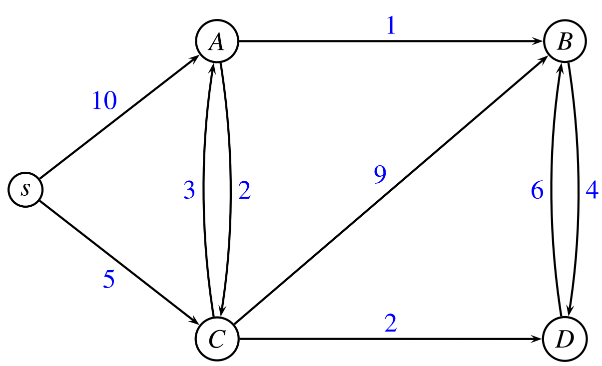
To be more formal, let e_{ij} be the exchange rate of currency i to currency j. Thus we are looking for a sequence of currencies $i, j, k, \ldots, m, n$ such that $e_{ij} \cdot e_{jk} \cdots e_{mn} > 1$.

Problem 9: (large) Formulate the problem as a network problem, and say which algorithm you would use to solve the problem. You do not need to actually solve the above problem. The formulation should be in general terms, assuming that there are N currencies available. ■

Hint: Use the fact that $\log(e_{ij} \cdot e_{jk} \cdots e_{mn}) = \log(e_{ij}) + \log(e_{jk}) + \cdots + \log(e_{mn})$.

Shortest path

In the following network the edge weights denote the distance between the end nodes.



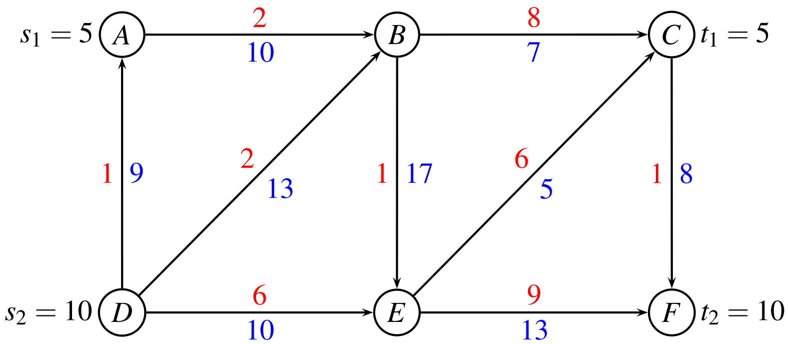
Problem 10: (medium) Use Dijkstra's algorithm to find a shortest path from s to all nodes. For every node report the shortest distance, and the shortest path. Give sufficient details so that the solution method can be followed. ■

Multi-commodity flow problem

An interesting variant of the multi-commodity flow problem is the one in which, in addition to having a capacity u_e for every edge e, we also have a capacity u_v for every node v. A flow x is feasible only if the flow out from a node v is less or equal to u_v for all nodes. The normal flow conservation constraints, and edge capacity constraints should still be satisfied.

Consider the following instance of the multi-commodity flow problem. Each edge e has a capacity u_e (blue), and a unit transportation cost c_e (red).

Assume that node B and E has a capacity $u_B = u_E = 12$, while all other nodes have unlimited capacity.

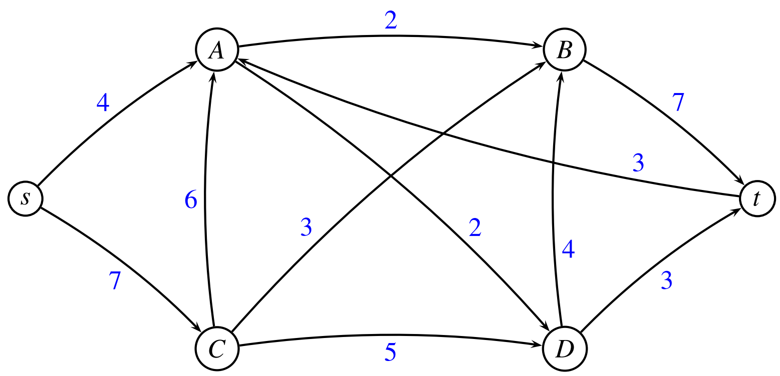


Problem 11: (small) Show how the problem can be transformed to a standard multi-commodity flow problem (where we only have capacities on edges). ■

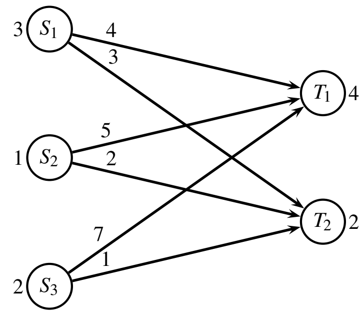
Problem 12: (small) Solve the resulting standard multi-commodity flow problem, assuming that all demands must be transported *unsplittable*. ■

end of assignment

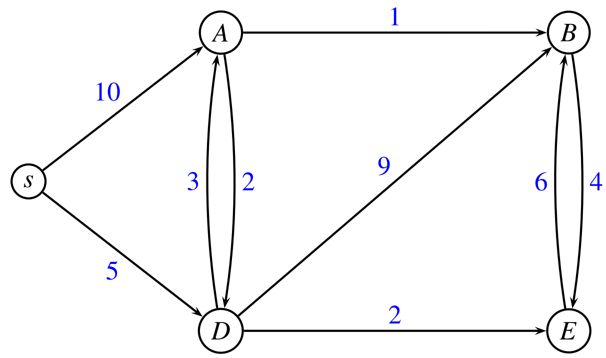
Maximum flow



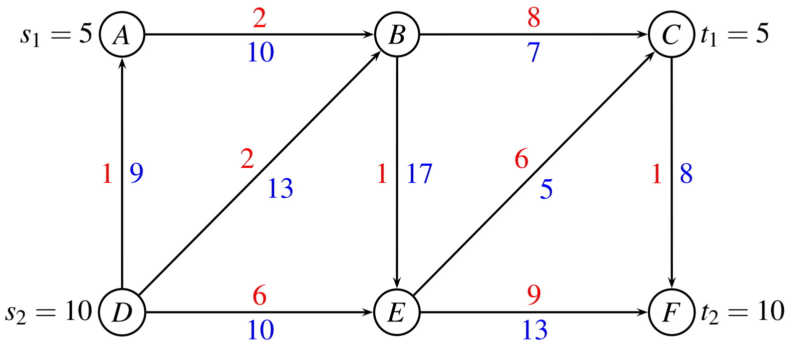
Fixed charge transportation



Shortest path



Multi-commodity flow problem



Written exam December 19, 2016.

Course name: Network Optimization **Course no:** 42115

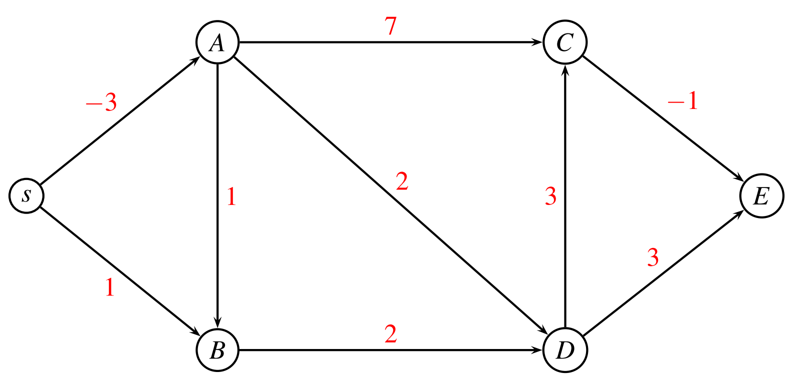
Rules: You are allowed to use your books, notes, slides, computer (but no internet access). You can write in Danish or English.

Assignment: The assignment consists of 12 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Points: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 35 points can be obtained.

Shortest path

In the following network the edge weights denote the distances. The problem is to find the shortest path from s to all nodes.



Problem 1: (small) Which algorithm would you use to solve the above problem? ■

Problem 2: (medium) Use the algorithm from previous question to solve the problem. Give sufficient details so that the solution method can be followed. ■

Critical path problem

A group of students following the Network Optimization course are solving the Project Assignment 2016. The assignment can be split into the following tasks:

Activity	Description	Predecessors	Duration
A	answer theoretical questions	E,B	5
B	buy color pencils		2
C	write theoretical model	E,B,G	7
D	write report	A,C	6
E	buy cake		4
F	install CPLEX	G	3
G	write up all paths in model		4
H	solve LP model	F	7

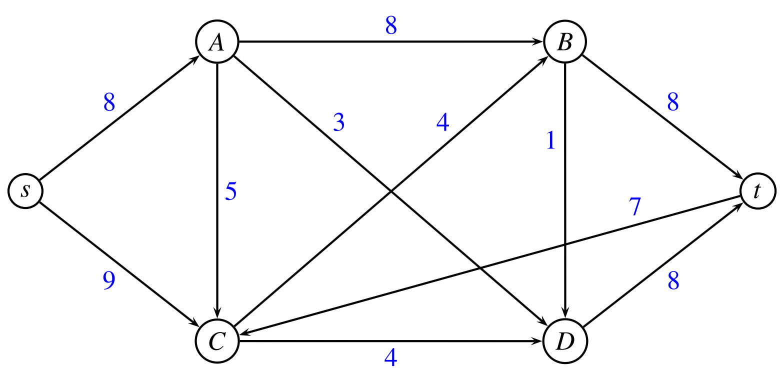
The duration of each task is given in the right-most column. A task cannot be started before the predecessor tasks have finished. Tasks can be carried out in parallel, by splitting the jobs between the group members.

Problem 3: (small) Order the tasks in topological order. ■

Problem 4: (medium) For each activity, find the earliest starting time and the latest starting time. Mark the critical path. ■

Maximum flow problem

Consider the following maximum flow problem where edge weights denote the capacity of an edge.

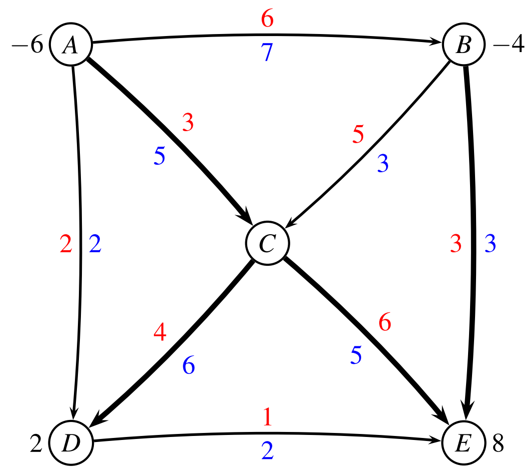


Problem 5: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. ■

Problem 6: (small) When the algorithm terminates, show the residual graph, indicate a minimum cut, and report the capacity of the cut. ■

Minimum cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i (black) in every node is positive for a sink, and negative for a source.



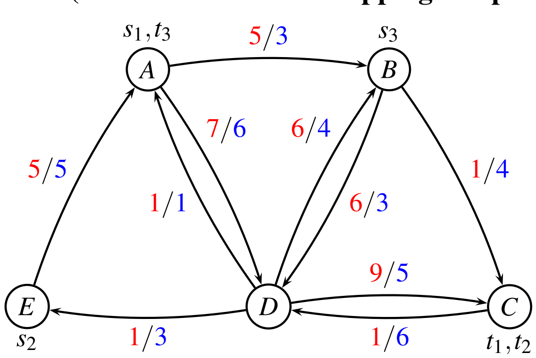
Having solved a maximum flow problem to find a feasible solution we are told that the tree solution T is defined by the bold edges in the above figure. Moreover, $L = \{(A,B)\}$ and $U = \{(A,D),(B,C),(D,E)\}$. The flow on edges L is at the lower bound, while the flow on edges in U is at the upper bound.

Problem 7: (small) Calculate the flow corresponding to the above feasible solution defined by T, L, U. ■

Problem 8: (large) Use the *network simplex* algorithm to solve the minimum cost flow problem to optimality. In each iteration report the tree solution, potentials on nodes, and reduced costs of edges. In the last iteration explain the stop criteria. ■

Column generation

MEDITLINE **(Route net of a liner shipping company)**



k	s_k	t_k	d_k
1	A	C	3
2	E	C	1
3	B	A	3

For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. Cities are A = Alexandria, B = Barcelona, C = Casablanca, D = Dubrovnik, E = Ephesus.

The above instance of MEDITLINEis the same as in the Project Assignment 2016, except that the demand for commodity $k = 3$ has been changed to $d_k = 3$. Costs of flowing a container is indicated with red, and capacity of an edge is indicated with blue.

This means that all paths you found in the Project Assignment are the same:

number	commodity k	route r	d_k	$\sum_{(ij)\in r} c_{ij}$	$d_k \sum_{(ij)\in r} c_{ij}$
1	1	ABC	3	6	18
2	1	ABDC	3	20	60
3	1	ADBC	3	14	42
4	1	ACD	3	16	48
5	2	EABC	1	11	11
6	2	EABDC	1	25	25
7	2	EADBC	1	19	19
8	2	EADC	1	21	21
9	3	BCDEA	3	8	24
10	3	BDEA	3	12	36

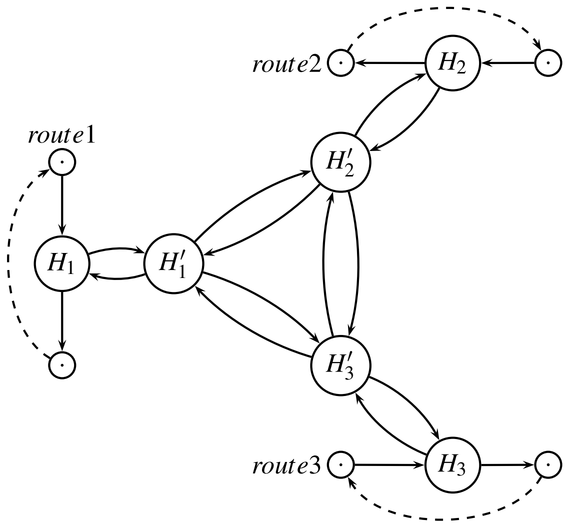
We solve the modified instance of MEDITLINE as an *unsplittable* multi-commodity flow problem through delayed column generation. After a number of iterations we have the following restricted master problem:

$$
\begin{array}{lrcrcrcrcrll}
\min & 18x_1 & + & 48x_4 & + & 11x_5 & + & 25x_6 & + & 36x_{10} & & \\
\text{s.t.} & 3x_1 & & & + & 1x_5 & + & 1x_6 & & & \leq 3 & (y_{AB}) \\
& & & 3x_4 & & & & & & & \leq 6 & (y_{AD}) \\
& 3x_1 & & & + & 1x_5 & & & & & \leq 4 & (y_{BC}) \\
& & & & & & + & 1x_6 & + & 3x_{10} & \leq 3 & (y_{BD}) \\
& & & & & & & & & & \leq 6 & (y_{CD}) \\
& & & & & & & & & & \leq 1 & (y_{DA}) \\
& & & & & & & & & & \leq 4 & (y_{DB}) \\
& & & 3x_4 & & & + & 1x_6 & & & \leq 5 & (y_{DC}) \\
& & & & & & & & & 3x_{10} & \leq 3 & (y_{DE}) \\
& & & & & 1x_5 & & & + & 3x_{10} & \leq 5 & (y_{EA}) \\
& & & & & & & & & & & \\
& x_1 & + & x_4 & & & & & & & \geq 1 & (\overline{y}_1) \\
& & & & & x_5 & + & x_6 & & & \geq 1 & (\overline{y}_2) \\
& & & & & & & & & x_{10} & \geq 1 & (\overline{y}_3) \\
& & & & & & & & & & & \\
& \multicolumn{11}{l}{x_1, x_4, x_5, x_6, x_{10} \geq 0}
\end{array}
\tag{1}
$$

Solving the above LP-model we find the dual solution to (1) as $y_{AB} = -10$, while all other $y_{ij} = 0$. Moreover, we have $\overline{y}_1 = 48$, $\overline{y}_2 = 21$, $\overline{y}_3 = 36$.

Problem 9: (large) Calculate the reduced costs of all paths that are not used in model (1) and indicate which column should be added to the restricted master problem. ■

Modeling a hub



Consider Model 5 from the Project Assignment 2016. The graph is used to model a hub port, where three routes meet and transship containers between the vessels. We model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H_1', H_2' and H_3'. Edges are added between all pairs of nodes H_1', H_2', and H_3'.

In this problem we are *only considering the crane capacities*. The crane capacity for loading/unloading is $u = 8$ containers for routes 1 and 2, while the capacity for loading/unloading is $u = 5$ containers for route 3 since it is operated by a smaller crane.

Problem 10: (medium) In the Project Assignment there were separate cranes for loading and unloading, each having capacity u. Assume now that loading and unloading is made by the *same* crane, having overall capacity u for both loading and unlaoding. How would you modify the graph to take this into account? The resulting graph should still be a multi-commodity flow problem. ■

Census rounding

You are consulting for an environmental statistics firm. They collect statistics and publish the collected data in a book. The statistics are about populations of different regions in the world and are recorded in multiples of one million. Examples of such statistics would look like the following table.

Table 1

Country	A	B	C	Total
grown-up men	11,6	9,4	3,0	24,0
grown-up women	12,7	10,5	3,8	27,0
children	1,7	2,1	0,2	4,0
total	26,0	22,0	7,0	55,0

We will assume here for simplicity that our data is such that all row and column sums are integers. The Census Rounding Problem is to round all data to integers without changing any row or column sum. Each fractional number can be rounded either up or down. For example, a good rounding for our table data would be as follows.

Table 2

Country	A	B	C	Total
grown-up men	11	10	3	24
grown-up women	13	10	4	27
children	2	2	0	4
total	26	22	7	55

Problem 11: (large) Consider first the special case when all data are between 0 and 1. So you have a matrix of fractional numbers between 0 and 1 as illustrated in the Table 3, and your problem is to round each fraction that is between 0 and 1 to either 0 or 1 without changing the row or column sums. Formulate the rounding problem in Table 3 as a network problem, and explain how a solution should be used to define a rounding of the matrix. Moreover, explain that the solution always will be integral. (You do not need to solve the specific problem). ■

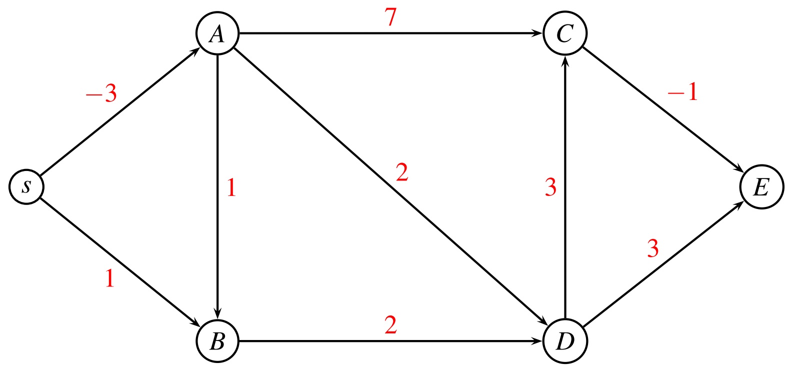
Table 3

Country	A	B	C	Total
grown-up men	0,1	0,7	0,2	1,0
grown-up women	0,4	0,6	0,0	1,0
children	0,5	0,7	0,8	2,0
total	1,0	2,0	1,0	4,0

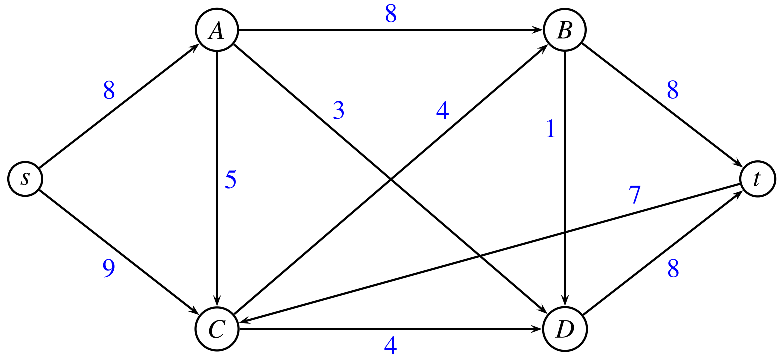
Problem 12: (medium) Consider now the general Census Rounding Problem as defined above, where row and column sums are integers, and you want to round each fractional number a to either $\lfloor a \rfloor$ or $\lceil a \rceil$. Formulate the rounding problem in Table 1 as a network problem, and explain how the solution should be used to define a rounding of the matrix. Moreover, explain that the solution always will be integral. (You do not need to solve the specific problem). ■

——— end of assignment ———

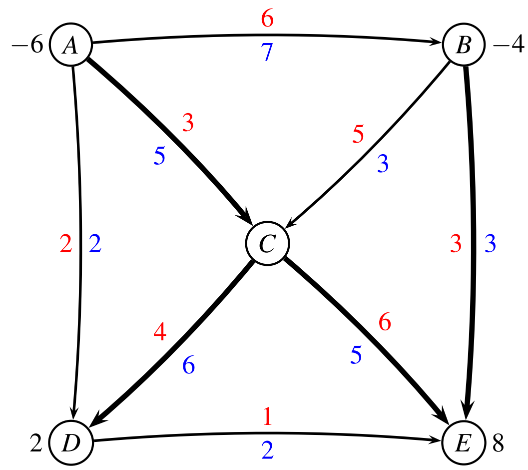
Shortest path



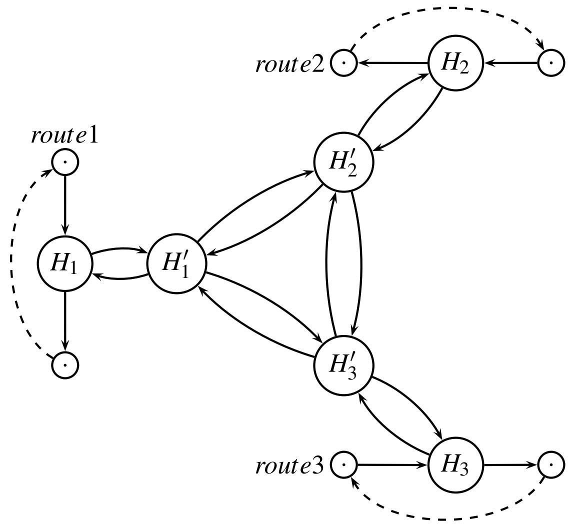
Maximum flow



Minimum cost flow



Modeling a hub



Written exam 20 December, 2017.

Course title: Network Optimization **Course no:** 42115

Aids allowed: Written aids permitted (textbooks, notes, old assignments etc.)

Exam duration: 4 hours

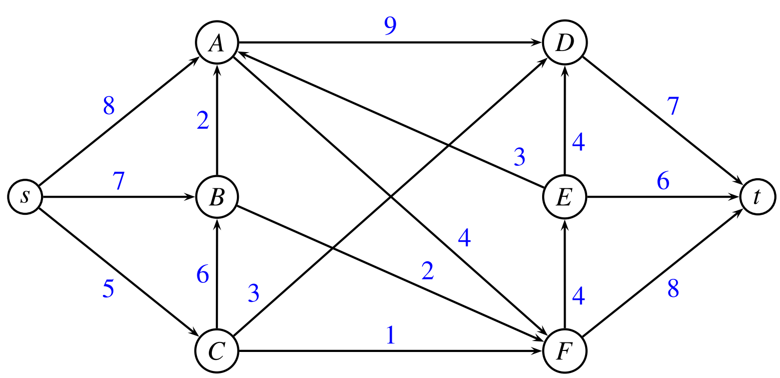
Weighting: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 42 points can be obtained.

Language: You can write in Danish or English.

Figures: All figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Maximum flow

Consider the following maximum flow problem where edge weights denote the capacity of an edge.



Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration report which augmenting path was used and the capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 2: (medium) Draw the residual graph when the algorithm terminates. Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Transportation problem

Consider the following transportation problem. We have three suppliers A_1, A_2, A_3 and two customers B_1, B_2. The cost of transporting one unit from A_i to B_j is given in the following table

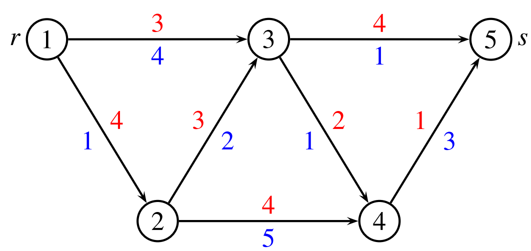
	B_1	B_2	supply
A_1	2	4	9
A_2	1	5	2
A_3	6	2	4
demand	7	8	

The supply from the A_i nodes is stated in the right column, while the demand in the B_j nodes is stated in the last row.

Problem 3: (large) Starting from a greedy solution, solve the problem using the network simplex algorithm. In each iteration report the dual variables and reduced costs. ■

Resource constrained shortest path problem

Consider the following directed acyclic graph. Every edge (i, j) has an associated cost c_{ij} (red) and time t_{ij} (blue). We are searching for a shortest path (with respect to cost) from r to s. There is a time constraint T imposed on the path. We would like to find the shortest path for every possible value of T.

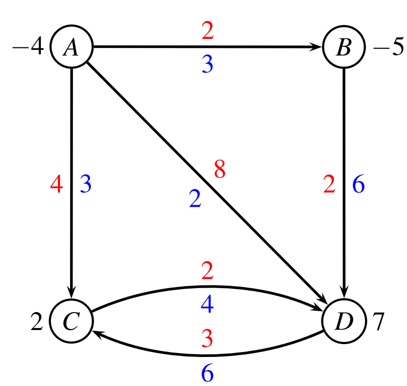


Problem 4: (small) Describe the algorithm you will use to solve the problem. ■

Problem 5: (large) Solve the problem, and report a table with shortest path from r to s for each possible value of T. ■

Min cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i (black) in every node is positive for a sink, and negative for a source.



Problem 6: (small) Find an initial solution by solving a maximum flow problem. ■

Problem 7: (large) Use the augmenting cycle algorithm to solve the min cost flow problem to optimality. In each iteration report the augmenting cycle, the cost of the cycle, and the new flow. In the last iteration explain the stop criteria. Also report the cost of the optimal solution. ■

Critical path problem

You have bought a new bicycle on the internet. However, it arrives in several pieces, with detailed instructions for assembling. Fortunately, you have many friends who are willing to help you, so several tasks can be carried out in parallel. The following table shows in which order the parts should be assembled, and how long time it takes:

Activity	Description	Predecessors	Duration
A	Chain	B,C	2
B	Crankset	E,H	7
C	Chainring	D	1
D	Handlebar	B,F,H	2
E	Spindle		2
F	Wheel	H,I	5
G	Pedals	C,J	4
H	Fork		3
I	Rear gear		3
J	Brakes	F	4

Problem 8: (small) Order the tasks in topological order. ■

Problem 9: (large) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan. ■

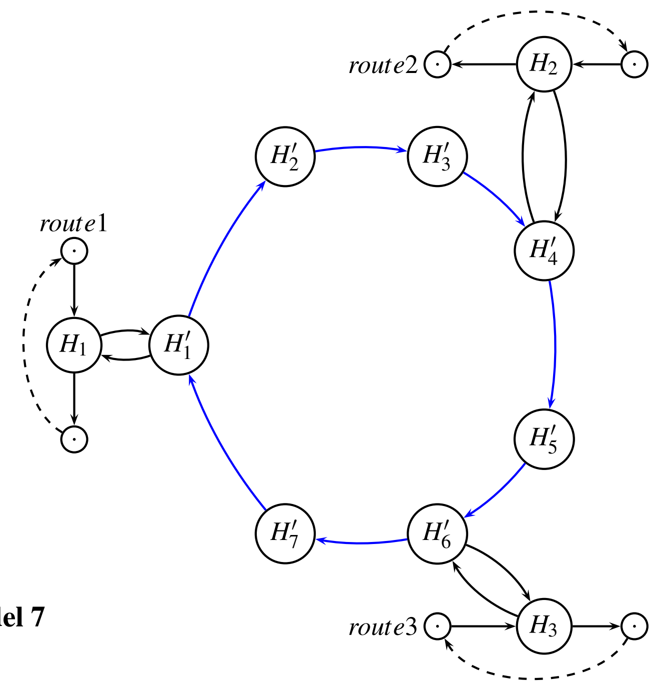
Problem 10: (small) Mark all critical paths. How many paths are there? ■

Arranging a dinner

Several families go out to dinner together. To increase their social interaction, they would like to sit at tables so that no two members of the same family are at the same table. Assume that the dinner contingent has p families, and that the i'th family has a_i members. Also assume that q tables are available and the j'th table has a seating capacity of b_j.

Problem 11: (large) Show how to formulate the problem as a maximum flow problem, and describe how the solution can be used to answer whether a seating is possible. You can use a small figure to illustrate your idea. ■

Modeling a hub



Model 7

We model the hub port H by using an extra node H_1, H_2 and H_3 for each route. Moreover, we introduce additional nodes $H'_1, H'_2, \ldots, H'_7$ representing the seven week days (1=Monday, 2=Tuesday, …, 7=Sunday). The latter nodes are connected in a cycle.

The crane capacity for loading/unloading is 10 containers for routes 1 and 2, while the capacity for loading/unloading is 7 containers for route 3 since it is operated by a smaller crane. There are separate cranes for loading and unloading, having the same capacity of 10 or 7.

The cost of a transshipment is 5$ for route 1 and 2, while it is 3$ for route 3. This mean that if you transship from route 1 to route 3 you pay 5$ for unloading the container at H and 3$ for loading the container.

Finally, the port has a limited storage capacity of $\overline{c} = 12$ containers on shore at any time. This means that the number of containers waiting one or more days for transshipment cannot exceed $\overline{c}$. The shipping company must pay 1$ for each container each day it is stored in the port.

It is assumed that, `route1` arrives Monday, `route2` arrives Thursday, and `route3` arrives Saturday.

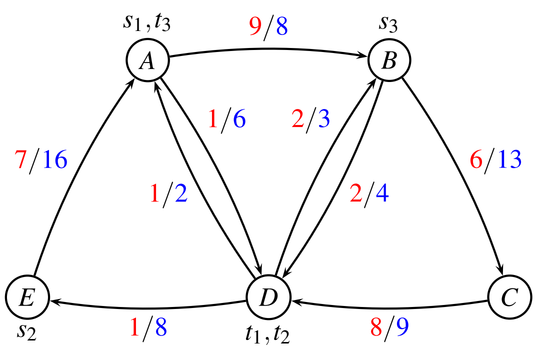
Consider Model 7 from Project Assignment 2017. The graph is used to model a hub port, where three routes meet and transship containers between the vessels.

It is now assumed that *all containers need to go through custom declaration immediately after unloading*. It takes two days to get a container through the custom declaration, so if the vessel arrives at day t the container will be available day $t+2$. The custom has infinite capacity for storing cotainers in a separate stockpile, and it costs 1$ per day in storage cost. Containers that stay on board a vessel do not need to go through custom declaration.

Problem 12: (large) Modify the above graph to reflect the times for loading/unloading. Is it possible to model transshipment costs, transshipment time, crane capacities, storage capacities, and storage costs? For those parameters where you answer yes, indicate the values on the graph. For those parameters where you answer no, argue why not. ■

Flowing containers in a network

MEDITLINE **(Route net of a Mediterranean liner shipping company)**



k	s_k	t_k	d_k	w_k
1	A	D	3	1
2	E	D	4	2
3	B	A	2	4

For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. Cities are $A =$ Alexandria, $B =$ Barcelona, $C =$ Casablanca, $D =$ Dubrovnik, $E =$ Ephesus.

Consider the above weighted multi-commodity flow problem. All demands and costs are exactly as in the Project Assignment 2017, hence we have the possible routes r outlined in the below table. The cost of sending 1 unit of size w_k along a route r is calculated as $w_k \sum_{(i,j)\in r} c_{ij}$.

number	commodity k	route r	w_k	$\sum c_{ij}$	cost
1	1	$ABCD$	1	23	23
2	1	ABD	1	11	11
3	1	AD	1	1	1
4	2	$EABCD$	2	30	60
5	2	$EABD$	2	18	36
6	2	EAD	2	8	16
7	3	$BCDEA$	4	22	88
8	3	$BDEA$	4	10	40

We solve the problem through column generation. Instead of starting from some dummy routes, we guess a part of the solution, by using route $r = 3$ for commodity 1, and routes $r = 7, 8$ for commodity 3. Hence, we only need to have a dummy route for commodity 2. This dummy route is denoted route 0, and it goes directly from E to A without using any capacity.

This leads to the following LP-model, the (*restricted master problem*):

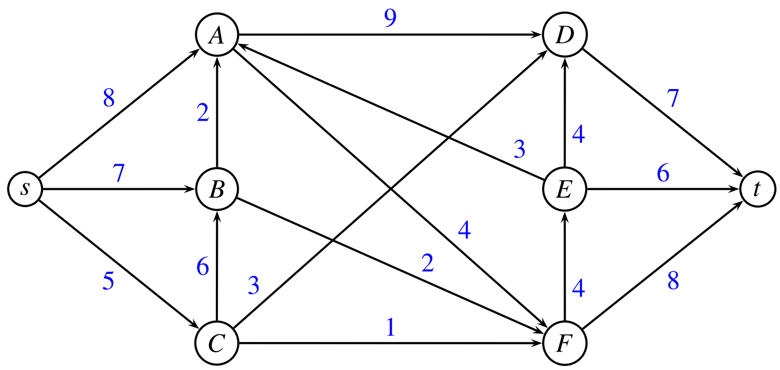
$$
\begin{array}{llllllll}
\min & 1x_3 & + \ 88x_7 & + \ 40x_8 & + \ 100x_0 & & & \\
\text{s.t.} & & & & & \leq 8 & (y_{AB}) & \\
 & x_3 & & & & \leq 6 & (y_{AD}) & \\
 & & x_7 & & & \leq 13 & (y_{BC}) & \\
 & & & x_8 & & \leq 4 & (y_{BD}) & \\
 & & x_7 & & & \leq 9 & (y_{CD}) & \\
 & & & & & \leq 2 & (y_{DA}) & \\
 & & & & & \leq 3 & (y_{DB}) & (1) \\
 & & x_7 \ + & x_8 & & \leq 8 & (y_{DE}) & \\
 & & x_7 \ + & x_8 & & \leq 16 & (y_{EA}) & \\
 & x_3 & & & & \geq 3 & (\overline{y}_1) & \\
 & & & & x_0 & \geq 4 & (\overline{y}_2) & \\
 & & x_7 \ + & x_8 & & \geq 2 & (\overline{y}_3) & \\
\multicolumn{8}{l}{x_3, x_7, x_8, x_0 \geq 0}
\end{array}
$$

An optimal solution to the above problem is $x_3 = 3$, $x_7 = 1$, $x_8 = 1$, $x_0 = 4$ with objective $z = 531$. The dual variables are $y_{BD} = -12$, $\overline{y}_1 = 1$, $\overline{y}_2 = 100$, $\overline{y}_3 = 88$. All other dual variables are zero.

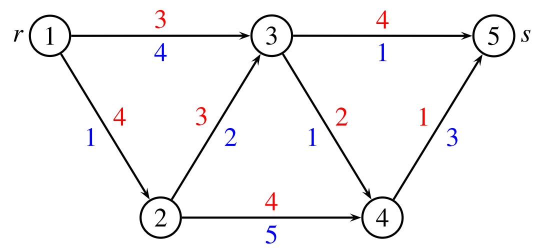
Problem 13: (large) Calculate the reduced costs of all routes not in the restricted master problem, and indicate which route should be added to the problem in the next iteration. ∎

end of assignment

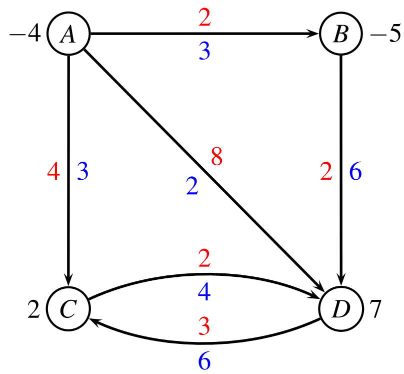
Maximum flow



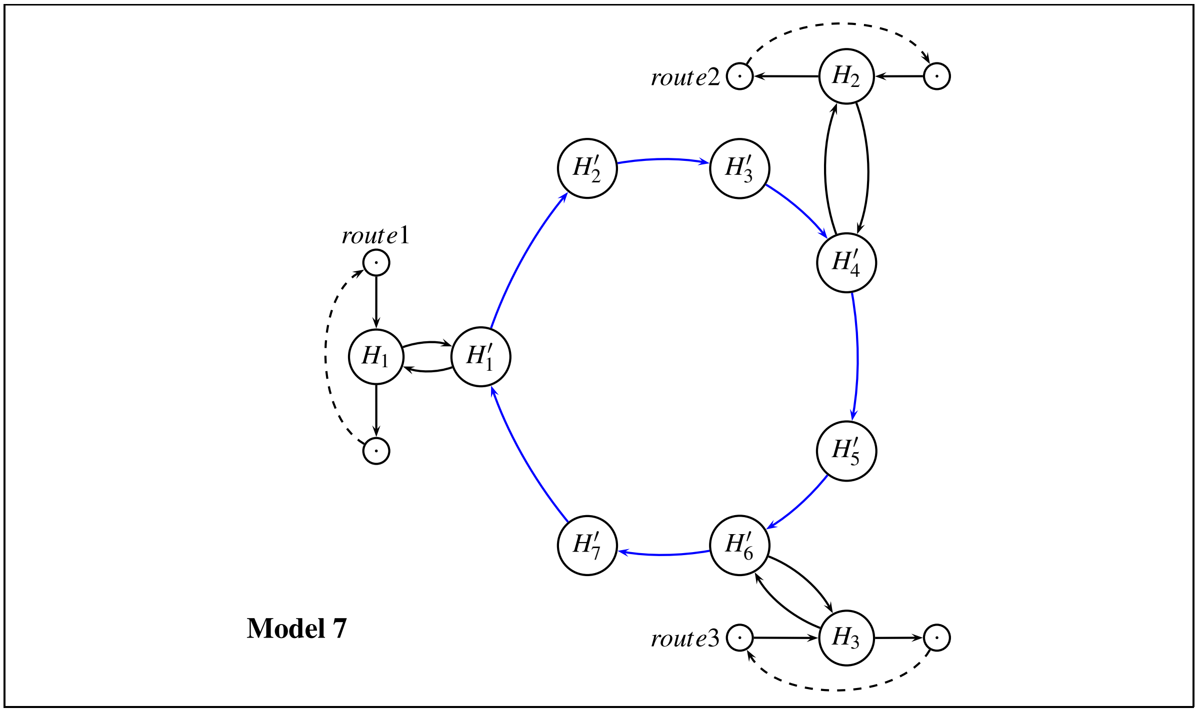
Resource constrained shortest path problem



Min cost flow



Modeling a hub



Written exam 19 December, 2018

Course title: Network Optimization **Course no:** 42115

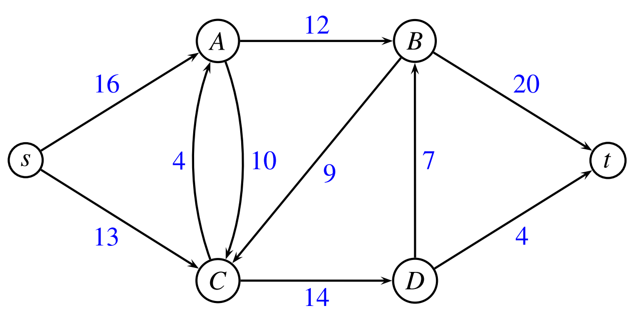
Aids allowed: All

Exam duration: 4 hours

Weighting: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 40 points can be obtained.

Language: You can write in Danish or English.

Maximum Flow problem



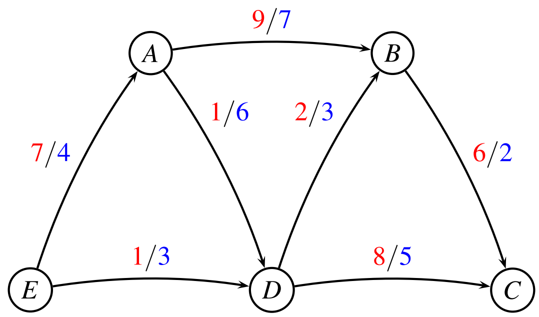
Problem 1: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration report which augmenting path was used and the capacity. (You do not need to draw the residual graph in each iteration) ■

Problem 2: (medium) Draw the residual graph when the algorithm terminates. Indicate a minimum cut, and report the capacity of the cut and the value of the flow. ■

Empty container repositioning

Consider the following network of a small shipping company, MEDITLINE. Nodes correspond to ports, and edges correspond to connections. The cost of shipping one container is indicated with red, while the capacity of the connection is indicated with blue.

MEDITLINE**: Route net of a Mediterranean liner shipping company**



node	*A*	*B*	*C*	*D*	*E*
demand	−2	4	3	0	−5

Due to the imbalance in trade, empty containers tend to accumulate at various ports, while other ports have a defecit. For MEDITLINE, the demands for empty containers are stated in the table next to the network. Positive values denote a demand, while negative values denote a surplus of containers. It is assumed that all containers are identical.

Problem 3: (small) Describe how you can formulate the repositioning problem as a network problem, and state which algorithm you will use to solve it. ■

Problem 4: (large) Solve the MEDITLINE repositioning problem. Give sufficient details to make it possible to follow your calculations. Report the optimal solution and the cost of the solution. ■

Critical Path problem

Ship's biscuit was a hard piece of bread that sailors ate at nearly every meal. The biscuit was baked on land, stored on board the ship, and then sent out to sea with the sailors. Sailors soaked the rock-hard biscuit in their stew to soften it before taking a bite.

In order to bake ship's biscuit one has to go through the below list of activities. Each activity has a duration and some predecessor tasks that need to be finished before the activity can start.

Activity	Description	Predecessors	Duration
A	Start oven at 250 degrees	*D*, *B*	8
B	Weigh rye flour and wheat flour		9
D	Dissolve yeast in the water		7
C	Prepare a barrel for storage	*E*	5
E	Cut the dough into squares	*A*	9
F	Bake the bisciuits	*G*, *E*	6
G	Mix ingredients and add salt	*D*	12

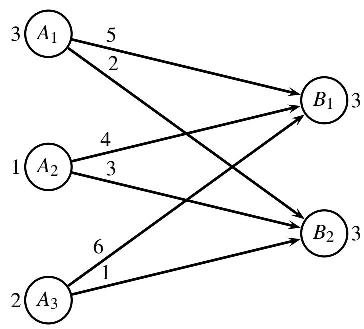
Problem 5: (small) Order the activities in topological order. ■

Problem 6: (large) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan. ■

Problem 7: (small) Mark the critical path. ■

Fixed charge transportation problem

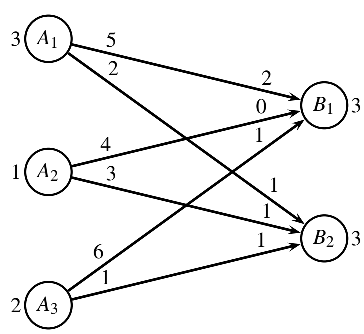
Three factories A_1, A_2, A_3 are transporting items to two customers B_1, B_2. The *transportation costs* of sending one item are indicated on the graph below. The capacity of the factories is respectively $p_1 = 3, p_2 = 1, p_3 = 2$, while the demand at the customers is $d_1 = 3, d_2 = 3$. There is a *fixed charge* of cost 4 associated with each transportation. The fixed charge is to be paid if one or more items are sent along an edge, however, the fixed charge is independant of the number of items sent.



Due to the fixed charge constraints, we use a Dantzig-Wolfe decomposed formulation, where we write up all patterns of sending the p_i items from node A_i. The cost of a pattern consists of the transportation costs, and the fixed charges for using edges.

Problem 8: (medium) Write up the Dantzig-Wolfe decomposed model. The first constraint should make sure that the d_1 items arrive at customer B_1, and the second constraint should make sure that the d_2 items arrive at customer B_2. ■

Since the Dantzig-Wolfe decomposed model may contain many columns for more complex problems, we use delayed column generation to solve the LP-relaxed master problem. We start with the initial flow:

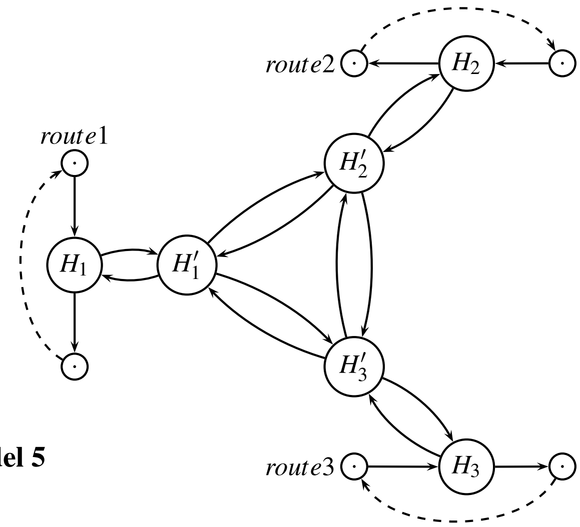


(numbers at start of edge denote costs, numbers at end of edge denote flow). The restricted master problem problem is now solved. Let u_i be the dual variables corresponding to the constraints saying that d_i items arrive at customer B_i. Let v_j be the dual variables corresponding to the constraints saying that only one pattern may be used for factory A_j.

The dual variables for the restricted master problem are $u_1 = 0$, $u_2 = 0$, $v_1 = 20$, $v_2 = 7$, $v_3 = 15$.

Problem 9: (medium) Calculate the reduced costs of all patterns (that are not currently in the restricted master problem). Which pattern should be added to the restricted master problem (and why?). ■

Modeling a hub port



Model 5

Consider Model 5 from Project Assignment 2018. The graph is used to model a hub port, where three routes meet and transship containers between the vessels. We model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H_1', H_2' and H_3'. Edges are added between all pairs of nodes H_1', H_2', and H_3'.

Problem 10: (medium) Assume that two vessels (route1 and route2) arrive the same day. Therefore, they have to share the crane capacity for loading and unloading. This means that the number of containers unloaded from vesels at route1 *and* route2 in total should not exceed $u = 8$. Similarly, the number of containers loaded to the two routes togther should not exceed $u = 8$. There is sufficient time to transship containers between route1 and route2 during the same day. Modify the above graph to handle these constraints. Indicate capacities on edges where relevant. The resulting graph should still be a multicommodity flow problem. ■

Flooding

Network flow issues come up in dealing with natural disasters and other crises, since major unexpected events often require the movement and evacuation of large numbers of persons in a short amount of time.

Consider the following scenario: Due to large-scale flooding in a region, paramedics have identified a set of n injured persons distributed across the region who need to be rushed to hospitals. There are k hospitals in the region, and each of the n persons needs to be brought to a hospital that is within a half-hour's driving time of their current location (so different persons will have different options for hospitals, depending on where they are right now).

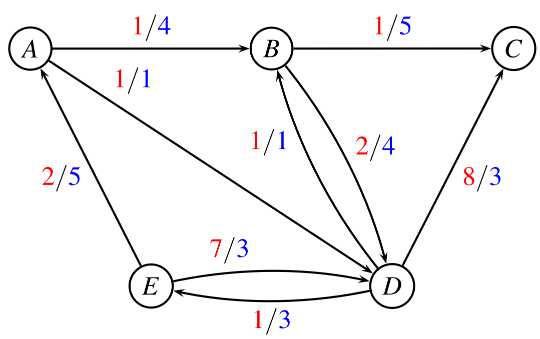
At the same time, one doesn't want to overload any one of the hospitals by sending too many patients its way. The paramedics are in touch by cell phone, and they want to collectively work out whether they can choose a hospital for each of the injured persons in such a way that the load on the hospitals is balanced: Each hospital receives at most $\lceil n/k \rceil$ persons.

Problem 11: (large) Given information about the persons's locations, we would like to know whether the assignment is possible. Formulate the problem as a network optimization problem that can be solved in polynomial time. ■

Unsplittable multicommodity flow problem

We are considering the *unsplittable multicommodity flow problem* from Project Assignment 2018. We have a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost $c_{ij} \geq 0$ and a capacity $u_{ij} \geq 0$. Moreover we are given a number of commodities $K = \{1, \ldots, m\}$ each given by a triple (s_k, t_k, d_k), where s_k denotes the source of commodity k, t_k denotes the target of commodity k, and d_k denotes how many units to be sent from s_k to t_k without splitting the cargo.

ASIANET **(Route net of a liner shipping company)**



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. Cities are $A =$ Manilla, $B =$ Shanghai, $C =$ HongKong, $D =$ Xiamen, $E =$ Singapore.

Like in the project assignment, we have three commodities with the following demand:

k	s_k	t_k	d_k
1	E	D	2
2	D	C	2
3	E	C	1

Let R_k denote the set of possible routes from s_k to t_k allowing d_k units of capacity. Let moreover $a^k_{r,ij} \in \{0, 1\}$ denote if route r for commodity k is using edge (i, j), and let $\hat{c}^k_r$ denote the cost for sending d_k containers along route $r \in R_k$. Let the binary variable x^k_r for each route $r \in R_k$ indicate whether or not the route is used for commodity k.

We can now write up the *path formulation* of the *unsplittable multicommodity flow problem*:

$$
\begin{aligned}
\min \quad & \sum_{k\in K}\sum_{r\in R_k} \hat{c}_r^k x_r^k && && (1)\\
\text{s.t} \quad & \sum_{k\in K}\sum_{r\in R_k} d_k a_{r,ij}^k x_r^k \le u_{ij} && (i,j)\in E && (2)\\
& \sum_{r\in R_k} x_r^k \ge 1 && k\in K && (3)\\
& x_r^k \in \{0,1\} && k\in K, r\in R_k && (4)
\end{aligned}
$$

Constraint (2) ensures that the capacity of each edge is respected, while (3) ensures that at least one route from s_k to t_k is selected. Finally, (4) ensures that the demand d_k is not split.

We solve the LP-relaxed path formulation of ASIANET, through column generation, starting with the following routes:

commodity k	route	cost
1	EABD	10
2	DC	16
3	EADBC	5
3	EDC	15

This leads to the following *restricted master problem*:

$$
\begin{array}{lccccccccccl}
\min & 10x_1 & + & 16x_2 & + & 5x_3 & + & 15x_4 & & & & \\
\text{s.t.} & 2x_1 & & & & & & & \le & 4 & & (y_{AB})\\
 & & & & & x_3 & & & \le & 1 & & (y_{AD})\\
 & & & & & x_3 & & & \le & 5 & & (y_{BC})\\
 & 2x_1 & & & & & & & \le & 4 & & (y_{BD})\\
 & & & & & x_3 & & & \le & 1 & & (y_{DB})\\
 & & & 2x_2 & & & + & x_4 & \le & 3 & & (y_{DC})\\
 & & & & & & & & \le & 3 & & (y_{DE}) \qquad (5)\\
 & & & & & x_3 & & & \le & 5 & & (y_{EA})\\
 & 2x_1 & & & & & + & x_4 & \le & 3 & & (y_{ED})\\
 & x_1 & & & & & & & \ge & 1 & & (\bar{y}_1)\\
 & & & x_2 & & & & & \ge & 1 & & (\bar{y}_2)\\
 & & & & & x_3 & + & x_4 & \ge & 1 & & (\bar{y}_3)
\end{array}
$$

$$0 \le x_1, x_2, x_3, x_4 \le 1$$

The dual solution to the model is: $y_{AD} = -10$, $\bar{y}_1 = 10$, $\bar{y}_2 = 16$, $\bar{y}_3 = 15$, while all other dual variables are zero.

Problem 12: (medium) Formulate the problem of finding a column with most negative reduced cost as a series of shortest path problems. For each commodity draw the graph and describe how you would solve the pricing problem. ■

Problem 13: (large) Find the column with most negative reduced cost, and add it to the above model. Give sufficient details to follow your calculations. ■

Danmarks Tekniske Universitet

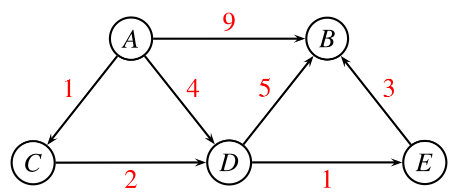
Written exam 18 December, 2019

Course title: Network Optimization **Course no:** 42115

Aids allowed: All
Exam duration: 4 hours
Weighting: Points are indicated on the assignments. A maximum of 44 points can be obtained.
Language: You can write in Danish or English.

Shortest path problem

Consider the following directed graph where distances w_{ij} are marked with red.



Assume that you should find the shortest path from A to all other nodes. If two solutions have the same length you would prefer the one visiting fewest edges.

Problem 1: (2 point) Describe how you would transform the distances, to make it possible to solve the problem using an ordinary shortest-path algorithm. Your transformation should work for *any* graph, not only the above. ■

Problem 2: (3 point) Solve the problem. State which algorithm you choose, and give sufficient details to make it possible to follow the algorithm. Report the shortest-path tree and optimal distances. ■

Critical path with time windows

Consider a critical path problem, in which each task i has an earliest start time e_i.

Problem 3: (2 point) Show how the critical path problem with start times can be transformed to an ordinary critical path problem. ■

Problem 4: (4 point) Solve the below problem:

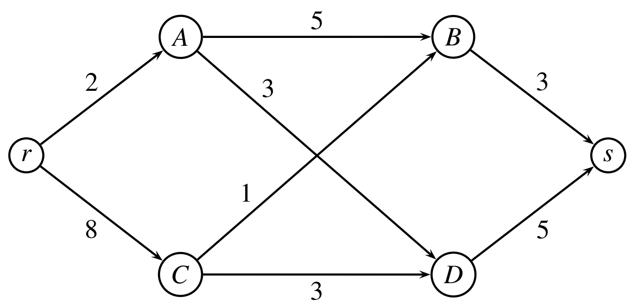
activity i	processing time p_i	predecessors	earliest start e_i
A	2	C	1
B	6		0
C	4		3
D	3	B,C	5

■

Problem 5: (2 point) Indicate the critical path, and argue how you found it. Will the critical path only be on tasks A, B, C, D or can it be on non-tasks? ■

Maximum flow problem

Consider the following maximum flow problem where the capacity u_{ij} is indicated on edge $(i, j) \in E$.



Problem 6: (5 point) Solve the problem using the *preflow-push* algorithm. (You can use the figures on the last page, and enclose them to the solution. Remember to write what you are doing.) ■

Problem 7: (2 point) Find a minimum cut and the corresponding capacity of the cut. (Explain how you find the cut). ■

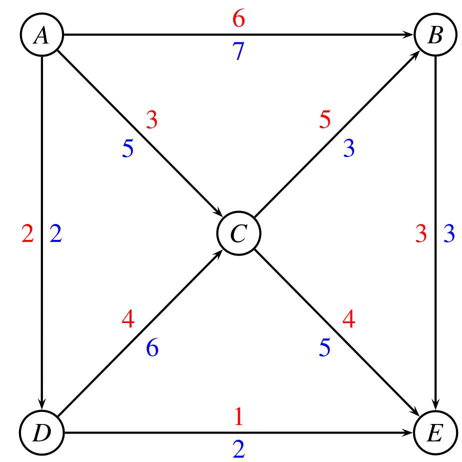
Workforce plan

A building company has to handle 3 projects, P1, P2, P3, over the next 4 months. Project P1 can start in the beginning of month 2, and must be completed at the end of month 3. Project P2 and P3 can start in the beginning of month 1, and must be completed, respectively, at the end of month 4 and 2. The projects require, respectively, 9, 10, and 11 man-months. Each month, 8 engineers are available. Due to the internal structure of the company, no more than 6 engineers can work, at the same time, on the same project. Determine whether it is possible to complete the three projects within the time constraints and, if it is possible, find a feasible workforce plan.

Problem 8: (5 point) Describe how this problem can be reduced to the problem of finding a maximum flow in an appropriate graph. (Hint: Look for a feasible flow of value 30.) ■

Repositioning containers

Consider the following liner shipping network where costs w_{ij} are indicated with red, and capacities u_{ij} (measured in TEU) are indicated with blue.



Due to imbalance in trade, some ports have a surplus of containers, while other ports are in demand of containers. The following table states the surplus for each port (negative values indicate a demand):

Port i	A	B	C	D	E
Surplus 20" containers σ_{1i}		-1		2	-1
Surplus 40" containers σ_{2i}	1	-1		1	-1
Surplus high-cube containers σ_{3i}	1				-1

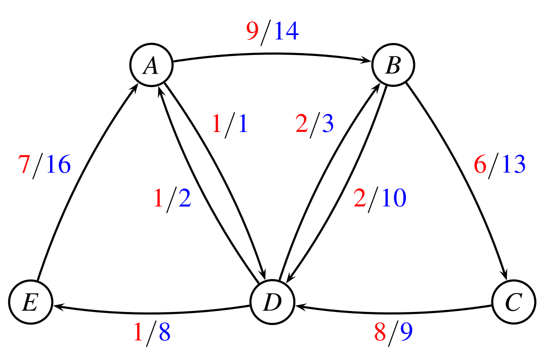
A 20" container takes up $w_1 = 1$ TEU space, a 40" container takes up $w_2 = 2$ TEU space, while a high-cube container takes up $w_3 = 4$ TEU space.

Problem 9: (2 point) We would like to re-balance the containers. Calculate the demand d_i in each port i measured in TEU. ■

Problem 10: (5 point) Solve the resulting minimum cost flow problem using demands d_i, and report the cost. ■

Multicommodity-flow problem

MEDITLINE **(Route net of a Mediterranean liner shipping company)**



For edge (i,j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij} (in TEU).

Note: The route net is the same as in Project Assignment 2019, except that edge (A,B) has capacity 14, edge (A,D) has capacity 1, edge (B,D) has capacity 10.

The capacity is measured in TEU (*Twenty-foot Equivalent Units*). A 20” container takes up $w_1 = 1$ TEU space, a 40” container takes up $w_2 = 2$ TEU space, while a high-cube container takes up $w_3 = 4$ TEU space. Containers may not be split. We have the demands

k	s_k	t_k	d_k	w_k
1	A	D	3	1
2	E	D	4	2
3	B	A	2	4

Commodity $k = 1$ is shipping three 20” containers ($w_k = 1$) from A to D, commodity $k = 2$ is shipping four 40” containers ($w_k = 2$) from E to D, and commodity $k = 3$ is shipping two 40” high-cube containers ($w_k = 4$) from B to A.

Consider the LP-relaxed path formulation of MeditLine, in which we have added the three dummy routes: AD, ED, BA of cost 100. To ensure a feasible solution, the dummy routes do not use any capacity in the existing network.

After a few iterations of column generation we have added routes ABD and AD for commodity 1, and $EABD$ for commodity 2. This leads to the following *restricted master problem*:

$$
\begin{array}{lcccccccccccccc}
\min & 100x_0^1 & + & 100x_0^2 & + & 100x_0^3 & + & 11x_1 & + & 1x_2 & + & 36x_3 & & & \\
\text{s.t.} & & & & & & + & 1x_1 & & & + & 2x_3 & \leq & 14 & (y_{AB}) \\
 & & & & & & & & + & 1x_2 & & & \leq & 1 & (y_{AD}) \\
 & & & & & & & & & & & & \leq & 13 & (y_{BC}) \\
 & & & & & & + & 1x_1 & & & + & 2x_3 & \leq & 10 & (y_{BD}) \\
 & & & & & & & & & & & & \leq & 9 & (y_{CD}) \\
 & & & & & & & & & & & & \leq & 2 & (y_{DA}) \\
 & & & & & & & & & & & & \leq & 3 & (y_{DB}) \\
 & & & & & & & & & & & & \leq & 8 & (y_{DE}) \\
 & & & & & & & & & & + & 2x_3 & \leq & 16 & (y_{EA}) \\
 & x_0^1 & & & & & + & x_1 & + & x_2 & & & \geq & 3 & (\overline{y}_1) \\
 & & & x_0^2 & & & & & & & + & x_3 & \geq & 4 & (\overline{y}_2) \\
 & & & & & x_0^3 & & & & & & & \geq & 2 & (\overline{y}_3) \\
 & \multicolumn{14}{l}{x_0^1, x_0^2, x_0^3, x_1, x_2, x_3 \geq 0}
\end{array} \qquad (1)
$$

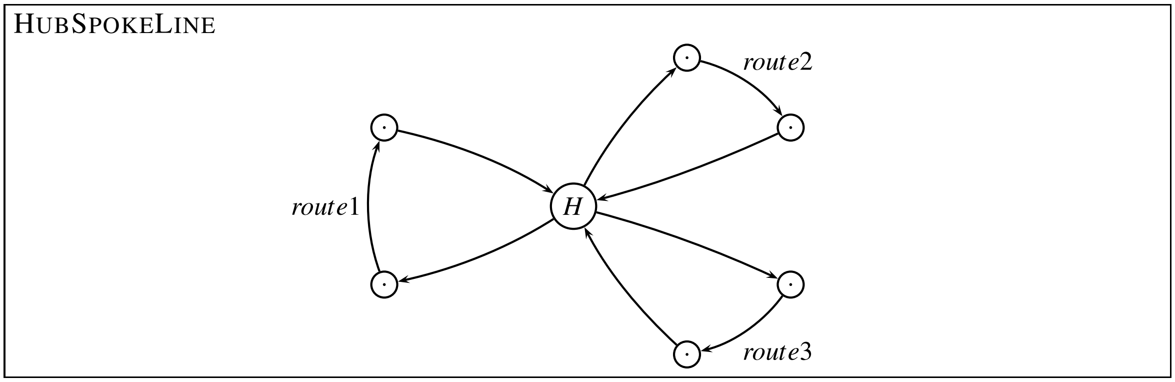
Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i, j), and let $\overline{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k.

The dual variables for the above model (1) are $y_{AD} = -10$ while $y_{ij} = 0$ for all other edges (i, j). Moreover, $\overline{y}_1 = 11$, $\overline{y}_2 = 36$, $\overline{y}_3 = 100$.

Problem 11: (5 point) Solve the pricing problem for each of the three commodities as a shortest-path problem. Show the graph, and give sufficient details to follow the calculations. ■

Problem 12: (2 point) Which column should be added to the restricted master problem, and why? ■

Modeling a hub port



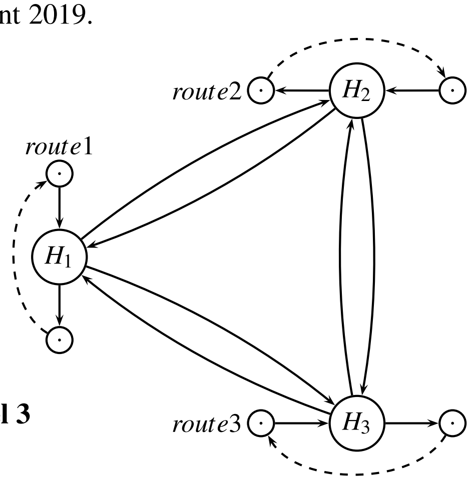
We will now study how the hub port H in a liner shipping network, HUBSPOKELINE, can be formulated in graph terms as part of a multicommodity flow network. We have three routes arriving to the hub as follows:

route	weekday
1	Wednesday
2	Wednesday
3	Friday

Note: The route net is the same as in Project Assignment 2019, except that Route 1 is changed so it arrives on a Wednesday instead of a Saturday. Since both route 1 and 2 arrive Wednesday, there is not sufficient time to transship containers from one vessel to the other, hence in this case transshipment takes 7 days.

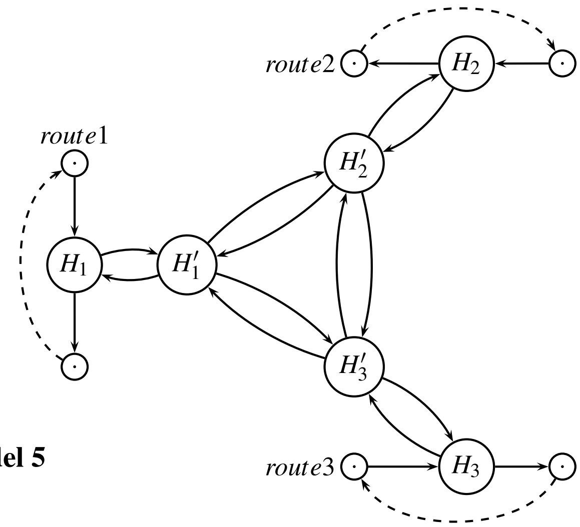
Crane capacity: The crane capacity u for loading/unloading is 10 containers per day for routes 1 and 2, while the capacity u for loading/unloading is 7 containers per day for route 3. There are separate cranes for loading and unloading both having the stated capacity. Each vessel has its own two cranes, meaning that route 1 and route 2 do not share the capacity u. *Transshipment cost:* The cost of loading/unloading is 5$ for route 1 and 2, while it is 3$ for route 3. *Transshipment time:* The time for transferring a container from one vessel to another can be calculated from the time table. If two vessels arrive the same day, transshipment takes 7 days. *Storage capacity:* The port has a limited storage capacity of $\overline{u} = 15$ containers on shore. *Storage cost:* The shipping company must pay $\overline{c} = 2\$$ for each container each night it is stored in the port.

We assume that each vessel is only one day in the port, and it is assumed that all loading and unloading is done during this day. We now consider three different models for the hub port H as known from the project assignment 2019.



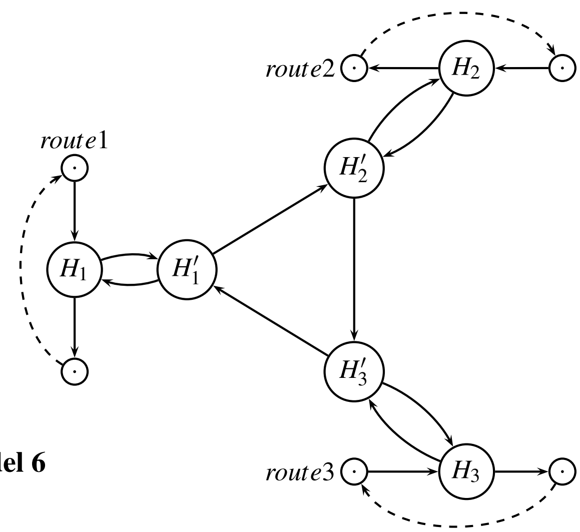
Model 3

(Model 3) A possible way of modeling the hub port H is to use an extra node H_1, H_2 and H_3 for each route. Edges are added between all pairs of nodes H_1, H_2, and H_3.



Model 5

(Model 5) Next, we may model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H_1', H_2' and H_3'. Edges are added between all pairs of nodes H_1', H_2', and H_3'.



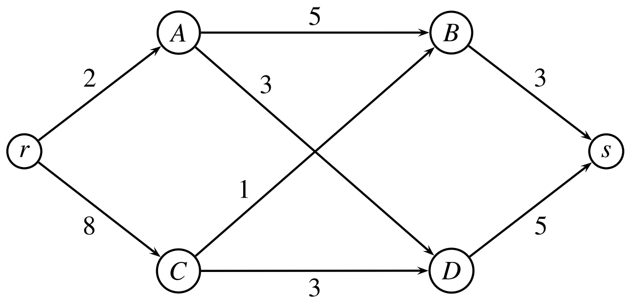
Model 6

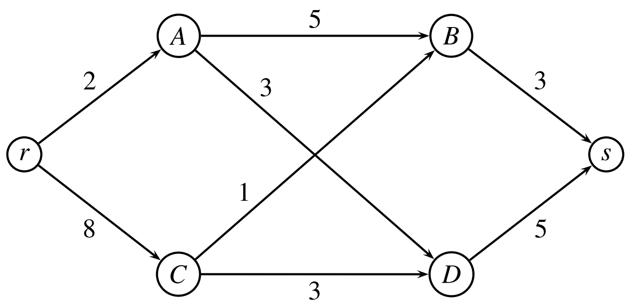
(Model 6) Finally, we may model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H_1', H_2' and H_3'. The latter nodes are connected in a cycle ordered according to the time of arrival.

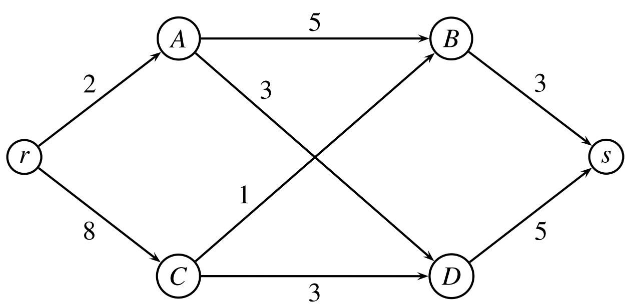
Problem 13: (5 point) Indicate on the models whether transshipment costs, transshipment time, crane capacity, storage capacity, and storage cost can be modeled as a multi-commodity flow problem using the three proposed models (Model 3, 5, 6). You can use the figures in the appendix. If the answer is positive, state the cost/capacity/time on the edges. If the answer is negative, argue why. ■

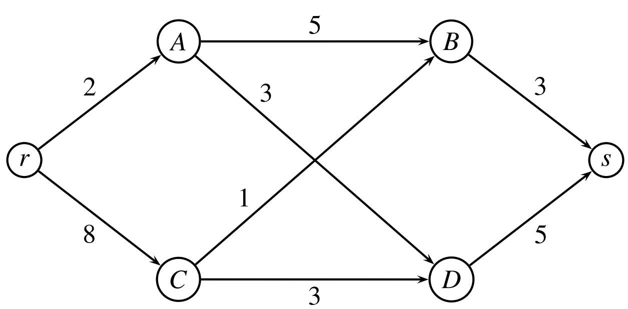
end of assignment

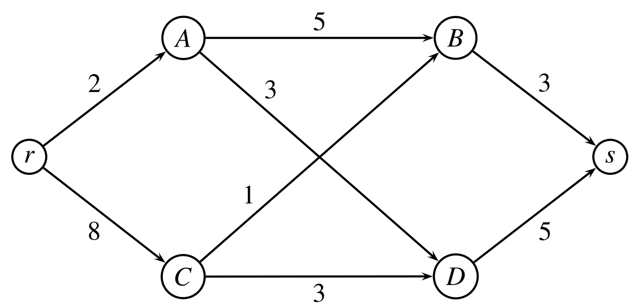
Problem 6,7 **Study number:**







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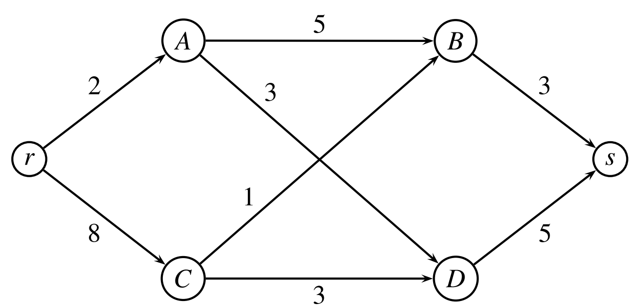
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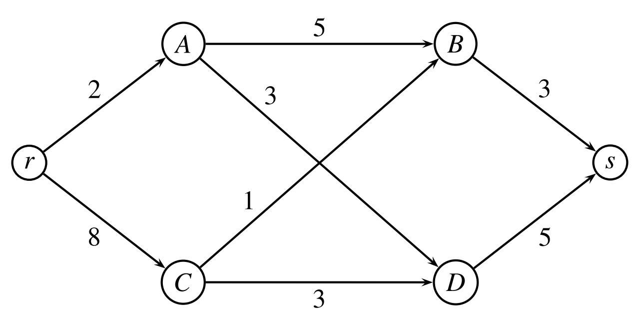
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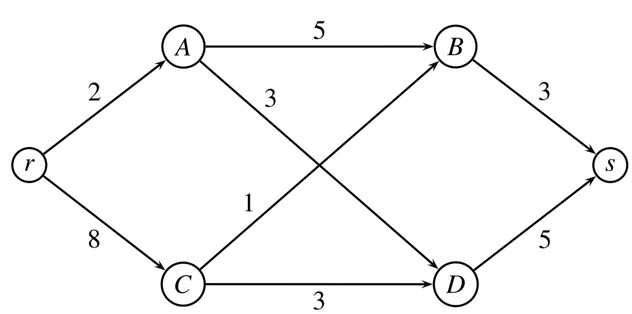
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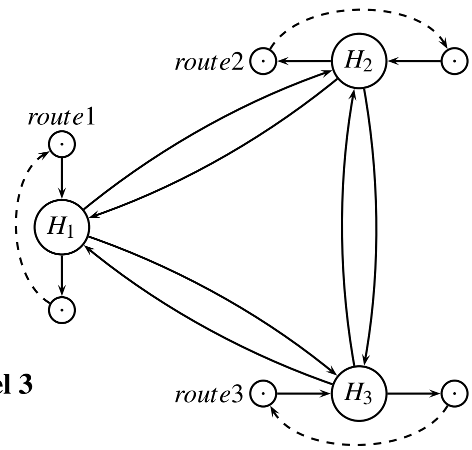
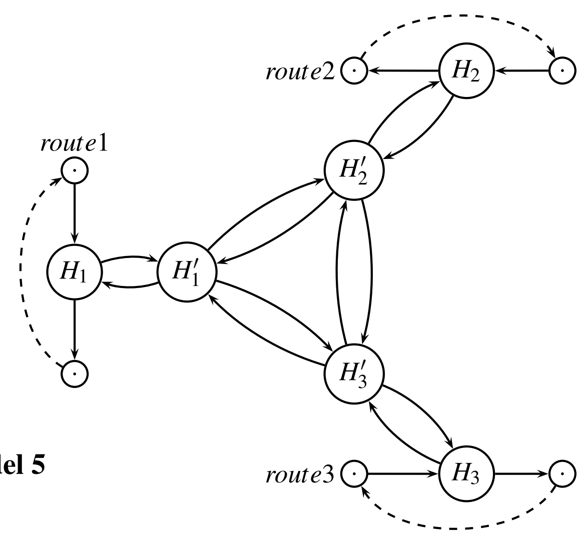
Problem 13 **Study number:**



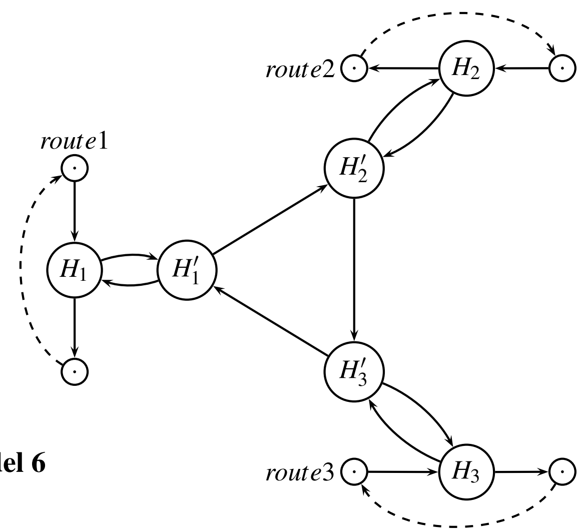
Model 3

Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
3					



Model 5

Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
5					



Model 6

Model	transshipment cost	transshipment time	crane capacity	storage capacity	storage cost
6					

Written exam 17 December, 2020

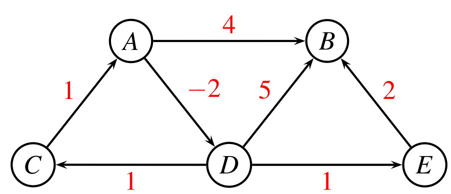
Course title: Network Optimization **Course no:** 42115

Aids allowed: All
Exam duration: 4 hours
Weighting: The points are indicated at each question
Language: You can write in Danish or English

Individual exam: You are not allowed to work together or contact any other people. All solutions that are suspiciously similar, will be reported to the study administration.

Shortest path problem

Consider the following directed graph where distances w_{ij} are marked with red. You are asked to find the shortest path from A to all other nodes.



Problem 1: (1 point) Argue whether the problem contains negative cycles, and state which algorithm you would use to solve the problem. ■

Problem 2: (3 point) Solve the problem. Give sufficient details to make it possible to follow the algorithm. ■

Project planning

Consider a critical path problem, in which each task i has a processing time p_i, and a list of predecessors P_i. The predecessors P_i need to finish before task i can begin.

task	predecessor(s)	processing time p_i
A	C	8
B	C, E	5
C		4
D	A, B, F	3
E		2
F	B, H	4
G	A	2
H	E	6

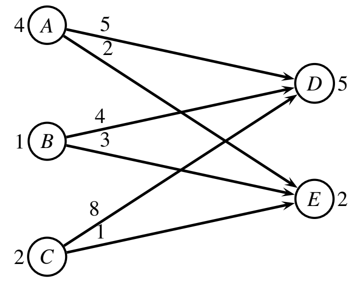
Problem 3: (2 point) Order the activities in topological order. ■

Problem 4: (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. ■

Problem 5: (1 point) Mark the critical path, and argue why the edges are on the critical path. ■

Transportation problem

Three factories A, B, C are transporting items to two customers D, E. The *transportation costs* of sending one item along an edge is indicated on the graph below. The production at factories are respectively $p_A = 4, p_B = 1, p_C = 2$, while the demands at the customers are $d_D = 5, d_E = 2$.



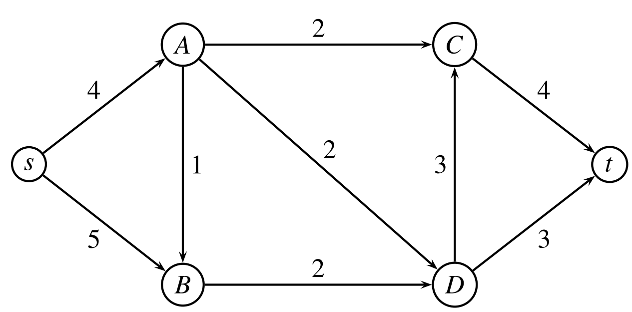
A greedy algorithm considers the factories in the order A, B, C, in each step sending the items to the customers in the cheapest possible way.

Problem 6: (2 point) Find the greedy solution, and report the cost of the solution. ■

Problem 7: (4 point) Use the network simplex algorithm to solve the problem to optimality. In each iteration report the dual variables, the reduced costs of edges. Report the cost of the optimal solution. ■

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t.



Problem 8: You can answer *one* of the following questions:

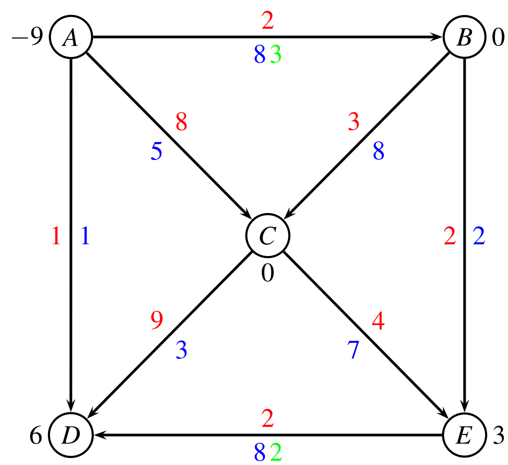
- (2 points) Use the augmenting path algorithm to solve the above problem.
- (4 points) Use the preflow-push (also called push-relabel) algorithm to solve the above problem.

In both cases, give sufficient details to make it possible to follow the algorithm. ■

Problem 9: (1 point) Describe how you find a minimum cut, and report the cut, and its capacity. ■

Repositioning of containers

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). Moreover, we have lower bounds ℓ_{ij} (green) on some of the edges. The demand b_i (black) in every node is positive for a sink, and negative for a source.



(∗)

Problem 10: (2 point) Transform the min-cost flow problem with upper and lower bounds to an ordinary min-cost flow problem, having only upper bounds. Describe the individual steps of the transformation. ■

Now, consider the transformed problem. Having solved a maximum flow problem to find a feasible solution we are told that $T = \{(A,B),(B,C),(C,E),(E,D)\}$ and $U = \{(A,D),(B,E),(C,D)\}$ and $L = \{(A,C)\}$. Edges in T form a tree. The flow on edges in U is at the upper bound, the flow on edges in L is at the lower bound (i.e. zero).

Problem 11: (2 point) Calculate the flow corresponding to the feasible solution defined by T, U, L. ■

Problem 12: You can answer *one* of the following questions:

- (3 point) Use the augmenting cycle algorithm to solve the problem to optimality.
- (4 point) Use the network simplex algorithm to solve the problem to optimality. In each iteration report the dual variables, the reduced costs of edges.

In each iteration report the selected cycle, and how much can be flown along the cycle. Report the cost of the optimal solution. ■

Problem 13: (2 point) Find the solution to the original problem (*). Report the flow on the edges, and the total cost. ■

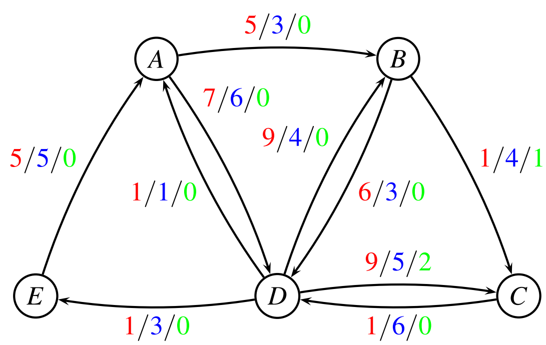
Flowing containers in a liner shipping network

The below instance, BALTICLINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

Note: The network is the same as in *Project Assignment 2020*, but the (source,terminal) pairs for each commodity is changed.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u, \ell)$ where each edge $(i, j) \in E$ has an associated cost (per container) $c_{ij} \geq 0$, a capacity $u_{ij} \geq 0$, and a lower bound $\ell_{ij} \geq 0$.

BALTICLINE **(Route net of a Baltic liner shipping company)**



For edge (i, j) the first number (red) is the cost c_{ij}, the second number (blue) is the capacity u_{ij}, and the third number (green) is the lower bound ℓ_{ij}.

We have the demands

k	s_k	t_k	d_k
1	D	C	2
2	D	B	1
3	A	E	3

Commodity $k = 1$ is shipping two containers from D to C, commodity $k = 2$ is shipping one container from D to B, and commodity $k = 3$ is shipping three containers from A to E. The commodities should be transported *unsplittable*, i.e. all d_k units should follow the same path.

Problem 14: (2 point) For each commodity k in BALTICLINE, write up a table with all possible simple routes from s_k to t_k through the graph $G = (V, E, c, u, \ell)$. A route may only use edges where the capacity is at least d_k, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the d_k containers. (**Hint:** There should be 9 distinct routes.) ■

Problem 15: (3 point) Write up the path formulation of BALTICLINE. ■

Now, assume that we want to solve the LP-relaxed version of the path formulation with column generation. After a few iterations of the algorithm (using dummy routes as initial columns) we have the following dual solution:

Dual variables for the upper bound constraints: $y_{AB} = -2$, $y_{DA} = -3$. Dual variables for the lower bound constraints: $y'_{BC} = 7$, $y'_{DC} = 2$. Dual variables for the commodity constraints: $\overline{y}_1 = 8$, $\overline{y}_3 = 2$. All other dual variables are zero.

Problem 16: (3 point) Formulate the pricing problem of commodity $k = 1$ as a shortest-path problem. Draw the corresponding graph, indicate the costs of the edges, and mark the root r and the sink s. ■

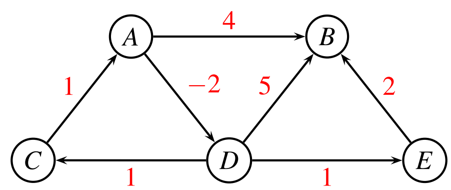
Organizing a conference

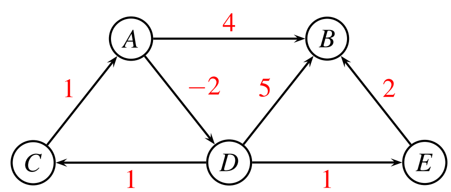
Suppose you are organizing a conference where researchers present articles they have written. Researchers who want to present an article send a paper to the conference organizers. The conference organizers have access to a committee of reviewers $R = \{r_1, \ldots, r_m\}$. Each reviewer r_i is willing to read up to a_i articles. There are $P = \{p_1, \ldots, p_n\}$ paper submissions, and each paper must be reviewed by 3 to 5 distinct reviewers (more reviewers is better). Moreover, each paper submission has a particular topic and each reviewer has a specialization for a set of topics, so papers on a given topic only get reviewed by those reviewers who are experts on that topic. The conference organizers need to decide which reviewers will review each article.

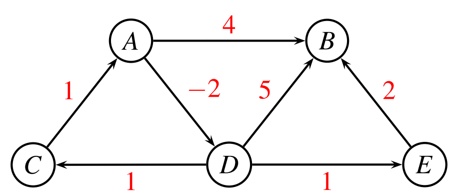
Problem 17: (4 point) Explain how the assignment of reviewers to papers can be formulated as a network optimization problem. Describe the network (nodes, edges, edge weights, demands) and state which type of problem it is. ■

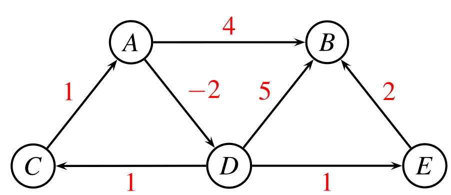
——— end of assignment ———

Shortest path problem

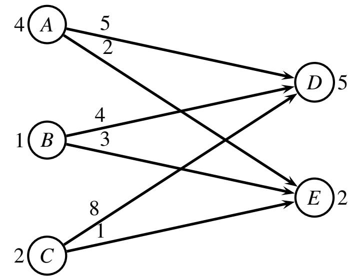


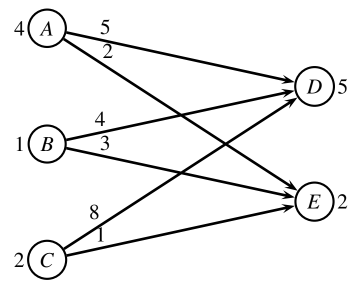


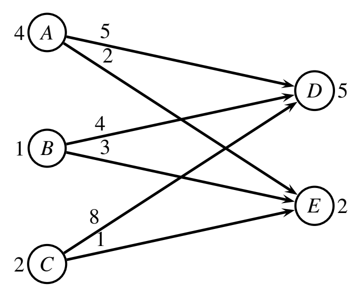


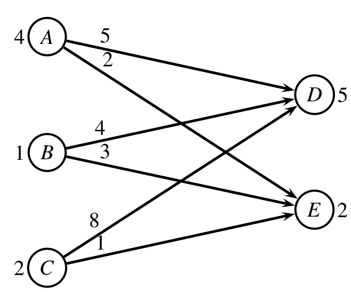


Transportation problem

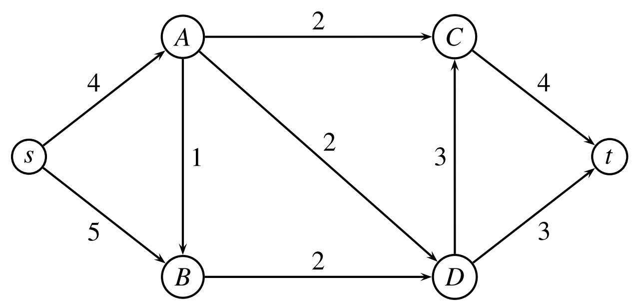


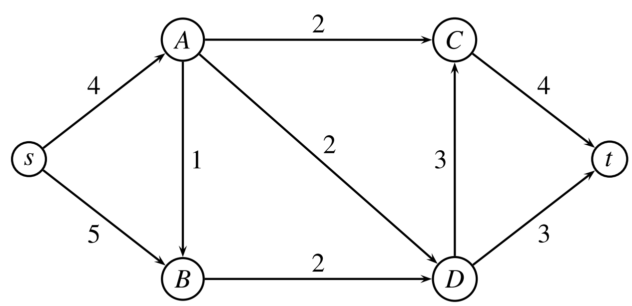


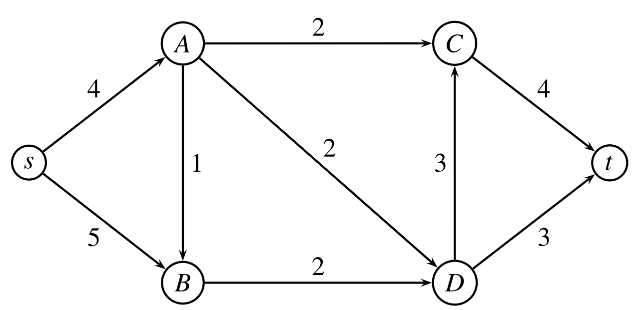


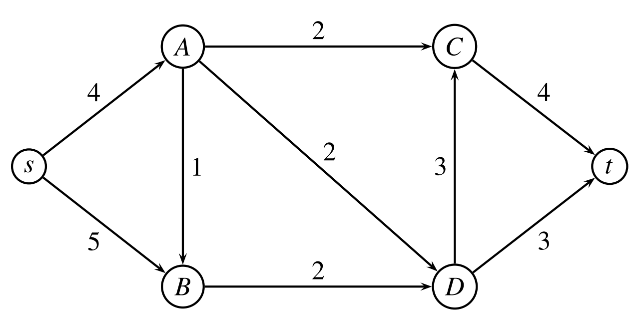


Maximum flow problem

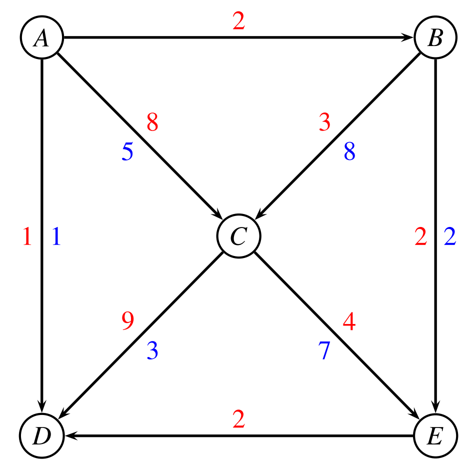


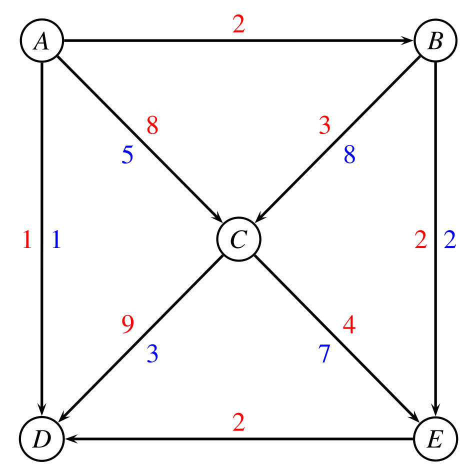


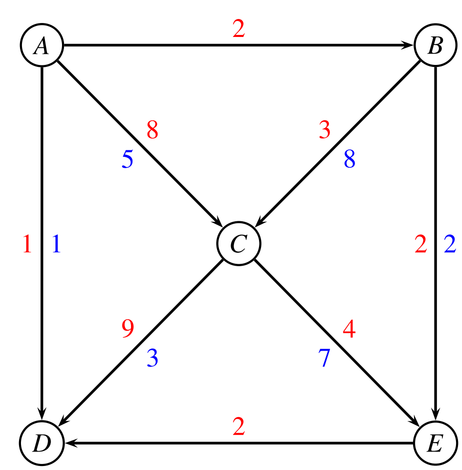




Repositioning of containers







DTU
Danmarks Tekniske Universitet

Written exam 15 December, 2021

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

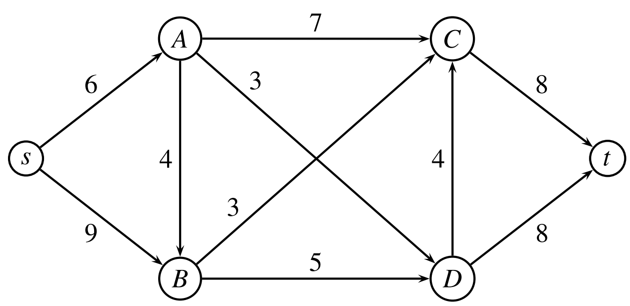
Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t.



Problem 1: (3 point) Use the augmenting path algorithm to solve the above problem. In every iteration use the augmenting path with largest capacity. Give sufficient details to make it possible to follow the algorithm. Report the maximum flow. ■

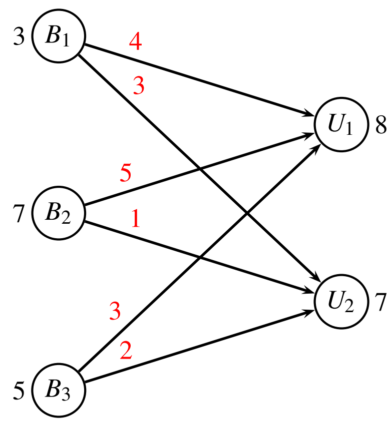
Problem 2: (2 point) Draw the residual graph, and describe how you find a minimum cut. Report the cut and its capacity. ■

Transportation problem

Three bakeries B_1, B_2, B_3 are delivering cakes to two universities U_1, and U_2. The bakeries are producing 3, 7 and 5 cakes respectively. University U_1 needs 8 cakes, while university U_2 needs 7 cakes. The profit of the cakes depends on the combination of bakery and university as given in the following matrix:

	B_1	B_2	B_3	demand
U_1	4	5	3	8
U_2	3	1	2	7
production	3	7	5	

We can formulate the problem as a transportation problem in maximization form:



Problem 3: (1 point) Show how you can transform the problem to an ordinary transportation problem in minimization form. ■

Problem 4: (2 point) Find a greedy solution for the transformed problem: Consider the nodes in the order B_1, B_2, B_3, always choosing the cheapest option. Report also the cost of the greedy solution. ■

Problem 5: (3 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criterium. ■

Problem 6: (1 point) Report the optimal solution to the original maximization problem and the objective value. ■

Critical path

Consider a critical path problem, in which each task i has a duration d_i, and a list of predecessors P_i. The predecessors P_i need to finish before task i can begin.

task i	duration d_i	predecessors P_i
A	6	C,H,I,J
B	7	C
C	4	E,H
D	5	A,B
E	4	
F	6	B,D,G
G	4	A
H	3	
I	2	E,K
J	6	E
K	5	

Problem 7: (2 point) Order the activities in topological order. ■

Problem 8: (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. ■

Problem 9: (1 point) Report the critical path, and argue why the edges are on the critical path. ■

Max flow and min cut

We are considering a maximum flow problem defined on a directed graph $G = (V, E)$ with edge capacities $u_{ij} \geq 0$. The solution vector is denoted (x_{ij}), and a cut is denoted $(R, \overline{R})$.

Problem 10: (3 point) Are the following statements correct? (Prove or disprove each statement. To disprove a statement it is sufficient to show an example.)

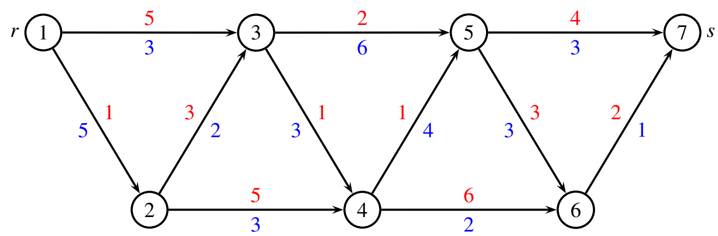
a) If all edge capacities are distinct, the max flow (x_{ij}) is unique.

b) If all edge capacities are multiplied by a positive integer k, the min cut $(R, \overline{R})$ remains unchanged.

■

Resource constrained shortest path (bidirectional method)

Consider the following resource constrained shortest path problem, where costs are red, and times are blue.



We want to find the shortest (cheapest) path from node $r = 1$ to node $s = 7$, with time constraint $T = 14$.

Since the resource constrained shortest path problem is computationally difficult to solve, we use the bidirectional method, where we use dynamic programming from both sides.

We first run forward dynamic programming where we stop extending states when their resource (time) is above $T/2$.

node 1		node 2		node 3		node 4		node 5		node 6		node 7	
cost	time	cost	time	cost	time	cost	time	cost	time	cost	time	cost	time
0	0	1	5	4	7	5	10	6	13	12	8		
				5	3	6	6	7	9				

We then run backward dynamic programming where we also stop extending states when their resource (time) is above $T/2$.

node 1		node 2		node 3		node 4		node 5		node 6		node 7	
cost	time	cost	time	cost	time	cost	time	cost	time	cost	time	cost	time
14	9	10	10	6	9	5	7	4	3	2	1	0	0
		12	8	9	6	8	3						
		13	6										

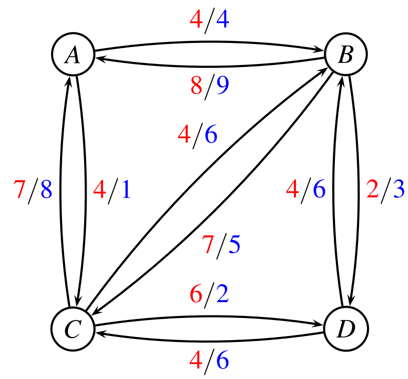
Problem 11: (2 point) For each node, combine the states from the forward and backward solution. Delete states that exceed the time constraint. Mark dominated (but feasible) states with parenthesis. ■

Problem 12: (1 point) What is the shortest (cheapest) path from node $r = 1$ to node $s = 7$, with time constraint $T = 14$? (You only need to report the cost and time, not the route). ■

Repositioning of empty containers

We are considering the liner shipping network ASIANET from the project assignment. The network, costs, and capacities are **unchanged**.

ASIANET



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}.

Due to the imbalance in trade, we need to reposition empty constainers. For ASIANET, we have the following demand for containers:

node	A	B	C	D
demand	1	2	-7	4

(*)

Negative demands denote a surplus of containers. It is assumed that all empty containers are equal. Furthermore, we have some **new lower bounds** on the edges: Edge (B,A) has a lower bound of $\ell_{BA} = 4$, and edge (D,B) has a lower bound of $\ell_{DB} = 1$.

Problem 13: (2 point) Formulte the ASIANET repositioning problem (with lower bounds) as a *normal* minimum-cost flow problem (only upper bounds on edges are allowed). ■

Problem 14: (4 point) Solve (by hand) the ASIANET repositioning problem, using the augmenting cycle or network simplex algorithm. When you find an initial solution using a maximum-flow algorithm, you should choose path *sCABt* as the first path. Give sufficient details to make it possible to follow your calculations. ■

Problem 15: (1 point) For the original problem (*) with lower bounds on edges, report the optimal solution (flow and cost). ■

Unsplittable Multi-commodity flow

We consider again the route net ASIANET, where **no lower bounds** are pending. We have three commodities $k = 1, \ldots, 3$ with **new demands** as follows:

k	s	t	d
1	C	B	4
2	A	D	2
3	D	A	6

Thus, commodity $k = 1$ is shipping four containers from C to B, commodity $k = 2$ is shipping two containers from A to D, and commodity $k = 3$ is shipping six containers from D to A.

Problem 16: (3 point) Write up the path formulation for the unsplittable multi-commodity flow problem. (Hint: there should be 7 paths). ■

We solve the LP-relaxed problem through column generation. After a few iterations we have the dual variables: $y_{DC} = -1$ (all other $y_{ij} = 0$), while $\bar{y}_1 = 16$, $\bar{y}_2 = 34$, $\bar{y}_3 = 72$.

Here, y_{ij} denote the dual variable corresponding to capacity constraints, and $\bar{y}_k$ denote the dual variable corresponding to commodity constraints.

Problem 17: (3 point) Write up an equation stating how the reduced cost of a route can be calculated. For each commodity k describe how a route with most negative reduced cost can be found as a shortest-path problem. Show the graph, and define root and sink of the problem. Describe how the edge costs are defined. Find an optimal solution (no details need to be given about the solution process). Report the reduced costs for each commodity, and select which route should be added to the master problem. ■

Research planning

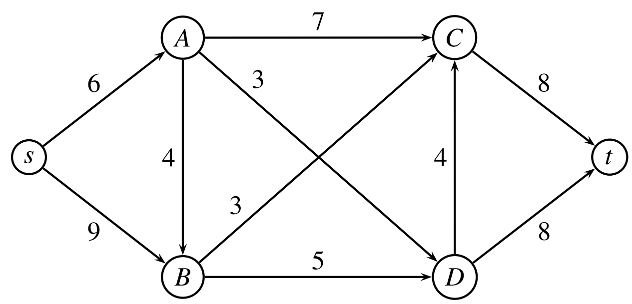
A research company has to handle 3 projects, P_1, P_2, P_3, during the next 5 months.

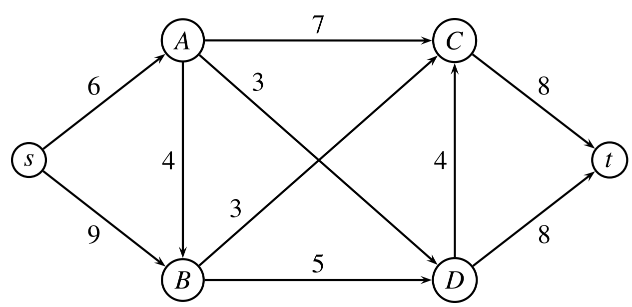
project	start month	end month	man-months needed
P_1	1	4	14
P_2	1	5	16
P_3	3	5	8

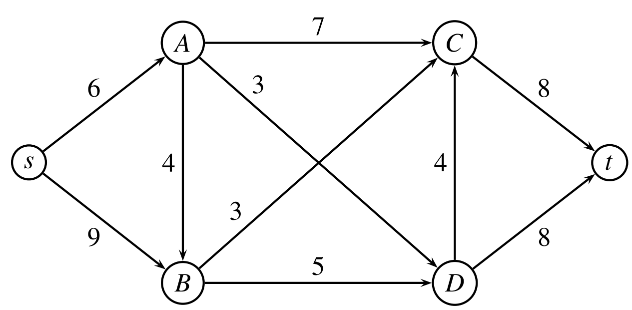
Project P_1 runs from month 1 to 4, P_2 runs from month 1 to 5, and P_3 runs from month 3 to 5. The projects require, respectively, 14, 16, and 8 man-months. Each month, 8 researchers are available, however not more than 5 researchers can work on the same project each month.

Problem 18: (5 point) Describe how this problem can be formulated as a maximum flow problem in an appropriate graph. ■

Extra figures







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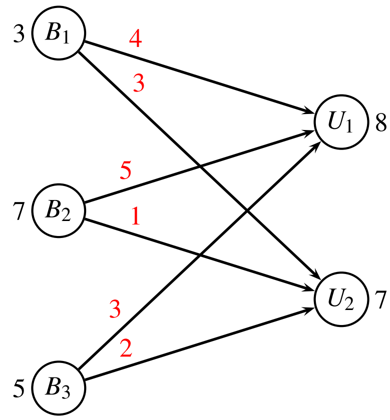

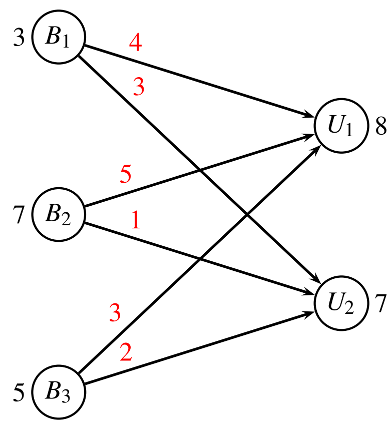

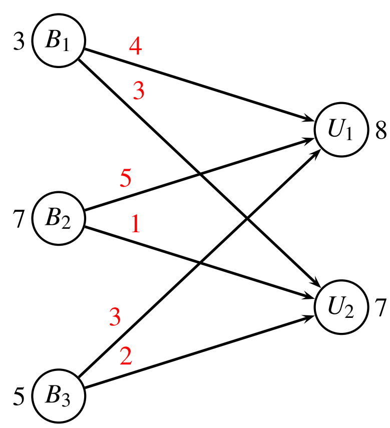

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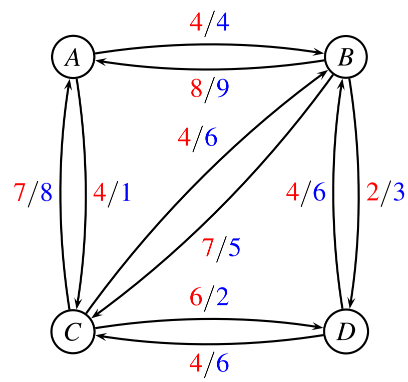
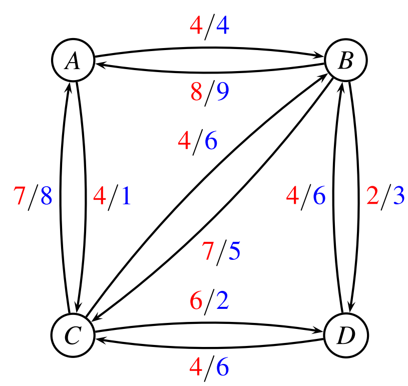
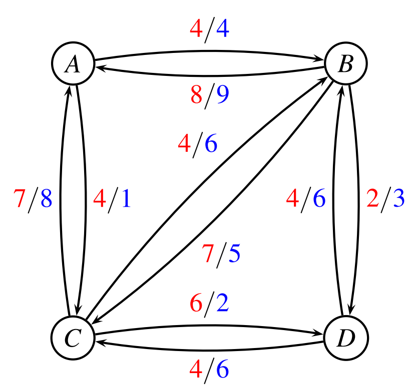






Written exam 16 December, 2022

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer.

The Pisinger cake

The Pisinger cake was created by Oscar Pischinger, a famous baker from Vienna, in the 19th century.

The preparation consists of the following eight tasks:

A. Leave the wafer-sheets to cool under load.
B. Coat the wafer-sheets with the obtained cream.
C. Using a round bowl, cut the wafers to get 10 round sheets.
D. Peel and fry almonds.
E. Cream: Beat eggs, chocolate, egg yolks, sugar in steam, when the mixture starts to thicken, cool it a bit, then mix in the frothy butter, lemon juice and ground almonds.
F. Chop almonds and mix them into the cream.
G. Pour the chocolate icing over the cooled cake and keep it cool.
H. Icing: Heat chocolate. Add oil and milk.

The duration of each task is given in the following table together with predecessors that need to be finished before the task can start

task	time (min)	predecessors
A	20	B
B	6	C,F
C	5	
D	12	C,E
E	15	
F	3	E,D
G	4	A,H
H	7	E,F,B

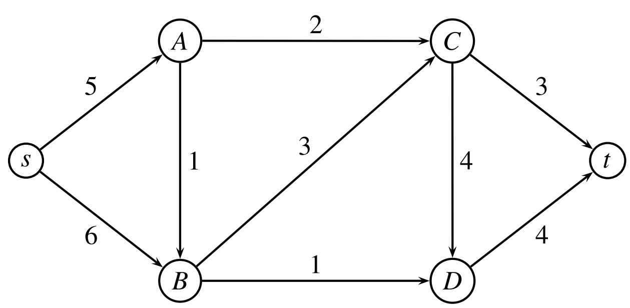
Problem 1 (1 point) Sort the tasks in topological order (either as graph, or as a list). ■

Problem 2 (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. ■

Problem 3 (1 point) Indicate the critical path, and argue why the edges are on the critical path. ■

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t.



Problem 4 (4 point) Use the *preflow-push* (also called *push-relabel*) algorithm to solve the above problem. Give sufficient details to make it possible to follow the algorithm. ■

Problem 5 (1 point) Describe how you find a minimum cut using the labels from previous question, and report the cut, and its capacity. ■

Max flow min cut

We are considering a maximum flow problem defined on a directed graph $G = (V,E)$ and edge capacities $u_{ij} \geq 0$. The solution vector is denoted (x_{ij}), and a cut is denoted $(R,\overline{R})$.

Problem 6 (4 point) Prove or disprove each statement. To disprove a statement it is sufficient to show an example.

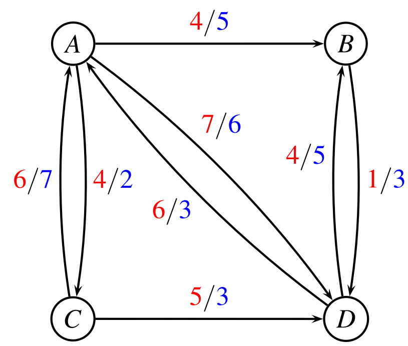
A) If all edge capacities in G are increased by an additive constant k, the min cut $(R,\overline{R})$ remains unchanged.

B) If all edge capacities in G are distinct, the min cut $(R,\overline{R})$ is unique.

■

Repositioning of empty containers

Consider the following network, where costs are red, and capacities are blue.



Due to the imbalance in trade, empty containers tend to accumulate at various ports. For the above network, one could have the following surplus of containers:

node i	A	B	C	D
surplus σ_i	8	-3	4	-9

We would like to reposition the containers in the cheapest possible way. It is assumed that all empty containers are equal.

Problem 7 (1 point) Formulate the problem of repositioning containers as a *minimum cost flow problem.* ■

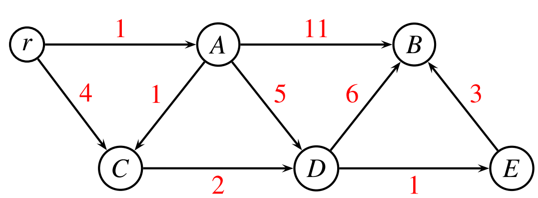
Problem 8 (2 point) Find an initial solution by solving a maximum flow problem. When solving the maximum flow problem always use the augmenting path with *largest capacity* in the residual graph. ■

Problem 9 (4 point) Use the *augmenting cycle* algorithm or *Network Simplex* to solve the problem. Report the optimal solution. ■

Shortest Path Problem in a dynamic network

In network communication there are several protocols to ensure that packages are sent to the correct destination. One of these, the *Open Shortest Path First* (OSPF), makes use of Dijkstra's algorithm to route packages in a dynamic network, where connections are added/removed.

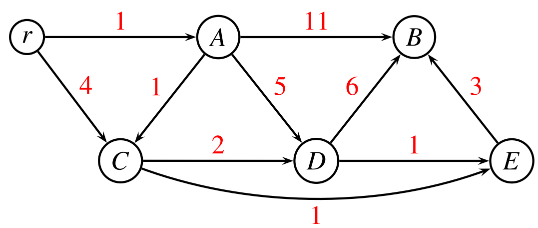
Consider the following directed graph where distances w_{ij} are marked with red.



Assume that you should find the shortest path from r to all other nodes.

Problem 10 (2 point) Solve the shortest path problem using Dijkstra's algorithm. Report the paths. ■

Now, assume that an edge (C,E) with distance 1 is added to the graph:



Problem 11 (2 point) How can we modify Dijkstra's algorithm to find the shortest path to all nodes, but somehow re-using calculations from previous questions? (You may assume that all calculations from previous questions are available). ■

Problem 12 (1 point) Use the modified algorithm to find the shortest path from r to all nodes in the new graph where edge (C,E) was added. Report the paths. ■

Transportation problem

Two producers A_1 and A_2 are delivering products to two customers B_1 and B_2. The producers can produce at most 8 and 7 items respectively. Customer B_1 has a demand of 5 items, while customer B_2 has a demand of 4 items. The transportation costs are given by the following table

	B_1	B_2	max production
A_1	4	5	8
A_2	1	6	7
demand	5	4	

Note, that the problem is not balanced.

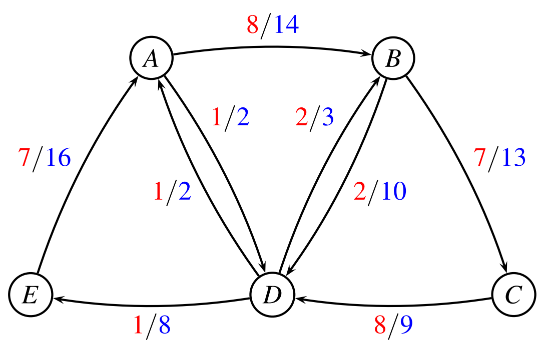
Problem 13 (1 point) Formulate the problem as a transportation problem, by specifying the network and all costs / demands. ■

Problem 14 (2 point) Find a greedy solution for the transformed problem: Consider the nodes in the order A_1, A_2 always choosing the cheapest option for transporting the production. ■

Problem 15 (4 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criteria, and report the optimal solution. ■

Flowing containers

MEDITLINE (Route net of a Mediterranean liner shipping company)



For edge (i,j) the first number is the cost c_{ij} and the second number is the capacity u_{ij} (in TEU).

We consider again the route net MEDITLINE from project assignment 2022. The network is *identical* to the one used in the project assignment, but we have *new demands* as follows:

k	s_k	t_k	d_k	w_k
1	D	C	3	2
2	A	D	2	4

Commodity $k=1$ is shipping three 40 foot containers (size $w_k = 2$ TEU) from D to C, and commodity $k=2$ is shipping two 40 foot high-cube containers (size $w_k = 4$ TEU) from A to D.

We are considering a *weighted multi-commodity flow problem*: For each commodity, a container of size w_k TEU has to follow the same path, but the d_k different containers can follow individual paths.

Problem 16 (2 point) For each commodity k in MEDITLINE, write up a table with all possible routes from s_k to t_k through the graph $G=(V,E,c,u)$. A route may only use edges where the capacity is at least w_k TEU, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the w_k TEU. (Hint: There should be 5 distinct routes.) ■

We solve the LP-relaxed problem through *column generation*. After a few iterations we have the *restricted master problem* on the form

$$
\begin{array}{llllllll}
\min & c_1x_1 & + & c_2x_2 & + & c_3x_3 & & & \\
\text{s.t.} & & & 2x_2 & + & 4x_3 & \leq & 14 & (y_{AB}) \\
& & & & & & \leq & 2 & (y_{AD}) \\
& 2x_1 & + & 2x_2 & & & \leq & 13 & (y_{BC}) \\
& & & & & 4x_3 & \leq & 10 & (y_{BD}) \\
& & & & & & \leq & 9 & (y_{CD}) \\
& & & & & & \leq & 2 & (y_{DA}) \\
& 2x_1 & & & & & \leq & 3 & (y_{DB}) \\
& & & 2x_2 & & & \leq & 8 & (y_{DE}) \\
& & & 2x_2 & & & \leq & 16 & (y_{EA}) \\
& x_1 & + & x_2 & & & \geq & 3 & (\bar{y}_1) \\
& & & & & x_3 & \geq & 2 & (\bar{y}_2) \\
& x_1, x_2, x_3 \geq 0 & & & & & & &
\end{array} \tag{1}
$$

Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i,j), and let $\overline{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k.

The dual variables for the above model are: $y_{DB} = -14$, while all other are $y_{ij} = 0$. Furthermore, $\overline{y}_1 = 46$, $\overline{y}_2 = 40$.

Problem 17 (4 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem, as well as the reduced cost. ■

Problem 18 (1 point) Which path should be added to the restricted master problem? ■

Chocolate factories

Two chocolate factories A and B are producing at most 500 chocolate-bars per month, each. Customer C demands 350 chocolate-bars, while customer D demands 400 chocolate-bars per month. Production costs are \$5 at factory A, while production costs are \$6 at factory B.

The factories can either deliver directly to the customers, or they can deliver through a warehouse W. Shipping costs per chocolate-bar are as follows:

From/to	W	C	D
A	1	5	6
B	2	4	5
W	-	3	2

The capacity of the shipping company is given by the following table:

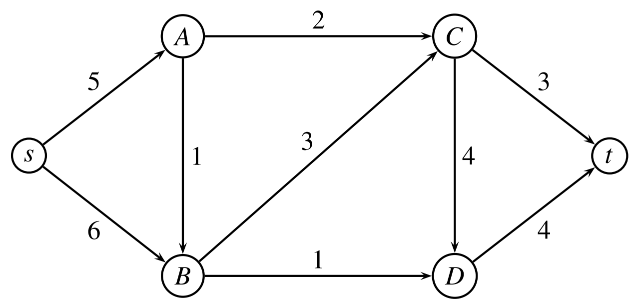
From/to	W	C	D
A	400	500	350
B	200	600	350
W	-	200	200

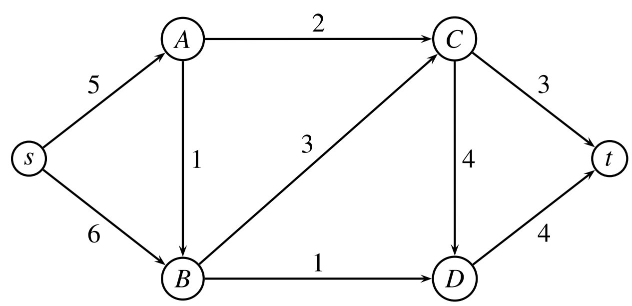
The warehouse has a limited capacity, so at most 300 chocolate-bars can be shipped through the warehouse. However, at least 80 chocolate-bars need to be shipped through the warehouse.

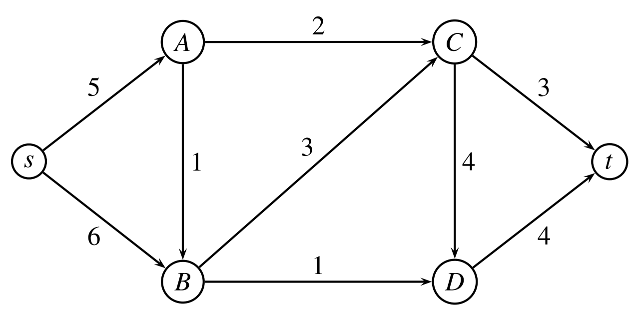
Problem 19 (5 point) Formulate the problem of deciding the company's production to minimize the total cost as a *minimum cost flow problem*. You should indicate the (cost, capacity) on *all* edges. Only upper bounds on the capacity are allowed. ■

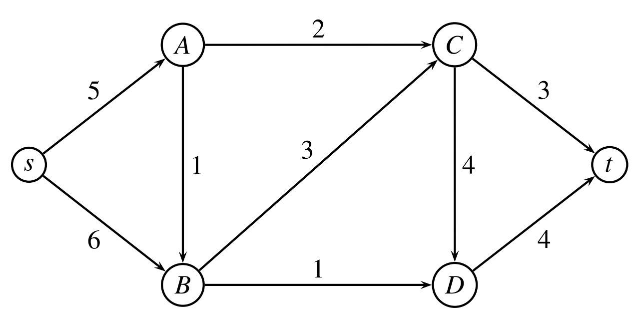
end of assignment

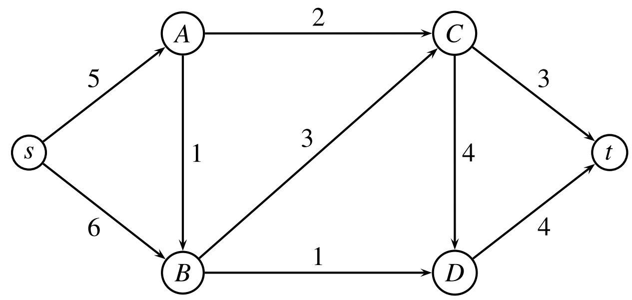
Maximum flow problem, Figures











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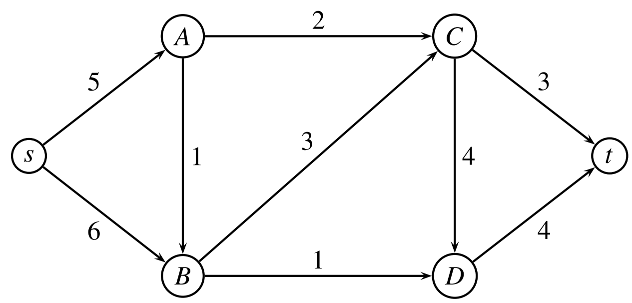

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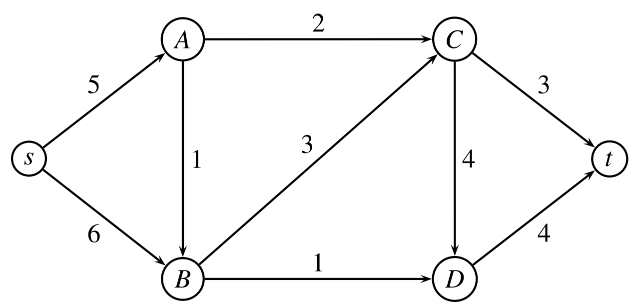

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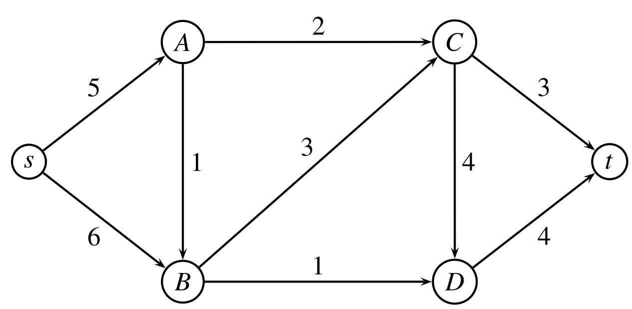

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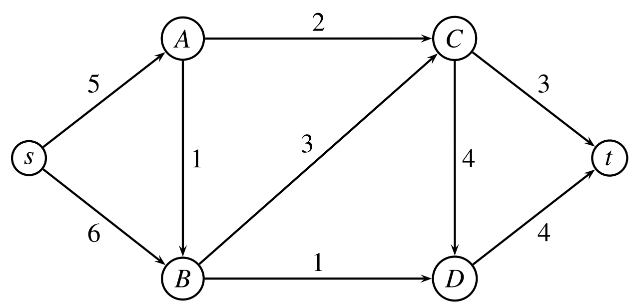
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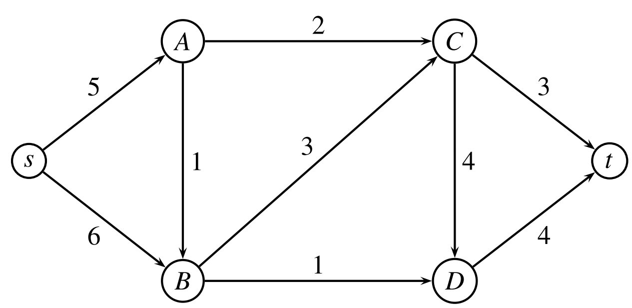
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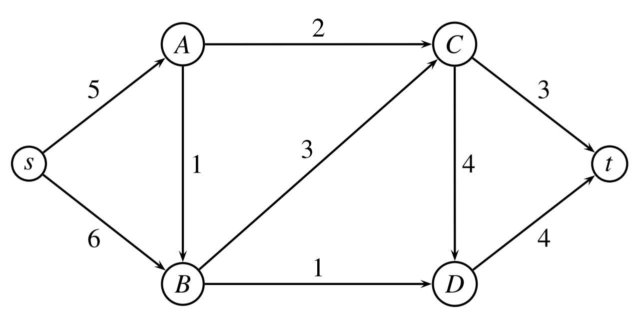
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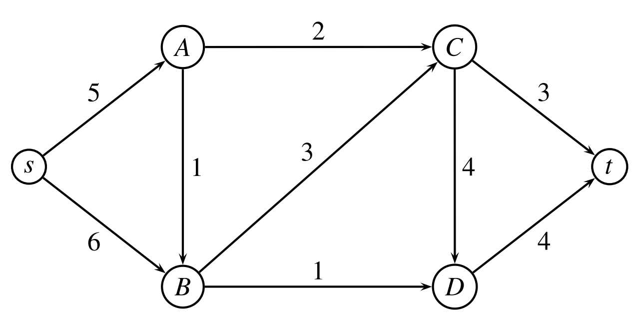
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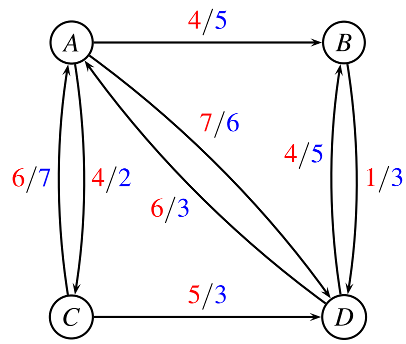
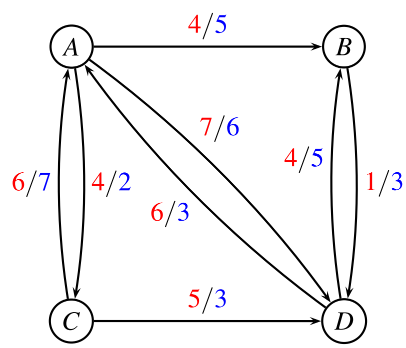
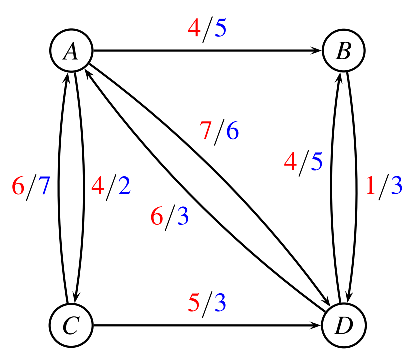
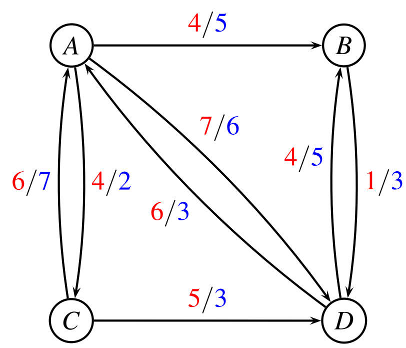
Repositioning of empty containers, Figures





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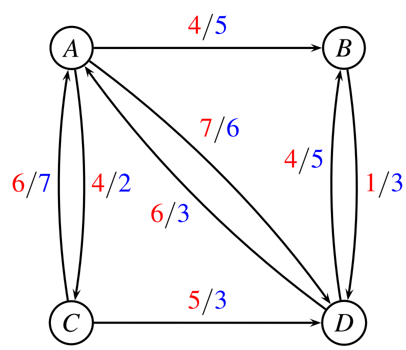
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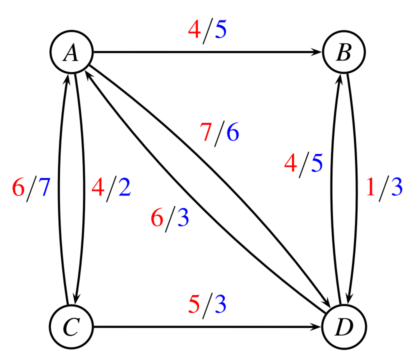
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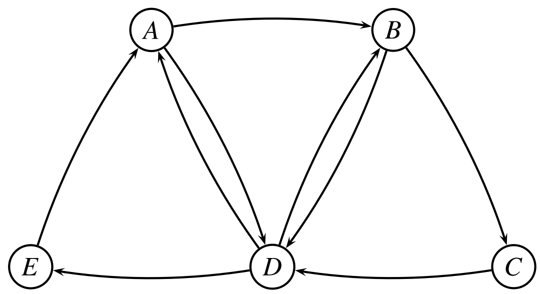
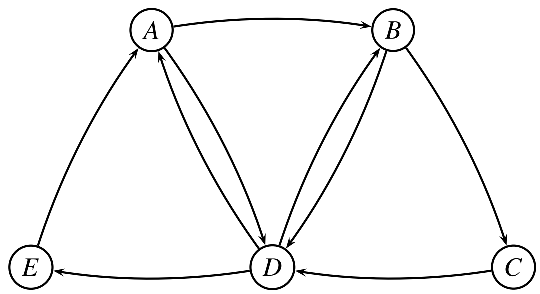
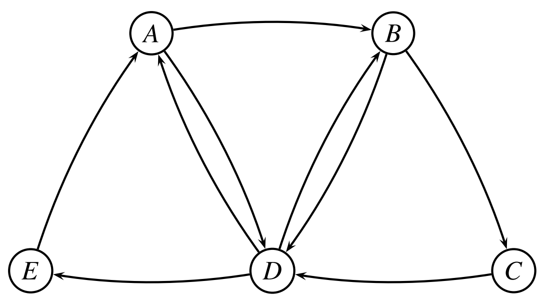
Flowing containers, Figures







Written exam 15 December, 2023

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer. Pull out the individual pages, and insert them in your answer.

Critical path with earliest start times

Thristur (þristur) is the favorite mini-candy in Iceland. Production consists of the following six steps:

activity	description	duration	predecessor	earliest starting time
A	Cut toffee into squares	2	F	8
B	Cut licorice into pieces	1		
C	Cover with chocolate	3	A,E	
D	Mix caramel cream	3		
E	Melt chocolate	4	B,F	7
F	Mix caramel cream and licorice	4	D,B	

Since some of the ingredients are made in remote factories, we have earliest starting times on some of the activities.

Problem 1 (2 point) Formulate the earliest starting times as tasks. Show the resulting table of tasks. ∎

Problem 2 (1 point) Sort the tasks in topological order (either as graph, or as a list). ■

Problem 3 (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. (**Note:** if you did not solve problem 1, then solve the original critical path problem). ■

Problem 4 (1 point) Indicate the critical path, and argue why the edges are on the critical path. ■

Transportation problem

Two bakeries B_1 and B_2 are delivering cakes to three courses C_1 (42115), C_2 (42114) and C_3 (42142). The bakeries are producing 6 and 4 cakes respectively. Course C_1 can eat at most 5 cakes, course C_2 can eat at most 3 cakes, and course C_3 can eat at most 4 cakes. All cakes produced need to be eaten. The transportation costs are given by the following table

	C_1	C_2	C_3	production
B_1	2	3	-	6
B_2	1	-	7	4
max demand	5	3	4	

A "-" indicates that transportation is not possible. The objective is to minimize overall transportation costs. **Note**, that the problem is not balanced.

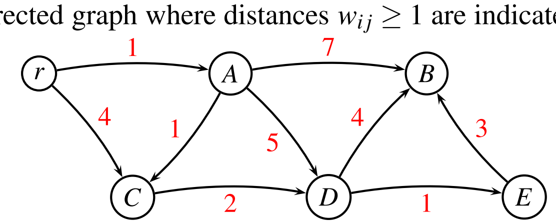
Problem 5 (1 point) Formulate the problem as a transportation problem, by specifying the network and all costs / demands. ■

Problem 6 (2 point) Find a greedy solution for the transformed problem: Consider the bakeries in the order $B_1, B_2, \ldots$ always choosing the cheapest option for transporting the cakes. ■

Problem 7 (4 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criteria, and report the optimal solution. ■

Shortest Path Problem

Consider the following directed graph where distances $w_{ij} \geq 1$ are indicated with red.



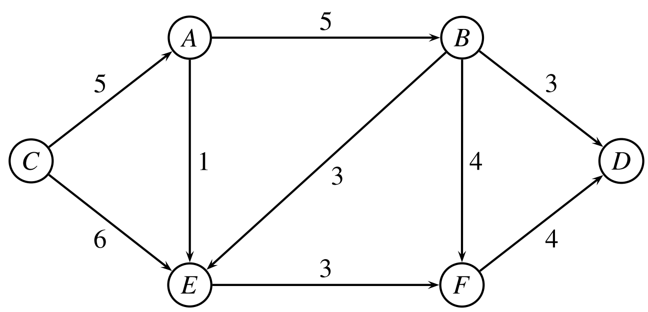
We want to find the shortest path from r to all nodes. If several paths have the same length, we are interested in the shortest path containing *most* edges.

Problem 8 (2 point) Formulate the problem as an ordinary shortest path problem. Show the resulting graph and edge costs. ■

Problem 9 (3 point) Solve the shortest path problem using Dijkstra's algorithm. Give sufficient detail to follow the calculations. Report the shortest path tree. (**Note:** if you could not solve the previous question, then solve the original shortest path problem). ■

Circulation problem / Maximum flow with demands

Given a directed graph $G = (V, E)$ and associated edge capacities u_{ij} for each edge (i, j). In the *Circulation Problem* we have multiple sources and multiple sinks. If a node has positive demand $d_i > 0$ it is called a sink. If a node has negative demand $d_i < 0$ it is called a source. Obviously, we need to have $\sum_{i \in V} d_i = 0$ to match supply and demand.



Assume that we have demands

node i	A	B	C	D	E	F
demand d_i	-2	0	-4	5	0	1

Problem 10 (2 point) Formulate the above problem as an ordinary maximum flow problem. (**Hint:** take inspiration from the min-cost flow problem.) ■

Assume that we have a lower bound on edge (B, E) of value $\ell_{BE} = 2$. This means that in a given solution, there should be at least 2 units flowing on the edge.

Problem 11 (2 point) Formulate the problem with lower bound on edge (B, E) as an ordinary maximum flow problem. ■

Problem 12 (4 point) Solve the resulting maximum flow problem. Provide sufficient details to follow your calculations. When choosing an augmenting path, always choose the path with largest possible capacity. (**Note:** if you could not solve the previous question, then solve the original problem.) ■

Problem 13 (1 point) Report the value of the maximum flow, and show a minimum cut. ■

Hungarian Algorithm

In the *Assignment problem* we are asked to assign a number of resources to an equal number of activities, so as to minimize total cost of allocation. Each resource should be assigned to exactly one activity, and each activity should have exactly one resource assigned.

The assignment problem is frequently solved with the *Hungarian Algorithm:* Given a cost matrix $C = \{c_{ij}\}$ saying how much it costs to assign resource i to activity j. The first step of the algorithm is to subtract the smallest number from each column, then subtract the smallest number from each row. (See example below).

$$
\begin{pmatrix}
- & 3 & 93 & 15 & 46 & 13 \\
6 & - & 79 & 46 & 36 & 22 \\
32 & 4 & - & 25 & 16 & 19 \\
53 & 104 & 92 & - & 83 & 25 \\
6 & 24 & 66 & 13 & - & 7 \\
3 & 88 & 18 & 48 & 105 & -
\end{pmatrix}
\rightarrow
\begin{pmatrix}
- & 0 & 75 & 2 & 30 & 6 \\
0 & - & 58 & 30 & 17 & 12 \\
29 & 1 & - & 12 & 0 & 12 \\
32 & 83 & 58 & - & 49 & 0 \\
3 & 21 & 48 & 0 & - & 0 \\
0 & 85 & 0 & 35 & 89 & -
\end{pmatrix}
$$

Next, we need to find the largest set of *independent* 0-values. Two 0-values are *independent* if they do not appear in the same row or column. For instances, in the first matrix below the bold-face 0-values are independent, but not in the second matrix

$$
\begin{pmatrix}
. & \mathbf{0} & . & . & . & . \\
\mathbf{0} & . & . & . & . & . \\
. & . & . & . & \mathbf{0} & . \\
. & . & . & . & . & \mathbf{0} \\
. & . & . & 0 & . & 0 \\
0 & . & \mathbf{0} & . & . & .
\end{pmatrix}
\qquad\qquad
\begin{pmatrix}
. & \mathbf{0} & . & . & . & . \\
0 & . & . & . & . & . \\
. & . & . & . & 0 & . \\
. & . & . & . & . & \mathbf{0} \\
. & . & . & 0 & . & \mathbf{0} \\
\mathbf{0} & . & \mathbf{0} & . & . & .
\end{pmatrix}
\tag{1}
$$

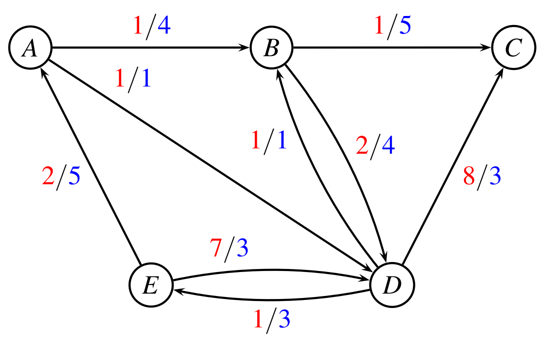
Problem 14 (4 point) Given a matrix with 0-values as in (1). Formulate the problem of finding the largest set of independent 0-values as a maximum flow problem. Explain how the graph is constructed, and how a solution should be interpreted. ■

Repositioning of empty containers

The below instance, MEDITLINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost (pr. container) $c_{ij} \geq 0$, and a capacity $u_{ij} \geq 0$.

MEDITLINE **(Route net of a liner shipping company)**



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}.

Consider the MEDITLINE network from the project assignment. In a given month, we have the following inventory of empty containers:

node	A	B	C	D	E
inventory	3	4	2	4	3

All empty containers are equal. We want to reposition the empty containers such that all containers are moved to C and D. Furthermore, the two nodes C and D should have the same amount of containers after repositioning.

Problem 15 (2 point) Formulate the repositioning of empty containers in MEDITLINE as a *minimum cost flow problem*. Define the graph and the demands in each node. ■

Problem 16 (5 point) Solve the MEDITLINE repositioning problem, using a network algorithm. State which algorithm you use, and give sufficient details to make it possible to follow your calculations. Report the optimal solution. ■

Flowing containers

We consider again the route net MEDITLINE from project assignment 2023, in which we have *new demands* as follows:

k	s_k	t_k	d_k
1	E	D	3
2	D	C	1
3	E	C	2

Commodity $k = 1$ is shipping three containers from E to D, commodity $k = 2$ is shipping one container from D to C, and commodity $k = 3$ is shipping two containers from E to C.

We are considering a *unsplittable multi-commodity flow problem* in which all the d_k containers must follow the same path.

Problem 17 (2 point) For each commodity k in MEDITLINE, write up a table with all possible routes from s_k to t_k through the graph $G = (V, E, c, u)$. A route may only use edges where the capacity is at least d_k, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the d_k units. (Hint: there should be 8 distinct routes.) ■

We solve the LP-relaxed problem through *column generation*. After a few iterations we have the *restricted master problem* on the form

$$
\begin{array}{lccccccccll}
\min & c_1x_1 & + & c_2x_2 & + & c_3x_3 & + & c_4x_4 & & & \\
\text{s.t.} & 3x_1 & & & & & + & 2x_4 & \leq & 4 & (y_{AB}) \\
 & & & & & & & & \leq & 1 & (y_{AD}) \\
 & & & & & & & & \leq & 5 & (y_{BC}) \\
 & 3x_1 & & & & & + & 2x_4 & \leq & 4 & (y_{BD}) \\
 & & & & & & & & \leq & 1 & (y_{DB}) \\
 & & & & & 1x_3 & + & 2x_4 & \leq & 3 & (y_{DC}) \\
 & & & & & & & & \leq & 3 & (y_{DE}) \\
 & 3x_1 & & & & & + & 2x_4 & \leq & 5 & (y_{EA}) \\
 & & & 3x_2 & & & & & \leq & 3 & (y_{ED}) \\
 & x_1 & + & x_2 & & & & & \geq & 1 & (\bar{y}_1) \\
 & & & & & x_3 & & & \geq & 1 & (\bar{y}_2) \\
 & & & & & & & x_4 & \geq & 1 & (\bar{y}_3) \\
 & \multicolumn{10}{l}{x_1, x_2, x_3, x_4 \geq 0}
\end{array}
$$

where c_1, c_2, c_3, c_4 are some costs we found in previous problem. Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i, j), and let $\overline{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k.

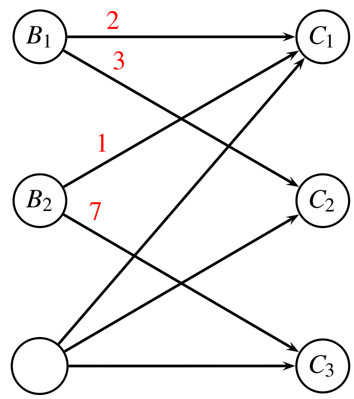
The dual variables for the above model are: $y_{AB} = -2$, while all other are $y_{ij} = 0$. Furthermore, $\overline{y}_1 = 21, \overline{y}_2 = 8, \overline{y}_3 = 30$.

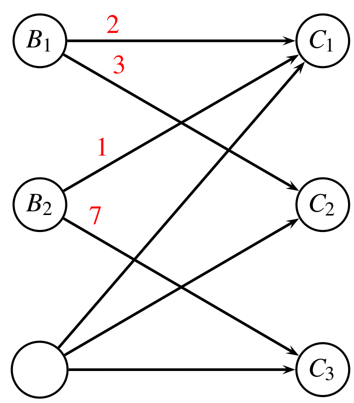
Problem 18 (4 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem, as well as the reduced cost. ■

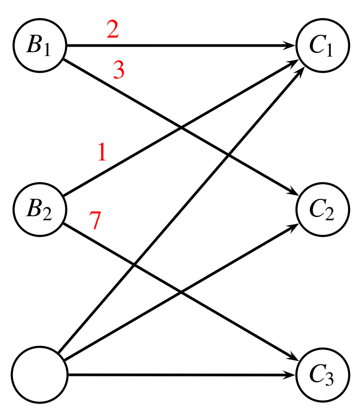
Problem 19 (1 point) Which path should be added to the restricted master problem? ■

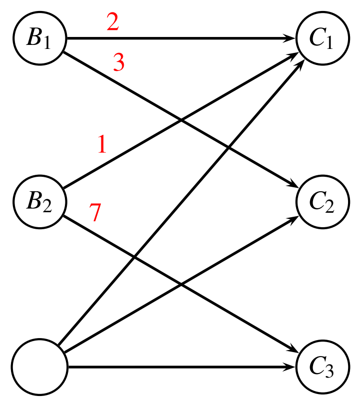
end of assignment

Transportation problem **Study number:**





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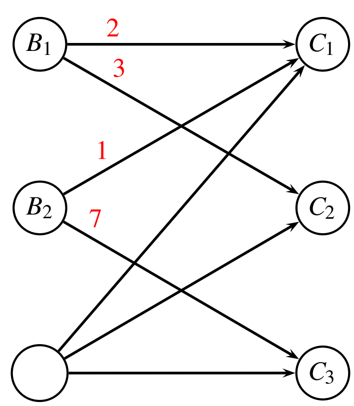
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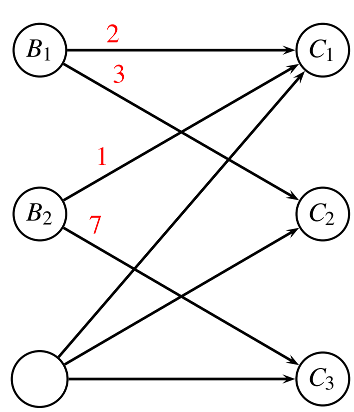
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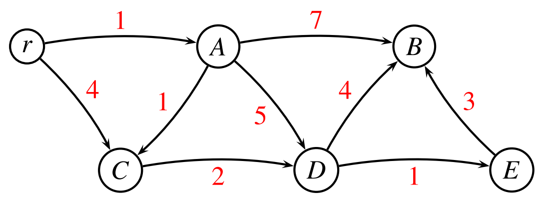
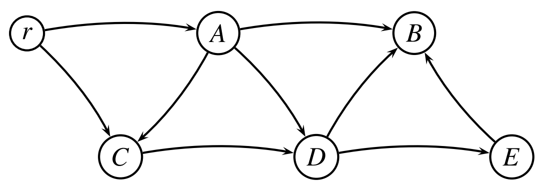
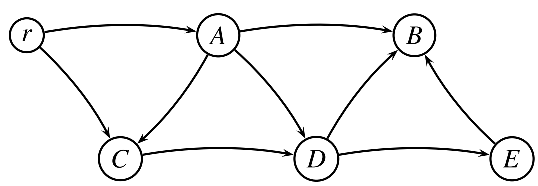
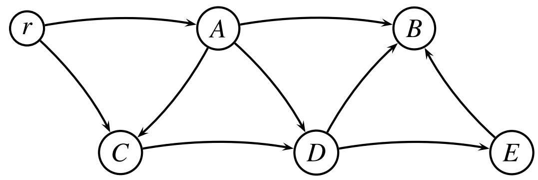
Shortest path problem **Study number:**





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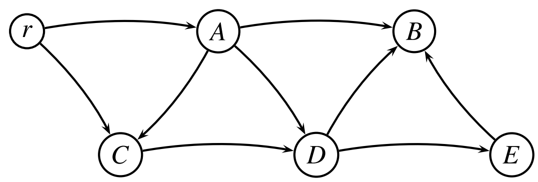
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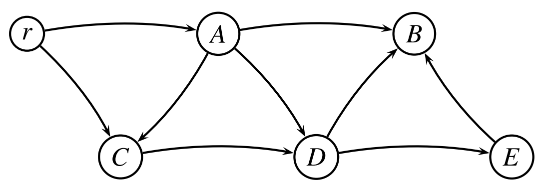
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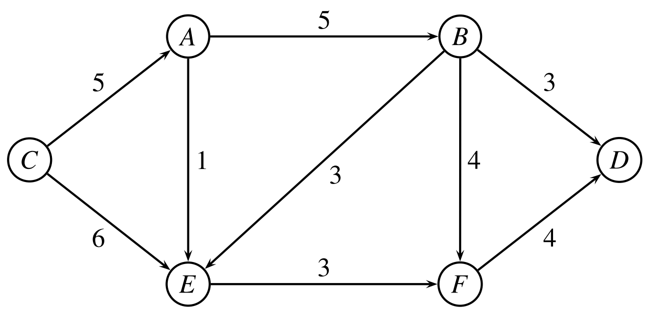
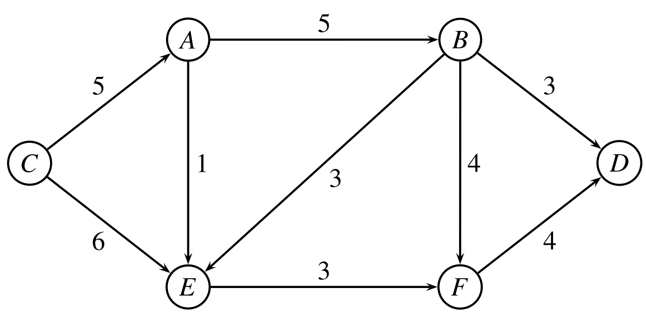
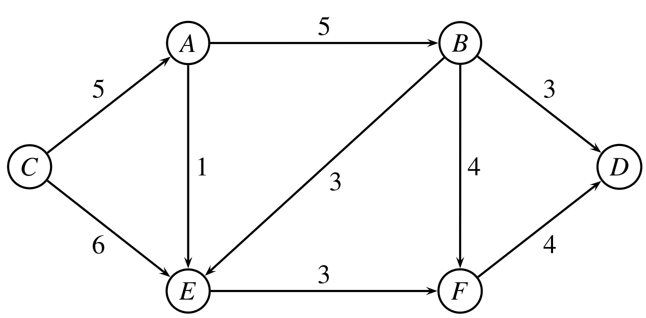
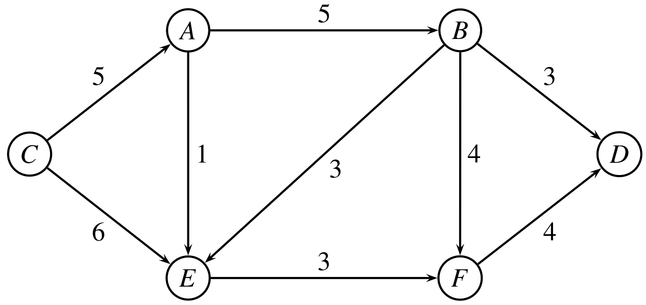
Circulation problem **Study number:**





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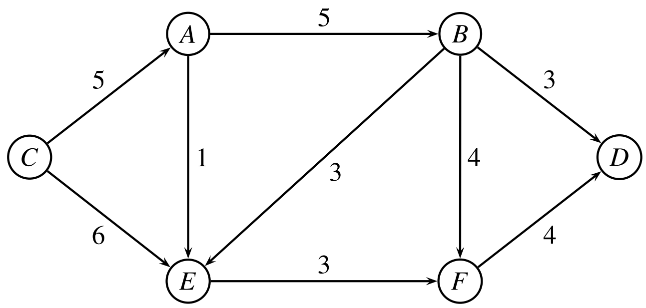
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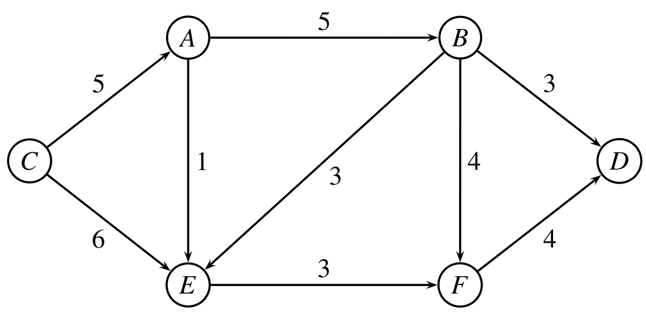
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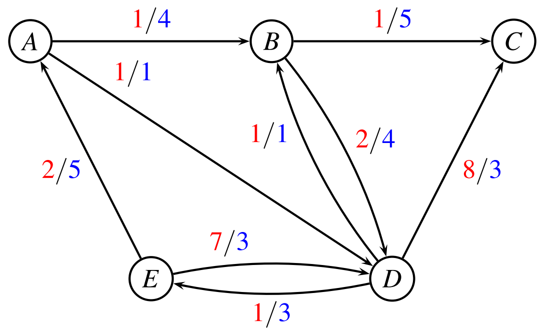
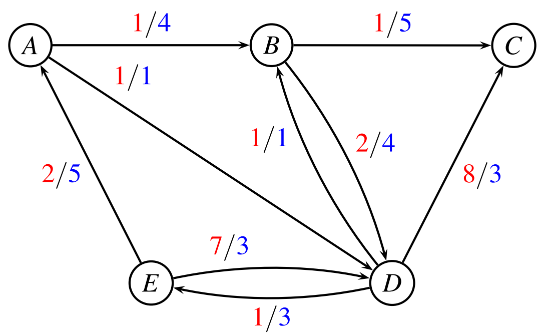
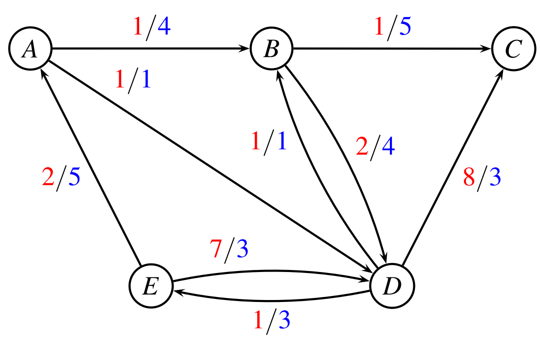
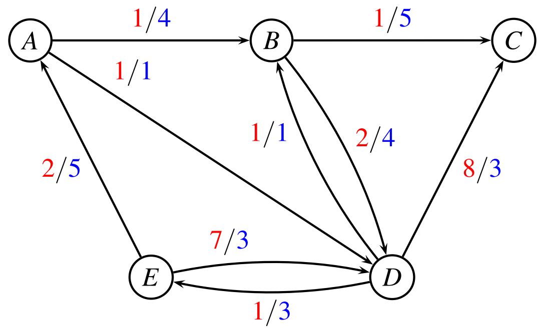
Repositioning of empty containers **Study number:**









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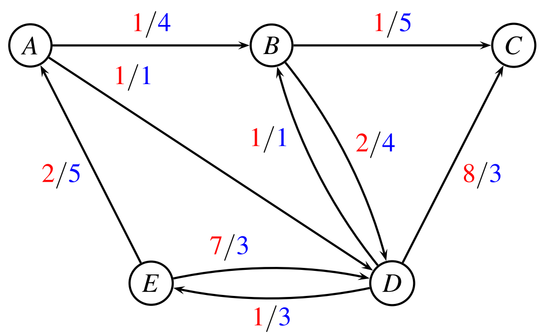

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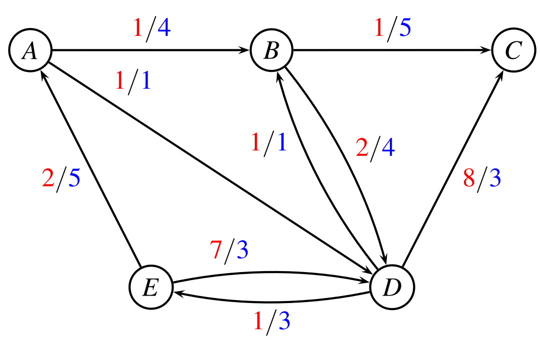

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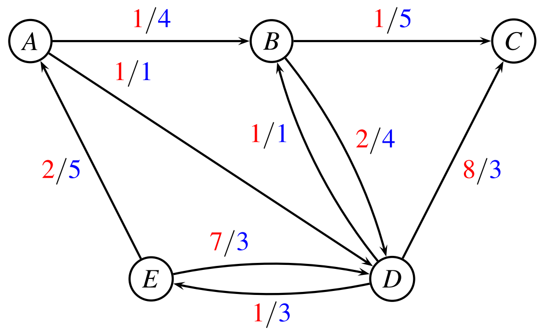
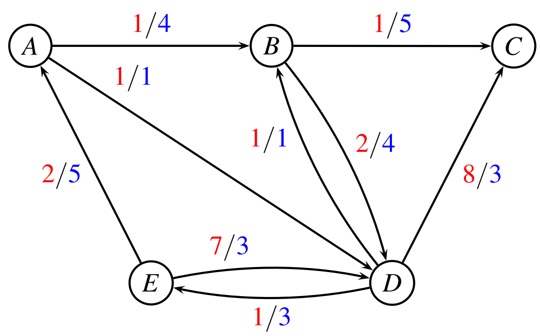
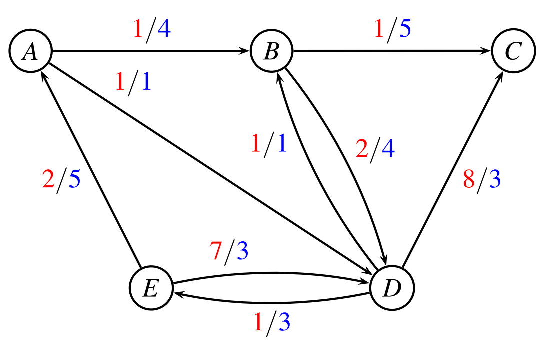
Flowing containers **Study number:**







Danmarks Tekniske Universitet

Written exam 17 December, 2024

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer. Pull out the individual pages, and insert them in your answer.

Critical path with time windows

Old-fashioned Danish apple cake (gammeldags æblekage) is a classic Danish dessert. Although it is called apple cake, it has more in common with a trifle or a crumble. It is a lovely dessert on top of a good meal, and then it is pretty easy to make. Æblekage is always served cold with a generous layer of whipped cream. Making the apple cake, consists of the below six steps $A, \dots, F$.

Now, assume that each task j has a corresponding time windows $[e_j, \ell_j]$ denoting the earliest and latest time when the task may start.

activity i	description	duration d_i	predecessor	e_j	ℓ_j
A	Boil apples	2	B, D	0	∞
B	Peel apples and cut into cubes	6		0	∞
C	whip cream	9	D, E	0	6
D	slice vanilla bean, scrape out seeds	3		2	3
E	fry bread-crumbs with butter	4		1	∞
F	mix apples and bread-crumbs	8	A, E	0	∞

Problem 1 (2 point) Sort the tasks in topological order (either as graph, or as a list) and find the earliest starting time U_i of each task i respecting time windows. Show the recursion you are using, and report the makespan T. ■

Problem 2 (3 point) Knowing the makespan T, find the latest starting time V_i of each task i, respecting the time windows. Show the recursion you are using. ■

Problem 3 (1 point) Indicate the critical path, and describe how you found it. ■

Assignment problem

In the *Assignment problem* we are asked to assign a number of resources A_i to an equal number of activities B_j, so as to minimize total cost of allocation. Each resource should be assigned to exactly one activity, and each activity should have exactly one resource assigned.

The below cost matrix $C = \{c_{ij}\}$ says how much it costs to assign resource A_i to activity B_j.

	B_1	B_2	B_3
A_1	2	6	1
A_2	3	1	4
A_3	9	3	2

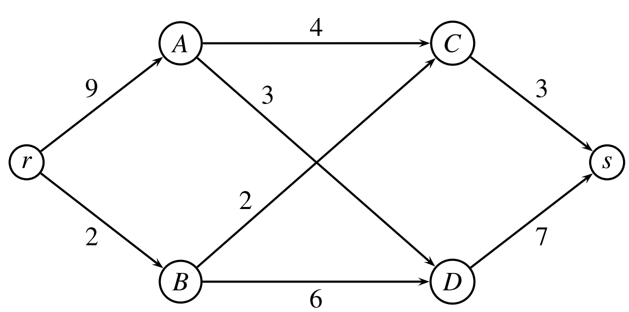
Problem 4 (2 point) Formulate the assignment problem as a transportation problem. ■

Problem 5 (1 point) Find a greedy solution for the transformed problem: Consider the rows in the order A_1, A_2, A_3, always choosing the cheapest option for transporting. ■

Problem 6 (4 point) Solve the problem for the above matrix C, using network simplex. **Hint:** If you have too few edges in basis, add some additional edges from A_3. ■

Maximum flow problem

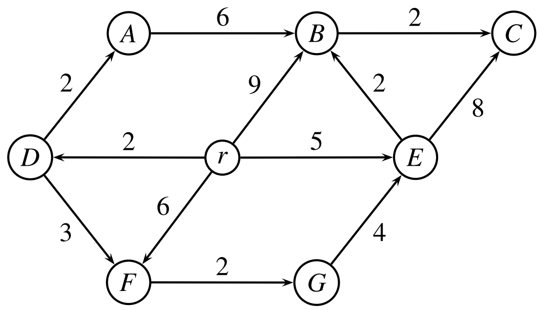
Consider the following maximum flow problem where the capacity u_{ij} is indicated on edge $(i, j) \in E$.



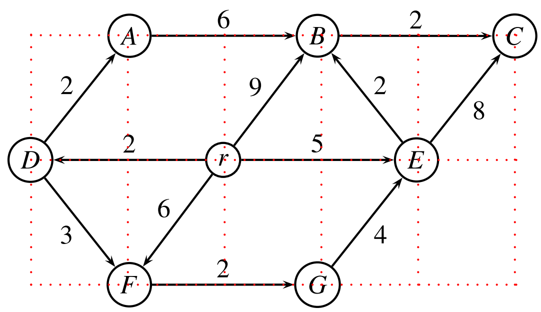
Problem 7 (5 point) Solve the problem using the *preflow-push* algorithm (also called *push-relabel*). You can use the figures on the last page, and enclose them to the solution. Remember to write what you are doing. ■

Problem 8 (2 point) Find a minimum cut and the corresponding capacity of the cut. (Explain how you find the cut). ■

A^* algorithm for shortest path



Problem 9 (3 point) Consider the above directed graph with r as root, and edge weights w_{ij}. Find the shortest distance from r to all nodes using Dijkstra's algorithm. Give sufficient details to follow your calculations. ■



If we only are interested in solving the shortest distance to a specific terminal node $t = C$, we can use the A^* algorithm which makes use of heuristic distances to guide the search.

The Manhattan distance between two nodes (x_1, y_1) and (x_2, y_2) is measured as $h = |x_1 - x_2| + |y_1 - y_2|$. For instance, the Manhattan distance from r to C is $h = 3 + 1 = 4$, while the Manhattan distance from F to C is $h = 4 + 2 = 6$.

Problem 10 (1 point) Calculate the Manhattan distance h_i from node i to terminal $t = C$ for all nodes. ■

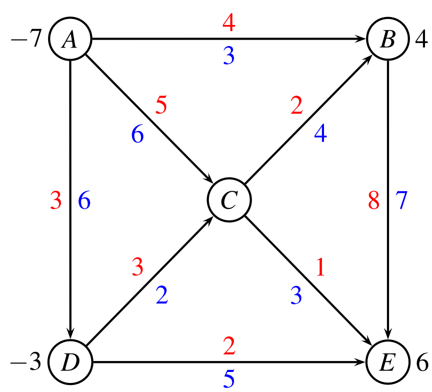
The following A^* *algorithm* finds the shortest path from r to t. It is identical to Dijkstra's algorithm, except that in every iteration we select the node i with lowest value of $y_i + h_i$. (The normal Dijkstra algorithm selects the node with lowest value of y_i). Furthermore, the algorithm stops when node t is added to T (i.e. marked "red"). All changes are written in blue.

$T \leftarrow \emptyset$;
$y_r \leftarrow 0$; $y_i \leftarrow \infty$ for all other i;
while $V \setminus T \neq \emptyset$ **do**
    select an $i \in V \setminus T$ minimizing $y_i + h_i$;
    for $\{(i,j) \in E : j \notin T\}$ **do**
        if $y_i + w_{ij} < y_j$ **then**
            $y_j \leftarrow y_i + w_{ij}$;
    $T \leftarrow T \cup \{i\}$;
    if $i = t$ **then**
        stop

Problem 11 (2 point) Use the above algorithm to find the shortest distance from r to $t = C$. Give sufficient details to follow your calculations. ■

Minimum cost flow problem

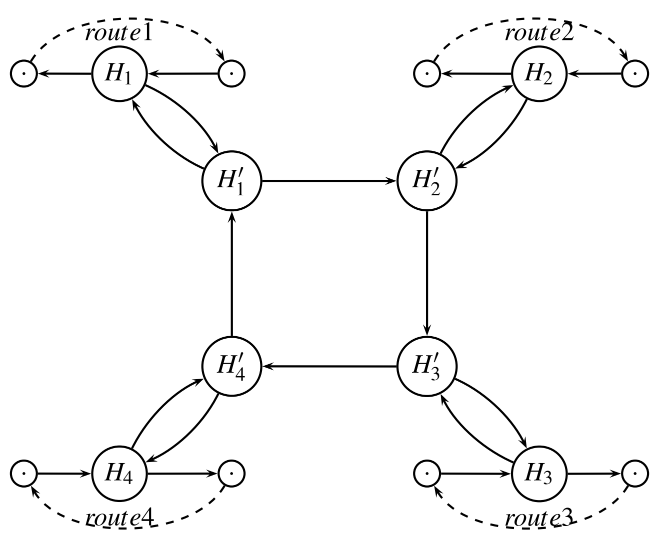
Consider the following minimum cost flow problem costs w_{ij} are indicated with red, capacities u_{ij} are indicated with blue, and demands d_i are indicated in black.



Problem 12 (2 point) Find an initial solution by solving a maximum flow problem. ■

Problem 13 (5 point) Solve the problem to optimality. ■

Modeling a hub



We study how the hub port H in a liner shipping network, HUBLINE, can be formulated in graph terms as part of a multi-commodity flow network. We have four routes arriving to the hub as follows:

route	1	2	3	4
weekday	Monday	Tuesday	Friday	Sunday

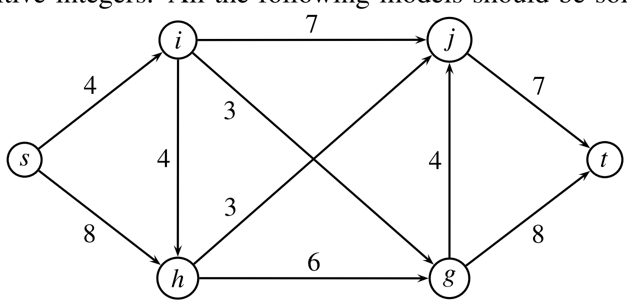
We assume that each vessel is only one day in the port, and it is assumed that all loading and unloading is done during this day.

Crane capacity: The crane capacity is $u = 5$ for unloading and $u = 4$ for loading a vessel. *Transshipment cost:* The cost of loading/unloading is 6 euro per container. *Transshipment time:* The time for transferring a container from one vessel to another can be calculated from the time table. *Storage capacity:* The port has a limited storage capacity of $\overline{u} = 9$ containers on shore. *Storage cost:* The shipping company must pay $\overline{c} = 2$ euro for each container each night it is stored in the port.

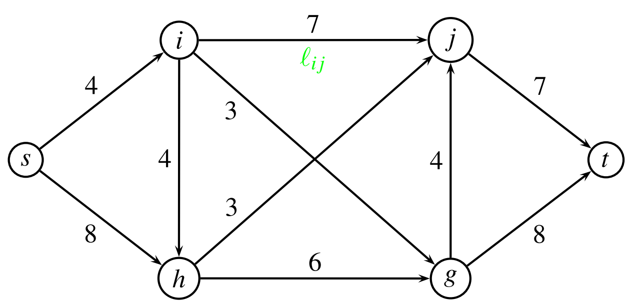
Problem 14 (3 point) Indicate the costs, capacities, times on the above graph. ■

Maximum flow with lower bound on edges

Given a directed graph $G = (V, E, u)$, root/sink (s, t) and a target for the flow k. We assume that all capacities u are positive integers. All the following models should be solvable as *maximum flow problems.*



Problem 15 (1 point) How would you check whether a feasible solution exists with flow $|f| = k$? (You can use the above graph as an example, but try to argue generally).
Hint: Formulate the problem as a circulation problem. ■



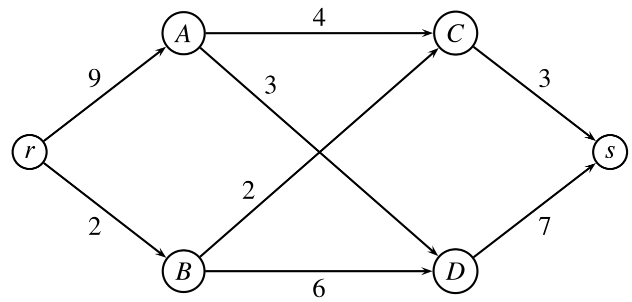
Problem 16 (2 point) Now, assume that there is a lower bound ℓ_{ij} on an edge (i, j). How would you check whether a feasible solution exists with flow $|f| = k$, respecting the lower bound? (You can use the above graph as an example, but try to argue generally. The method should also work if you have multiple edges with lower bound).
Hint: Formulate the problem as a circulation problem. ■

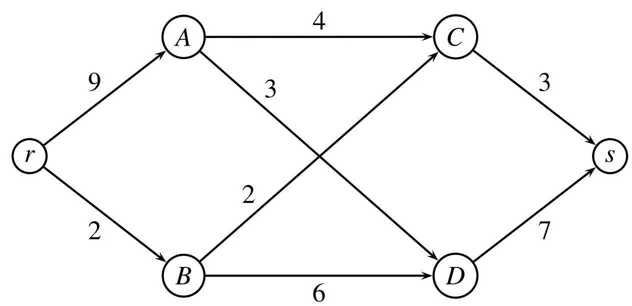
Problem 17 (2 point) Finally, consider the same problem as in previous question, but assume that k is not fixed. Assume that you would like to find a maximum flow, i.e. maximize k, How would you solve this problem?
Hint: You may solve several maximum-flow problems. But try to use as few as possible. ■

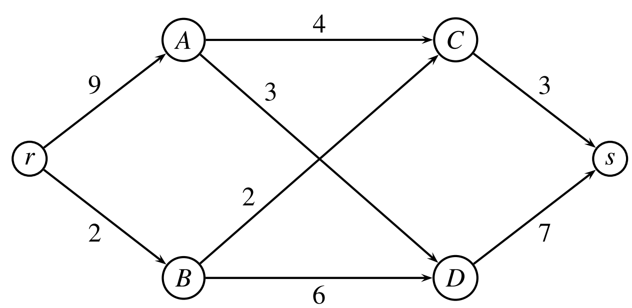
end of assignment

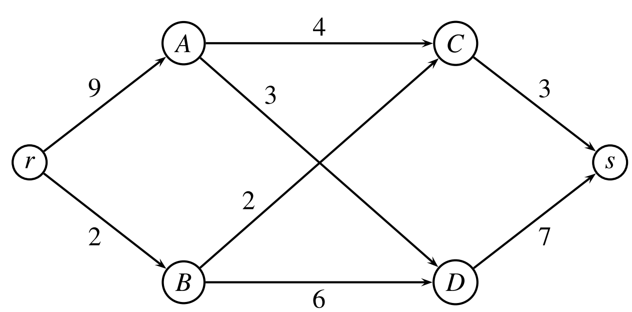
Maximum flow problem

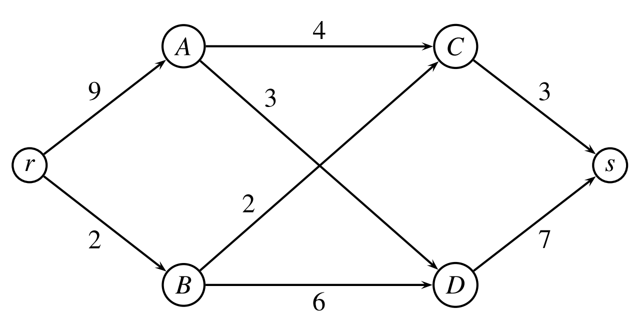
Study number:

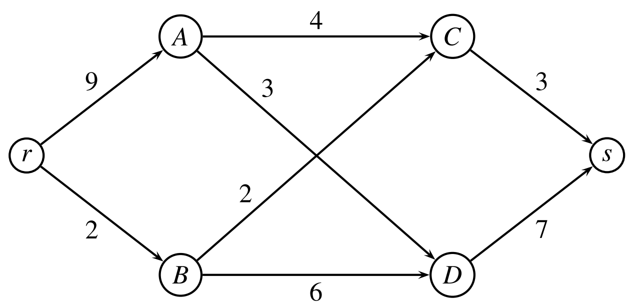








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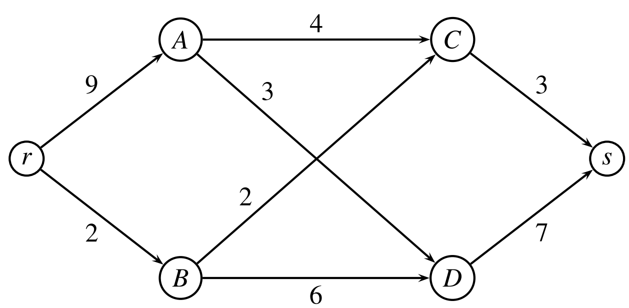
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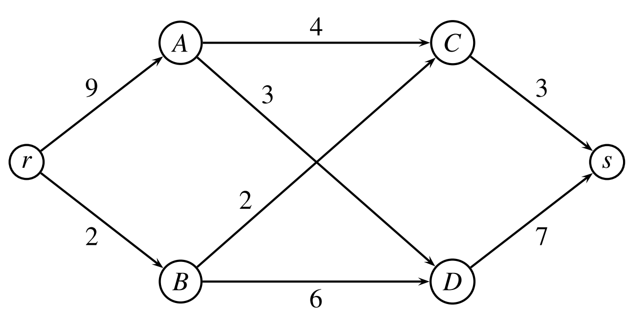
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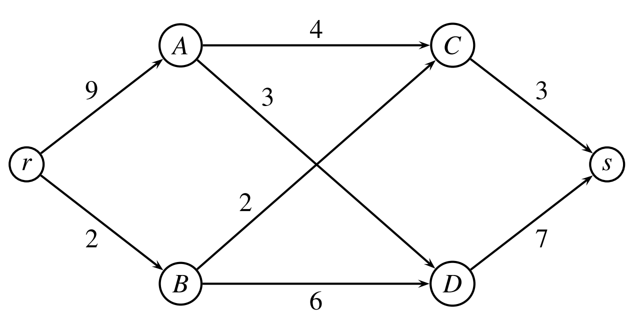
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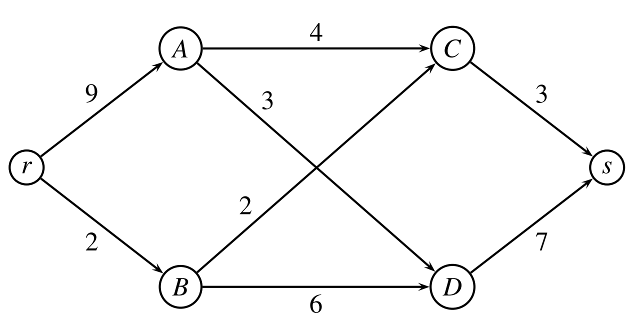
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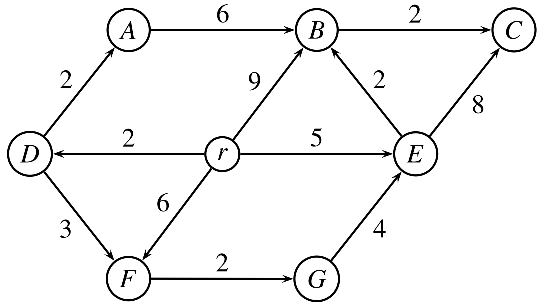
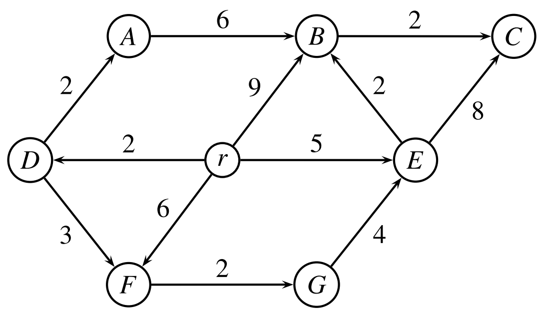
A^* algorithm for shortest path **Study number:**





Minimum cost flow problem

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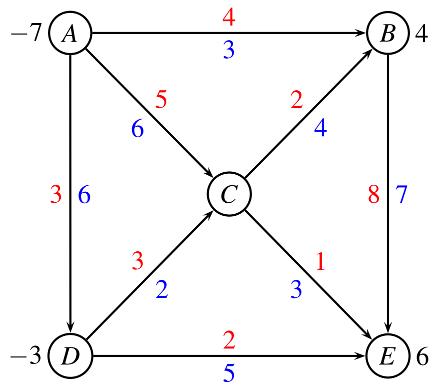

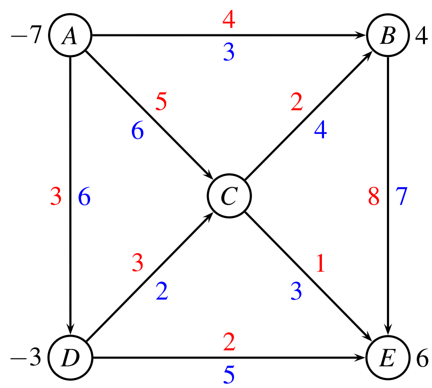

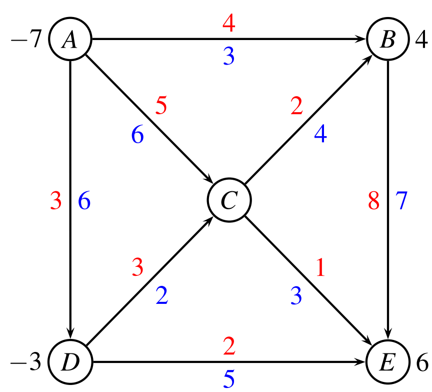

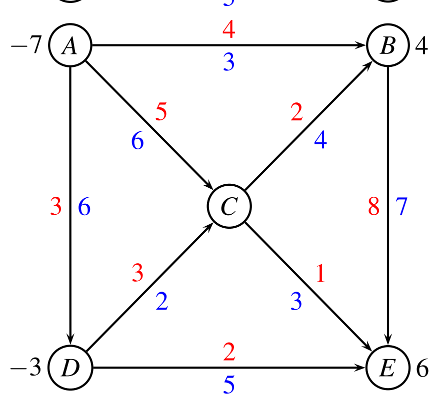

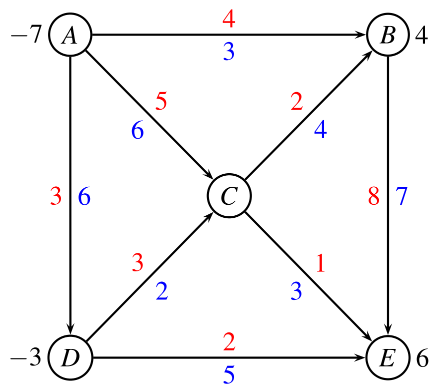
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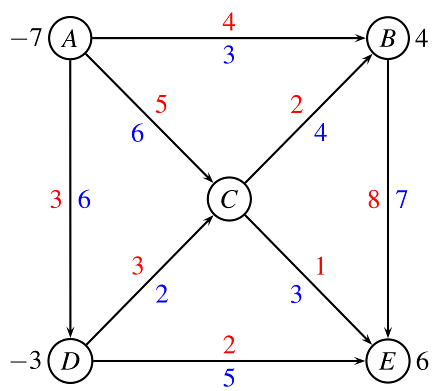
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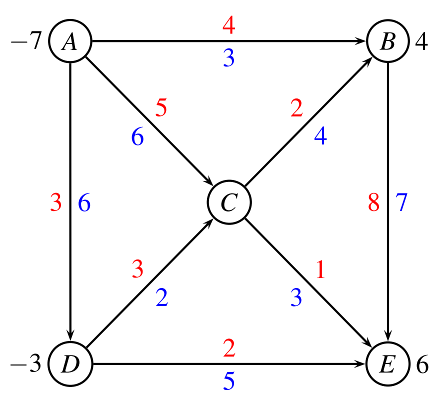
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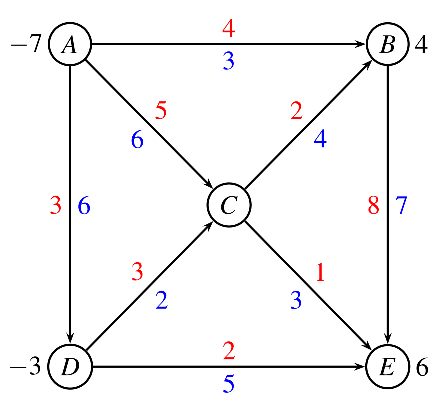
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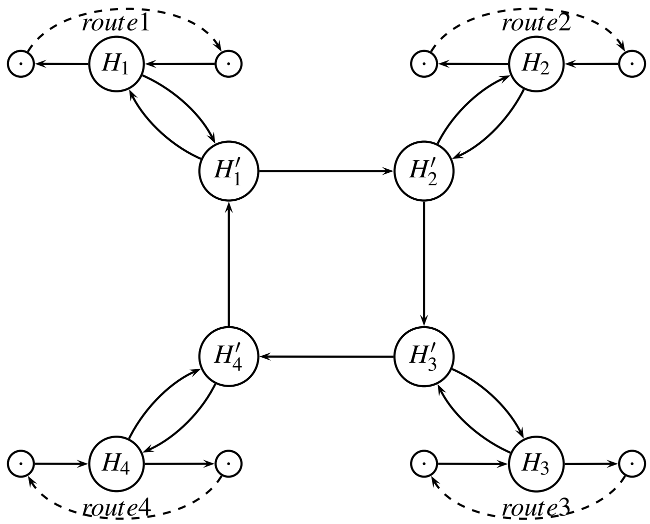
Modeling a hub **Study number:**



Maximum flow with lower bound on edges **Study number:**